

Numerical Methods — Final Exam Solutions

Islamic University of Technology (IUT)
Department of Computer Science and Engineering
Math 4641 / Math 4543

Worked solutions to four semester-final question papers

Compiled 24 August 2026

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1 How this document is organised

Four semester-final papers are solved here, grouped by the exam session in which they were set and ordered chronologically:

#	Session	Course	Semester	Date	Source file
A	CSE 6th Sem	Math 4641	Summer 2020–2021	15 April 2022	CSE final 2022.pdf
B	SWE 5th Sem	Math 4543	Winter 2021–2022	12 December 2022	SWE Final 2022.pdf
C	CSE 6th Sem	Math 4641	Summer 2021–2022	15 May 2023	final 2023.pdf
D	SWE 5th Sem	Math 4543	Winter 2024–2025	08 April 2026	Numericals Final.pdf

Every problem is restated in full before its solution. Marks, COs and POs from the right margin of the original papers are reproduced in square brackets.

1.1 Index of exact duplicates (solved once, skipped later)

Three questions are repeated verbatim across sessions. They are solved at their **first** chronological appearance and skipped afterwards with a pointer:

Repeated question	Exact duplicate of	Solved in
D — Q6(b)(i) LU decomposition of $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$	A — Q4(c)	Session A
D — Q6(c) Orders of error of Forward / Backward / Central Difference rules	A — Q1(a)	Session A
C — Q2(b) Variable transformation under linearization (Exponential, Power, Growth)	A — Q3(b)	Session A

1.1.1 Near-duplicates (distinct enough to be solved separately)

These overlap heavily but differ in data, in the point of evaluation, or in what is asked, so each is worked out in full with a cross-reference:

- **B — Q5(a)** (multi-segment Simpson's 3/8) vs. **A — Q4(b)**: same derivation, B additionally asks for the minimum number of points.
- **D — Q5(a)** (Trapezoidal by method of coefficients) vs. **B — Q3(a)**: B also asks for the calculus derivation and the geometric reading.
- **B — Q5(b)** vs. **C — Q1(a)(ii)**: identical data table, but different evaluation point and C also wants f'' .

- **B — Q4** vs. **A — Q6(a)**: identical Lagrange procedure, different data sets.
- **D — Q4(a)(ii)** vs. **A — Q5(b)**: both ask whether the least-squares constants give an absolute minimum.

1.2 Conventions used throughout

- $|\epsilon_a|$ — absolute relative approximate error, $\left| \frac{\text{present} - \text{previous}}{\text{present}} \right| \times 100\%$.
- $|\epsilon_t|$ — absolute relative true error, $\left| \frac{\text{true} - \text{approx}}{\text{true}} \right| \times 100\%$.
- Intermediate arithmetic is carried at full precision and reported to 4–7 significant figures.

2 Session A — Math 4641, Summer Semester 2020–2021 (15 April 2022)

B.Sc. in CSE, 6th Semester — Duration 3 hours — Full marks 150

2.1 Question 1(a) [9+5, CO1/CO3, PO1]

Problem. Use Taylor's polynomial to derive the orders of error for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for choosing one of these three.

2.1.1 Solution

Taylor's theorem. For f sufficiently smooth near x_i ,

$$f(x_i + h) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f'''(x_i)}{3!}h^3 + \dots \quad (1)$$

$$f(x_i - h) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 - \frac{f'''(x_i)}{3!}h^3 + \dots \quad (2)$$

Forward Difference. Rearranging (1),

$$f'(x_i) = \frac{f(x_i + h) - f(x_i)}{h} - \frac{f''(x_i)}{2}h - \frac{f'''(x_i)}{6}h^2 - \dots$$

Truncating after the first term and applying the mean-value form of the remainder,

$$f'(x_i) \approx \frac{f(x_i + h) - f(x_i)}{h}, \quad E_t = -\frac{f''(\xi)}{2}h = O(h)$$

Backward Difference. Rearranging (2),

$$f'(x_i) = \frac{f(x_i) - f(x_i - h)}{h} + \frac{f''(x_i)}{2}h - \frac{f'''(x_i)}{6}h^2 + \dots$$

$$f'(x_i) \approx \frac{f(x_i) - f(x_i - h)}{h}, \quad E_t = +\frac{f''(\xi)}{2}h = O(h)$$

Central Difference. Subtracting (2) from (1), the h^2 terms cancel identically:

$$f(x_i + h) - f(x_i - h) = 2f'(x_i)h + \frac{2f'''(x_i)}{6}h^3 + \dots$$

$$f'(x_i) \approx \frac{f(x_i + h) - f(x_i - h)}{2h}, \quad E_t = -\frac{f'''(\xi)}{6}h^2 = O(h^2)$$

2.1.2 Mathematical reasoning for the choice

Rule	Leading error term	Order	Nodes used
Forward	$-\frac{h}{2}f''(\xi)$	$O(h)$	$x_i, x_i + h$
Backward	$+\frac{h}{2}f''(\xi)$	$O(h)$	$x_i - h, x_i$
Central	$-\frac{h^2}{6}f'''(\xi)$	$O(h^2)$	$x_i - h, x_i + h$

For $h < 1$ we have $h^2 \ll h$, so the Central Difference truncation error is smaller in magnitude. More decisively, look at how the error *responds to refinement*. If h is halved:

$$\text{FD/BD: } E \rightarrow \frac{1}{2}E \quad \text{but} \quad \text{CD: } E \rightarrow \frac{1}{4}E$$

so the Central Difference gains accuracy four times faster per refinement. The structural reason is visible in the derivation: the forward and backward errors are $\pm \frac{h}{2}f''(\xi)$ — *equal in magnitude and opposite in sign* — so averaging the two stencils (which is exactly what the central difference does) annihilates the $O(h)$ term.

Therefore the Central Difference rule is chosen whenever data exist on both sides of x_i . (See Session C Q1(b) for why the one-sided rules are still indispensable at the ends of a data set.)

2.2 Question 1(b) [11, CO2, PO2]

Problem. A tissue paper company advertises that every roll of their toilet paper has at least 250 sheets. The probability that there are 250 or more sheets in the roll is given by

$$P(y \geq 250) = \int_{250}^{\infty} 0.3515 e^{-0.3881(y-252.2)^2} dy.$$

By theoretical reasoning the above integral calculates the probability up to infinite sheets. But practically it has been seen that the number of sheets rarely exceeds 270. Considering 270 as the maximum possible number of sheets, the integral can be approximated as

$$P(y \geq 250) = \int_{250}^{270} 0.3515 e^{-0.3881(y-252.2)^2} dy.$$

Use the **2-segment Trapezoidal rule** to find the probability that there are 250 or more sheets in a random toilet paper roll.

2.2.1 Solution

The 2-segment (composite) Trapezoidal rule is

$$I \approx \frac{h}{2} [f(a) + 2f(a+h) + f(b)], \quad h = \frac{b-a}{n} = \frac{270-250}{2} = 10.$$

Nodes $y_0 = 250$, $y_1 = 260$, $y_2 = 270$ with $f(y) = 0.3515 e^{-0.3881(y-252.2)^2}$.

i	y_i	exponent $-0.3881(y_i - 252.2)^2$	$f(y_i)$
0	250	-1.87840	5.372109×10^{-2}
1	260	-23.61200	1.955970×10^{-11}
2	270	-122.96560	1.388807×10^{-54}

$$I \approx \frac{10}{2} \left[5.372109 \times 10^{-2} + 2(1.955970 \times 10^{-11}) + 1.388807 \times 10^{-54} \right]$$

$$P(y \geq 250) \approx 0.26861$$

2.2.2 Remark on the quality of this estimate

The exact value of the definite integral is 0.97377, so the 2-segment estimate carries $|\epsilon_t| = 72.4\%$ error. The reason is visible in the table: the integrand is a very narrow Gaussian centred at $y = 252.2$ with an effective width of only $\approx 1/\sqrt{0.3881} \approx 1.6$ sheets, and it has already collapsed to $\sim 10^{-11}$ by $y = 260$. With $h = 10$ the rule places both interior/right nodes far out in the dead tail and completely misses the peak. Practically one would need $h \lesssim 0.5$ (i.e. $n \geq 40$ segments) here — a useful reminder that a low-order composite rule is only as good as its resolution of the integrand's features.

2.3 Question 2(a) [4+5+4, CO1, PO1]

Problem. Without using linearization transform, describe the individual solution approaches for the following nonlinear regression models: (i) Exponential model, (ii) Growth model, (iii) Polynomial model.

2.3.1 Solution

In every case the criterion is the same — minimise the sum of the squares of the residuals

$$S_r = \sum_{i=1}^n (y_i - f(x_i))^2$$

and set the partial derivatives with respect to each parameter to zero. What differs is the *nature of the resulting equations*.

(i) **Exponential model** $y = a e^{bx}$

$$S_r = \sum_{i=1}^n (y_i - a e^{bx_i})^2$$

$$\frac{\partial S_r}{\partial a} = -2 \sum (y_i - a e^{bx_i}) e^{bx_i} = 0 \implies a = \frac{\sum_{i=1}^n y_i e^{bx_i}}{\sum_{i=1}^n e^{2bx_i}} \quad (\text{A})$$

$$\frac{\partial S_r}{\partial b} = -2 \sum (y_i - ae^{bx_i}) a x_i e^{bx_i} = 0 \implies \sum y_i x_i e^{bx_i} - a \sum x_i e^{2bx_i} = 0 \quad (\text{B})$$

Substituting (A) into (B) eliminates a and leaves **one nonlinear equation in the single unknown b** :

$$g(b) = \sum_{i=1}^n y_i x_i e^{bx_i} - \frac{\sum y_i e^{bx_i}}{\sum e^{2bx_i}} \sum_{i=1}^n x_i e^{2bx_i} = 0$$

Solve $g(b) = 0$ by Bisection or Newton–Raphson; back-substitute into (A) for a .

(ii) **Growth (saturation-growth) model** $y = \frac{ax}{b+x}$

$$S_r = \sum_{i=1}^n \left(y_i - \frac{ax_i}{b+x_i} \right)^2$$

$$\frac{\partial S_r}{\partial a} = 0 \implies a = \frac{\sum \frac{x_i y_i}{b+x_i}}{\sum \frac{x_i^2}{(b+x_i)^2}} \quad (\text{C})$$

$$\frac{\partial S_r}{\partial b} = 0 \implies \sum \left(y_i - \frac{ax_i}{b+x_i} \right) \frac{ax_i}{(b+x_i)^2} = 0 \quad (\text{D})$$

Again substitute (C) into (D) to obtain a single nonlinear equation in b , solved by a root-finding method, then recover a from (C).

(iii) **Polynomial model** $y = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m$ This model is **nonlinear in x but linear in the parameters $a_0, \dots, a_m$** , which is what actually matters. Hence

$$\frac{\partial S_r}{\partial a_j} = -2 \sum_{i=1}^n \left(y_i - \sum_{k=0}^m a_k x_i^k \right) x_i^j = 0, \quad j = 0, 1, \dots, m$$

gives the **linear** normal equations

$$\begin{bmatrix} n & \sum x_i & \dots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \dots & \sum x_i^{m+1} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_i^m & \sum x_i^{m+1} & \dots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \vdots \\ \sum x_i^m y_i \end{bmatrix}$$

solved **directly** by Gaussian elimination or LU decomposition — **no iteration and no initial guess are required**, and the solution is unique.

Contrast. Models (i) and (ii) force an iterative root-finding step because the parameters enter nonlinearly; model (iii) does not.

2.4 Question 2(b) [3+3, CO3, PO2]

Problem. What is the drawback in directly using the first 3 terms of the Taylor series for deriving y_{i+1} in the Runge–Kutta 2nd order method? What is the alternative way of approximating the first 3 terms of the series?

2.4.1 Solution

The drawback. The direct Taylor route gives

$$y_{i+1} = y_i + f(x_i, y_i)h + \frac{1}{2!}f'(x_i, y_i)h^2$$

where, because y is itself a function of x , the chain rule forces

$$f'(x, y) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f(x, y).$$

So the method requires the **analytical partial derivatives** $\partial f/\partial x$ and $\partial f/\partial y$. This is a serious practical drawback:

1. For a complicated f the symbolic differentiation is laborious and error-prone, and it must be redone for every new ODE.
2. If f is available only as tabulated data, a black-box subroutine, or the output of another numerical procedure, the derivatives **do not exist in closed form at all**.
3. Higher-order methods would need still higher derivatives, so the approach does not scale.

The alternative. Approximate the same three terms by a **weighted average of two slope evaluations of f alone** — no derivatives:

$$y_{i+1} = y_i + (a_1 k_1 + a_2 k_2)h$$

$$k_1 = f(x_i, y_i), \quad k_2 = f(x_i + p_1 h, y_i + q_{11} k_1 h)$$

Expanding k_2 in a two-variable Taylor series and matching term-by-term with the three-term Taylor series yields the conditions

$$a_1 + a_2 = 1, \quad a_2 p_1 = \frac{1}{2}, \quad a_2 q_{11} = \frac{1}{2}$$

Three equations in four unknowns — one parameter is free, which is precisely why a *family* of RK2 methods exists (see Q2(c)).

2.5 Question 2(c) [6, CO1, PO1]

Problem. What are the different forms of the approximation of Runge–Kutta 2nd order method based on Heun’s method, Midpoint method and Ralston’s method?

2.5.1 Solution

Each choice of the free parameter a_2 in $a_1 + a_2 = 1$, $a_2 p_1 = a_2 q_{11} = \frac{1}{2}$ produces one named scheme.

Method	a_2	a_1	$p_1 = q_{11}$
Heun	1/2	1/2	1
Midpoint	1	0	1/2
Ralston	2/3	1/3	3/4

Heun's method ($a_2 = \frac{1}{2}$):

$$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h, \quad k_1 = f(x_i, y_i), \quad k_2 = f(x_i + h, y_i + k_1 h)$$

Interpretation: average of the slope at the left end and the slope at a predicted right end (predictor–corrector).

Midpoint method ($a_2 = 1$):

$$y_{i+1} = y_i + k_2 h, \quad k_1 = f(x_i, y_i), \quad k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right)$$

Interpretation: use the slope estimated at the midpoint of the interval only.

Ralston's method ($a_2 = \frac{2}{3}$):

$$y_{i+1} = y_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2\right)h, \quad k_1 = f(x_i, y_i), \quad k_2 = f\left(x_i + \frac{3}{4}h, y_i + \frac{3}{4}k_1 h\right)$$

Interpretation: the weights are chosen to **minimise the bound on the truncation error**, so Ralston's is generally the most accurate of the three.

2.6 Question 3(a) [12, CO2, PO1]

Problem. Find the Eigenvalues and Eigenvectors of the following matrix A :

$$A = \begin{bmatrix} 2 & 27 & 0 \\ 0 & 4 & 40 \\ 0 & 3 & 30 \end{bmatrix}$$

2.6.1 Solution

Eigenvalues. Expand $\det(A - \lambda I) = 0$ along the first column (only one nonzero entry):

$$\det \begin{bmatrix} 2 - \lambda & 27 & 0 \\ 0 & 4 - \lambda & 40 \\ 0 & 3 & 30 - \lambda \end{bmatrix} = (2 - \lambda) \left[(4 - \lambda)(30 - \lambda) - 120 \right] = 0$$

$$(4 - \lambda)(30 - \lambda) - 120 = 120 - 34\lambda + \lambda^2 - 120 = \lambda^2 - 34\lambda = \lambda(\lambda - 34)$$

$$\implies (2 - \lambda)\lambda(\lambda - 34) = 0 \quad \boxed{\lambda_1 = 2, \quad \lambda_2 = 0, \quad \lambda_3 = 34}$$

(Note $\lambda = 0$: A is singular, consistent with rows 2 and 3 being proportional in their trailing 2×2 block, $\det \begin{bmatrix} 4 & 40 \\ 3 & 30 \end{bmatrix} = 0$.)

Eigenvector for $\lambda_1 = 2$.

$$(A - 2I)\mathbf{v} = \begin{bmatrix} 0 & 27 & 0 \\ 0 & 2 & 40 \\ 0 & 3 & 28 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}$$

Row 1 $\Rightarrow 27y = 0 \Rightarrow y = 0$; Row 2 $\Rightarrow 40z = 0 \Rightarrow z = 0$; x free.

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Eigenvector for $\lambda_2 = 0$.

$$A\mathbf{v} = \mathbf{0} : \begin{cases} 2x + 27y = 0 \\ 4y + 40z = 0 \\ 3y + 30z = 0 \end{cases}$$

Rows 2 and 3 both give $y = -10z$. Row 1: $x = -\frac{27}{2}y = -\frac{27}{2}(-10z) = 135z$. Take $z = 1$:

$$\mathbf{v}_2 = \begin{bmatrix} 135 \\ -10 \\ 1 \end{bmatrix}$$

Eigenvector for $\lambda_3 = 34$.

$$(A - 34I) = \begin{bmatrix} -32 & 27 & 0 \\ 0 & -30 & 40 \\ 0 & 3 & -4 \end{bmatrix}$$

Row 2 $\Rightarrow -30y + 40z = 0 \Rightarrow z = \frac{3}{4}y$. Row 1 $\Rightarrow -32x + 27y = 0 \Rightarrow x = \frac{27}{32}y$. Take $y = 32$: $x = 27$, $z = 24$.

$$\mathbf{v}_3 = \begin{bmatrix} 27 \\ 32 \\ 24 \end{bmatrix}$$

$$\text{Verification. } A\mathbf{v}_3 = \begin{bmatrix} 2(27) + 27(32) \\ 4(32) + 40(24) \\ 3(32) + 30(24) \end{bmatrix} = \begin{bmatrix} 918 \\ 1088 \\ 816 \end{bmatrix} = 34 \begin{bmatrix} 27 \\ 32 \\ 24 \end{bmatrix} \checkmark$$

2.7 Question 3(b) [2+2+2, CO1, PO1]

Problem. Describe how linearization transform is applied on the following nonlinear models:
(i) Exponential model, (ii) Power model, (iii) Growth model.

■ Session C Q2(b) is an exact duplicate of this question and is skipped there.

2.7.1 Solution

The aim in every case is to force the model into the **straight-line form**

$$Y = A_0 + A_1 X$$

so the ordinary least-squares formulae

$$A_1 = \frac{n \sum X_i Y_i - \sum X_i \sum Y_i}{n \sum X_i^2 - (\sum X_i)^2}, \quad A_0 = \bar{Y} - A_1 \bar{X}$$

can be applied directly, after which the original constants are recovered.

(i) **Exponential model** $y = ae^{bx}$ Take natural logarithms of both sides:

$$\ln y = \ln a + bx$$

transformed quantity	definition
Y	$\ln y$
X	x
A_0	$\ln a$
A_1	b

Recover: $a = e^{A_0}$, $b = A_1$.

(ii) **Power model** $y = ax^b$ Take natural logarithms:

$$\ln y = \ln a + b \ln x$$

transformed quantity	definition
Y	$\ln y$
X	$\ln x$
A_0	$\ln a$
A_1	b

Recover: $a = e^{A_0}$, $b = A_1$.

(iii) **Growth (saturation-growth) model** $y = \frac{ax}{b+x}$ Take reciprocals:

$$\frac{1}{y} = \frac{b+x}{ax} = \frac{b}{a} \cdot \frac{1}{x} + \frac{1}{a}$$

transformed quantity	definition
Y	$1/y$
X	$1/x$

transformed quantity	definition
A_0	$1/a$
A_1	b/a

Recover: $a = \frac{1}{A_0}$, $b = \frac{A_1}{A_0}$.

Important caveat. Linearization does **not** minimise $\sum (y_i - f(x_i))^2$; it minimises the sum of squared residuals *in the transformed variables*. The constants it produces therefore differ from the true nonlinear least-squares constants of Q2(a). See Session D Q4(c) for a full discussion.

2.8 Question 3(c) [2+5, CO3, PO2]

Problem. Why do we normally use polynomials as interpolating functions? Can single polynomials always be used to accurately interpolate data?

2.8.1 Solution

Why polynomials.

1. **Weierstrass approximation theorem** — any continuous function on a closed interval $[a, b]$ can be approximated uniformly to arbitrary accuracy by some polynomial. Polynomials are therefore *sufficient* as a class.
2. **Existence and uniqueness** — through $n+1$ distinct data points there passes exactly one polynomial of degree $\leq n$ (proved in Session D Q3(a)), so the interpolant is well-defined regardless of which construction (Lagrange, Newton's divided differences, direct method) is used.
3. **Cheap and exact arithmetic** — evaluation needs only $+$, $-$, $\times$ (Horner's scheme costs n multiplications and n additions), which every processor does natively; no transcendental function evaluations are involved.
4. **Closed under calculus** — the derivative and the antiderivative of a polynomial are again polynomials, obtained by trivial coefficient manipulations. This is what makes polynomial-based numerical differentiation and integration (Trapezoidal, Simpson) possible at all.

Can a single polynomial always interpolate accurately? — No.

1. **Runge's phenomenon.** For $f(x) = \frac{1}{1+25x^2}$ on $[-1, 1]$ with equispaced nodes, the interpolating polynomial oscillates with *increasing* amplitude near the ends as the degree n grows; the error diverges, $\max |f - P_n| \rightarrow \infty$ as $n \rightarrow \infty$. Adding data makes matters worse, not better.
2. **Ill-conditioning.** The direct-method (Vandermonde) system becomes catastrophically ill-conditioned for large n ; its condition number grows exponentially, so round-off swamps the coefficients.
3. **Global sensitivity.** A single high-degree polynomial is a *global* object: perturbing one data value alters the curve everywhere, and a small measurement error can produce large excursions far from the perturbed point.

4. **Wrong shape.** Data with a corner, a jump, or a sharp local feature (such as the thermocline of Q6(a)) simply is not polynomial-like, and no single smooth polynomial reproduces it faithfully.

Practical resolution. Use **piecewise low-order interpolation** — linear or quadratic segments, or better, **cubic splines**, which keep each piece cubic while enforcing C^2 continuity at the knots. This retains the arithmetic advantages of polynomials while eliminating the oscillation and the conditioning problem. (Chebyshev nodes are the other classical cure if a single global polynomial must be retained.)

2.9 Question 4(a) [9, CO1, PO1]

Problem. For approximating $I = \int_a^b f(x) dx$ using Simpson's 3/8 rule, how are the points chosen over the interval $[a, b]$, and what are their exact coordinates in terms of a and b ? Provide necessary illustrations.

2.9.1 Solution

Simpson's 3/8 rule replaces f by the **unique cubic** passing through **four equally spaced points** spanning $[a, b]$. The interval is divided into 3 equal sub-intervals of width

$$h = \frac{b - a}{3}$$

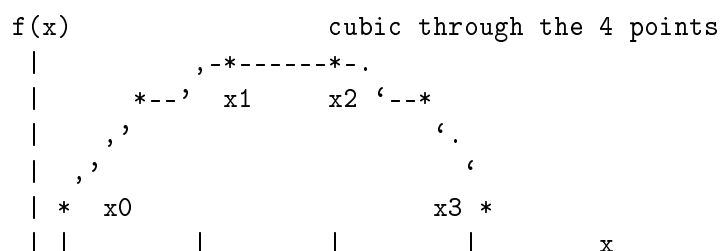
and the nodes are taken at the sub-interval boundaries, including both endpoints:

$x_0 = a$ $x_1 = a + h = a + \frac{b - a}{3} = \frac{2a + b}{3}$ $x_2 = a + 2h = a + \frac{2(b - a)}{3} = \frac{a + 2b}{3}$ $x_3 = a + 3h = b$
--

Both endpoints are used (so the rule is a **closed** Newton–Cotes formula) and the spacing is uniform, which is what allows the weights 1, 3, 3, 1 to be constants independent of f :

$$I \approx \frac{3h}{8} \left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) \right] = \frac{b - a}{8} \left[f(a) + 3f\left(\frac{2a+b}{3}\right) + 3f\left(\frac{a+2b}{3}\right) + f(b) \right]$$

Illustration.



$$\begin{array}{cccc}
a & 2a+b & a+2b & b \\
& \text{---} & \text{----} & \\
& 3 & 3 & \\
|<--h-->|<--h-->|<--h-->| & h = (b-a)/3
\end{array}$$

The shaded area under the cubic between a and b is the estimate of I .

Truncation error. $E_t = -\frac{(b-a)^5}{6480}f^{(4)}(\xi)$, so the rule is **exact for all polynomials of degree ≤ 3** .

2.10 Question 4(b) [8, CO1, PO2]

Problem. The minimum number of segments for Simpson's $3/8$ rule is 3. Given that the 3-segment formula is $I = \frac{3h}{8}[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$, derive the general n -segment formula.

■ *Session B Q5(a) asks for the same derivation with an extra part; both are written out.*

2.10.1 Solution

Let n be the number of segments, **necessarily a multiple of 3**, with

$$h = \frac{b-a}{n}, \quad x_i = a + ih, \quad i = 0, 1, \dots, n.$$

Step 1 — split the integral into groups of three segments.

$$\int_a^b f dx = \int_{x_0}^{x_3} f dx + \int_{x_3}^{x_6} f dx + \dots + \int_{x_{n-3}}^{x_n} f dx = \sum_{j=1}^{n/3} \int_{x_{3j-3}}^{x_{3j}} f dx$$

Step 2 — apply the given 3-segment formula to each group. Each group spans exactly three sub-intervals of width h , so

$$\int_{x_{3j-3}}^{x_{3j}} f dx \approx \frac{3h}{8} [f(x_{3j-3}) + 3f(x_{3j-2}) + 3f(x_{3j-1}) + f(x_{3j})]$$

Step 3 — sum and collect like nodes.

$$I \approx \frac{3h}{8} \sum_{j=1}^{n/3} [f(x_{3j-3}) + 3f(x_{3j-2}) + 3f(x_{3j-1}) + f(x_{3j})]$$

Now observe the bookkeeping:

- x_0 and x_n each appear **once**, with weight 1.
- Nodes with index $\not\equiv 0 \pmod{3}$ appear **once**, with weight 3.
- Interior nodes with index $\equiv 0 \pmod{3}$ (i.e. $x_3, x_6, \dots, x_{n-3}$) are the right end of one group and the left end of the next, so they appear **twice** with weight 1 each $\Rightarrow$ total weight 2.

$$I \approx \frac{3h}{8} \left[f(x_0) + 3 \sum_{\substack{i=1 \\ i \not\equiv 0 \pmod{3}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=3 \\ i \equiv 0 \pmod{3}}}^{n-3} f(x_i) + f(x_n) \right], \quad h = \frac{b-a}{n}$$

Written out explicitly, the weight pattern is

$$1, 3, 3, 2, 3, 3, 2, \dots, 2, 3, 3, 1$$

Sanity check ($n = 3$). There are no interior multiples of 3, so the formula collapses to $\frac{3h}{8}[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] \checkmark$

Error. Summing the $n/3$ group errors gives $E_t = -\frac{(b-a)h^4}{80} \overline{f^{(4)}} = O(h^4)$.

2.11 Question 4(c) [8, CO2, PO1]

Problem. What is the LU decomposition of A , where

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

■ *Session D Q6(b)(i) is an exact duplicate of this question and is skipped there.*

2.11.1 Solution

Use **Doolittle's method**: $A = LU$ with L unit-lower-triangular (the multipliers of Gaussian elimination) and U the resulting upper-triangular matrix.

Step 1 — eliminate column 1.

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1: \quad R_2 \leftarrow R_2 - 1 \cdot R_1$$

$$\begin{bmatrix} 1 & -2 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -3 & 4 \end{bmatrix}$$

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2: \quad R_3 \leftarrow R_3 - 2 \cdot R_1$$

$$\begin{bmatrix} 2 & 3 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 3 \end{bmatrix}$$

$$A \longrightarrow \begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 2 — eliminate column 2.

$$m_{32} = \frac{1}{-3} = -\frac{1}{3}: \quad R_3 \leftarrow R_3 - \left(-\frac{1}{3}\right) R_2$$

$$\begin{bmatrix} 0 & 1 & 3 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 3 + \frac{4}{3} \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{13}{3} \end{bmatrix}$$

Result.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -\frac{1}{3} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & \frac{13}{3} \end{bmatrix}$$

Verification $LU = A$:

$$LU = \begin{bmatrix} 1 & 1 & -1 \\ 1(1) + 1(0) & 1(1) + 1(-3) & 1(-1) + 1(4) \\ 2(1) + (-\frac{1}{3})(0) & 2(1) + (-\frac{1}{3})(-3) & 2(-1) + (-\frac{1}{3})(4) + \frac{13}{3} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = A \checkmark$$

By-product. $\det A = \det L \cdot \det U = 1 \cdot (1 \times (-3) \times \frac{13}{3}) = -13$.

2.12 Question 5(a) [5+5, CO3, PO2]

Problem. What is the motivation behind the use of LU decomposition? Provide a scenario where the use of LU decomposition can enhance the performance of a numerical analysis system.

2.12.1 Solution

Motivation — the cost structure of repeated solves.

Solving $A\mathbf{x} = \mathbf{b}$ by Gaussian elimination costs

$$\underbrace{\frac{n^3}{3} + O(n^2)}_{\text{forward elimination}} + \underbrace{O(n^2)}_{\text{back substitution}} \text{ flops}$$

and the expensive elimination phase **operates on $\mathbf{b}$ as well**. If the right-hand side changes, the whole $O(n^3)$ elimination must be repeated even though A has not changed at all — the factorisation work is thrown away.

LU decomposition separates the two:

$$A\mathbf{x} = \mathbf{b} \implies LU\mathbf{x} = \mathbf{b} \implies \begin{cases} L\mathbf{z} = \mathbf{b} & \text{forward substitution, } O(n^2) \\ U\mathbf{x} = \mathbf{z} & \text{back substitution, } O(n^2) \end{cases}$$

The $O(n^3/3)$ factorisation depends **only on A** and is done **once**; each additional right-hand side then costs only $O(n^2)$.

Comparison for m right-hand sides sharing the same A :

Approach	Cost
Gaussian elimination repeated	$m \left(\frac{n^3}{3} \right)$
Compute A^{-1} , then multiply	$\approx n^3 \text{ (inverse)} + m n^2$

Approach	Cost
LU decomposition	$\frac{n^3}{3} + m n^2$

LU also beats explicit inversion: forming A^{-1} is about **three times** the work of an LU factorisation (it is n LU-solves), it destroys sparsity, and it is numerically less stable. Two further by-products come free: $\det A = \prod u_{ii}$, and A^{-1} itself can be obtained (when genuinely needed) by solving $LU \mathbf{x}_j = \mathbf{e}_j$ for $j = 1..n$.

Scenario. Consider the missile-velocity problem of **Session C Q3(b)**: two missiles are tracked and a cubic $v = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ must be fitted through the *same* four times $t = 15, 20, 22.5, 30$ for each missile. The direct method produces

$$\underbrace{\begin{bmatrix} 1 & 15 & 225 & 3375 \\ 1 & 20 & 400 & 8000 \\ 1 & 22.5 & 506.25 & 11390.625 \\ 1 & 30 & 900 & 27000 \end{bmatrix}}_{\text{identical for both missiles}} \mathbf{a} = \mathbf{v}_1 \text{ or } \mathbf{v}_2$$

Factor once, then run two cheap forward/back substitutions. In a real tracking system with hundreds of targets sampled at a fixed schedule, this turns an $O(mn^3)$ workload into $O(n^3 + mn^2)$.

Other classic scenarios. Implicit ODE/PDE time-stepping (the same stiffness matrix is used at every time step with a new load vector), structural analysis of one structure under many load cases, and iterative refinement of a computed solution.

2.13 Question 5(b) [9+6, CO1, PO2]

Problem. Given n data pairs $(x_1, y_1), \dots, (x_n, y_n)$, do the values of the constants a_0 and a_1 in the least square regression line model $y = a_0 + a_1 x$ correspond to the absolute minimum of the sum of the squares of residuals? Are the constants of regression unique?

2.13.1 Solution

Yes to both. Here is the proof.

Let

$$S_r(a_0, a_1) = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2.$$

Step 1 — stationary point.

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i) = 0, \quad \frac{\partial S_r}{\partial a_1} = -2 \sum (y_i - a_0 - a_1 x_i) x_i = 0$$

giving the normal equations

$$n a_0 + a_1 \sum x_i = \sum y_i, \quad a_0 \sum x_i + a_1 \sum x_i^2 = \sum x_i y_i$$

with solution

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}, \quad a_0 = \bar{y} - a_1 \bar{x}.$$

Step 2 — the stationary point is a minimum (second-derivative test).

$$\frac{\partial^2 S_r}{\partial a_0^2} = 2n, \quad \frac{\partial^2 S_r}{\partial a_1^2} = 2 \sum x_i^2, \quad \frac{\partial^2 S_r}{\partial a_0 \partial a_1} = 2 \sum x_i$$

Hessian

$$H = 2 \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

- $H_{11} = 2n > 0$.
- $\det H = 4 \left[n \sum x_i^2 - (\sum x_i)^2 \right] = 4n \sum_{i=1}^n (x_i - \bar{x})^2 > 0$

using the identity $n \sum x_i^2 - (\sum x_i)^2 = n \sum (x_i - \bar{x})^2$, which is strictly positive **provided the x_i are not all identical** — i.e. provided at least two distinct abscissae exist, which is exactly the condition for a line to be determinable at all.

H is therefore **positive definite**, so the stationary point is a strict local minimum.

Step 3 — from local to absolute. S_r is a **quadratic form** in (a_0, a_1) with positive-definite Hessian — a convex paraboloid. For such a function every local minimum is the **global (absolute) minimum**, and there can be exactly one stationary point (the linear normal equations have a unique solution because their determinant is the nonzero $n \sum (x_i - \bar{x})^2$).

The constants give the absolute minimum of S_r , and they are unique.

When uniqueness fails. Only in the degenerate case $x_1 = x_2 = \dots = x_n$, where $\det H = 0$: the data give a vertical line, infinitely many (a_0, a_1) pairs are equally good, and the model $y = a_0 + a_1 x$ is inappropriate.

2.14 Question 6(a) [6+7+3, CO2/CO3, PO3]

Problem. To maximize the catch of bass in a lake, it is suggested to throw the line to the depth of the thermocline. You can infer that there is a thermocline at a particular depth by noticing the sudden change in temperature. The following table provides the depth vs temperature data for the lake:

Depth, h (m)	Temperature, T (°C)
0	19.1
−1	19.1
−2	19.0
−3	18.8
−4	18.7
−5	18.3
−6	18.2
−7	17.6

Depth, h (m)	Temperature, T ($^{\circ}\text{C}$)
-8	11.7
-9	9.9
-10	9.1

It is noticed that there is a large difference in temperature from $h = -7$ to $h = -8$, so the thermocline must be somewhere between these two depths. The intermediate temperature data between these two depths is missing. Determine the temperature at $h = -7.5$ using a **first order Lagrange polynomial** and then a **second order Lagrange polynomial** and find the absolute relative approximate error.

2.14.1 Solution

First order (linear) Lagrange polynomial Choose the two nodes bracketing $h = -7.5$: $h_0 = -7$ ($T = 17.6$), $h_1 = -8$ ($T = 11.7$).

$$T_1(h) = L_0(h)T(h_0) + L_1(h)T(h_1), \quad L_0 = \frac{h - h_1}{h_0 - h_1}, \quad L_1 = \frac{h - h_0}{h_1 - h_0}$$

At $h = -7.5$:

$$L_0 = \frac{-7.5 - (-8)}{-7 - (-8)} = \frac{0.5}{1} = 0.5, \quad L_1 = \frac{-7.5 - (-7)}{-8 - (-7)} = \frac{-0.5}{-1} = 0.5$$

(Check: $L_0 + L_1 = 1$ ✓)

$$T_1(-7.5) = 0.5(17.6) + 0.5(11.7) = 8.80 + 5.85$$

$$\boxed{T_1(-7.5) = 14.65 \text{ }^{\circ}\text{C}}$$

Second order (quadratic) Lagrange polynomial The two nearest nodes are again -7 and -8 ; the next nearest are -6 and -9 , both at distance 1.5. Taking $h_2 = -9$ (the choice that keeps the stencil on the cold side of the thermocline, where the data are changing fastest):

$$h_0 = -7 \text{ (17.6)}, \quad h_1 = -8 \text{ (11.7)}, \quad h_2 = -9 \text{ (9.9)}$$

$$\begin{aligned} L_0 &= \frac{(h - h_1)(h - h_2)}{(h_0 - h_1)(h_0 - h_2)} = \frac{(0.5)(1.5)}{(1)(2)} = 0.375 \\ L_1 &= \frac{(h - h_0)(h - h_2)}{(h_1 - h_0)(h_1 - h_2)} = \frac{(-0.5)(1.5)}{(-1)(1)} = 0.750 \\ L_2 &= \frac{(h - h_0)(h - h_1)}{(h_2 - h_0)(h_2 - h_1)} = \frac{(-0.5)(0.5)}{(-2)(-1)} = -0.125 \end{aligned}$$

(Check: $0.375 + 0.750 - 0.125 = 1$ ✓)

$$T_2(-7.5) = 0.375(17.6) + 0.750(11.7) - 0.125(9.9) = 6.600 + 8.775 - 1.2375$$

$$\boxed{T_2(-7.5) = 14.1375 \text{ }^{\circ}\text{C}}$$

Absolute relative approximate error

$$|\epsilon_a| = \left| \frac{T_2 - T_1}{T_2} \right| \times 100 = \left| \frac{14.1375 - 14.65}{14.1375} \right| \times 100 = \frac{0.5125}{14.1375} \times 100$$

$$\boxed{|\epsilon_a| = 3.625\%}$$

Since $|\epsilon_a| = 3.625\%$, at least **one** significant digit is correct ($|\epsilon_a| < 5\% = 0.5 \times 10^{-1}$).

Alternative stencil. Had $h_2 = -6$ (18.2) been used instead, the weights become $L_{-6} = -0.125$, $L_{-7} = 0.750$, $L_{-8} = 0.375$, giving $T_2(-7.5) = 15.3125^\circ\text{C}$ and $|\epsilon_a| = 4.327\%$. The large spread between the two admissible quadratics (14.14 vs 15.31) is itself the signature of the thermocline: the data are not locally polynomial there, which is precisely why the temperature is dropping so sharply.

2.15 Question 6(b) [5+4, CO1, PO1]

Problem. Derive Simpson's 1/3 rule using the method of coefficients. Using necessary illustrations, determine the geometric interpretation of this derivation.

2.15.1 Solution

The method of coefficients **Postulate** the quadrature formula as a weighted sum of function values at three prescribed nodes a , $m = \frac{a+b}{2}$, b :

$$\int_a^b f(x) dx \approx c_0 f(a) + c_1 f\left(\frac{a+b}{2}\right) + c_2 f(b) \quad (*)$$

and **determine** c_0, c_1, c_2 **by requiring** $(*)$ **to be exact for the three lowest monomials** $f = 1, x, x^2$ (three conditions, three unknowns).

Condition 1 ($f = 1$).

$$\int_a^b 1 dx = b - a \implies c_0 + c_1 + c_2 = b - a \quad (\text{i})$$

Condition 2 ($f = x$).

$$\int_a^b x dx = \frac{b^2 - a^2}{2} \implies c_0 a + c_1 \frac{a+b}{2} + c_2 b = \frac{b^2 - a^2}{2} \quad (\text{ii})$$

Condition 3 ($f = x^2$).

$$\int_a^b x^2 dx = \frac{b^3 - a^3}{3} \implies c_0 a^2 + c_1 \frac{(a+b)^2}{4} + c_2 b^2 = \frac{b^3 - a^3}{3} \quad (\text{iii})$$

Solving (i)–(iii). The algebra is cleanest after shifting the origin to the midpoint; put $t = x - m$ and $h = \frac{b-a}{2}$, so the nodes are $t = -h, 0, h$ and the interval is $[-h, h]$. The three conditions become

$$c_0 + c_1 + c_2 = 2h, \quad -c_0h + c_2h = 0, \quad c_0h^2 + c_2h^2 = \frac{2h^3}{3}$$

From the second, $c_0 = c_2$. From the third, $2c_0h^2 = \frac{2h^3}{3} \Rightarrow c_0 = c_2 = \frac{h}{3}$. From the first, $c_1 = 2h - \frac{2h}{3} = \frac{4h}{3}$.

$$c_0 = c_2 = \frac{h}{3} = \frac{b-a}{6}, \quad c_1 = \frac{4h}{3} = \frac{4(b-a)}{6}$$

Substituting back into (*):

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right], \quad h = \frac{b-a}{2}$$

which is Simpson's 1/3 rule. (The name comes from the leading factor $h/3$.)

Bonus — the rule is exact for cubics too. Testing $f = x^3$ with $t = x - m$: $\int_{-h}^h t^3 dt = 0$ and $\frac{h}{3}[(-h)^3 + 4(0) + h^3] = 0 \checkmark$. The formula was built to be exact to degree 2 but is *free* to degree 3, by the odd symmetry of the node set. This is why $E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\xi)$ involves the *fourth* derivative.

Geometric interpretation

```

f(x)
|
|      *-----*-----*      <- the unique parabola P2(x)
|      ,-'      f(m)          '-.
|      , '                      '*  f(b)
|      *  f(a)
|  ##### <- shaded area = the estimate
|  #####
|_ |-----|-----|----- x
   a           m           b
           m=(a+b)/2
|<-----h----->|<-----h----->|

```

The three exactness conditions ($f = 1, x, x^2$) say precisely that the formula must reproduce the integral of **every quadratic**. Since exactly one parabola $P_2(x)$ passes through the three points $(a, f(a))$, $(m, f(m))$, $(b, f(b))$, the formula is nothing other than

$$\int_a^b f(x) dx \approx \int_a^b P_2(x) dx$$

i.e. **the area under the interpolating parabola replaces the area under the true curve**. The weights $(\frac{1}{6}, \frac{4}{6}, \frac{1}{6})(b-a)$ are the exact areas contributed by each node, and their 1 : 4 : 1 ratio expresses that the midpoint ordinate carries four times the influence of either endpoint — geometrically, the midpoint is where a parabola deviates most from the chord.

Compare with the Trapezoidal rule, which integrates the interpolating **straight line** through only $(a, f(a))$ and $(b, f(b))$: Simpson's 1/3 rule is the natural next member of the family, buying one extra order of accuracy for one extra function evaluation.

3 Session B — Math 4543, Winter Semester 2021–2022 (12 December 2022)

B.Sc. Engg. SWE, 5th Semester — Duration 3 hours — Full marks 150

3.1 Question 1(a) [8+10, CO2, PO1]

Problem. A Trunnion of diameter 12.363" has to be cooled from a room temperature of 80 °F before it can be shrink-fit into a steel hub. The equation that gives the diametric contraction (in inches) of the trunnion in dry-ice (boiling point −108 °F) is:

$$\Delta D = 12.363 \int_{80}^{-108} (-1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6}) dT$$

Here, a positive ΔD indicates expansion and negative value indicates contraction. Find ΔD using the **2-segment and 4-segment Simpson's 1/3 rule**.

3.1.1 Solution

Write $g(T) = -1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6}$, with $a = 80$, $b = -108$ (note $b < a$, so h comes out **negative** — this is fine, the formula handles it and the sign is what makes ΔD negative).

Composite Simpson's 1/3 rule:

$$I \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i=1,3,5,\dots}^{n-1} f(x_i) + 2 \sum_{i=2,4,6,\dots}^{n-2} f(x_i) + f(x_n) \right], \quad h = \frac{b-a}{n}$$

2-segment ($n = 2$)

$$h = \frac{-108 - 80}{2} = -94$$

i	T_i	$g(T_i)$
0	80	6.4319408×10^{-6}
1	−14	5.9258775×10^{-6}
2	−108	5.2028374×10^{-6}

(sample: $g(80) = -1.2278 \times 10^{-11}(6400) + 6.194 \times 10^{-9}(80) + 6.015 \times 10^{-6} = -7.8579 \times 10^{-8} + 4.9552 \times 10^{-7} + 6.015 \times 10^{-6} = 6.43194 \times 10^{-6}$)

$$\begin{aligned} I_2 &= \frac{-94}{3} \left[6.4319408 \times 10^{-6} + 4(5.9258775 \times 10^{-6}) + 5.2028374 \times 10^{-6} \right] \\ &= -31.33333 \times [3.53382882 \times 10^{-5}] = -1.1072664 \times 10^{-3} \end{aligned}$$

$$\Delta D_2 = 12.363 \times (-1.1072664 \times 10^{-3})$$

$\Delta D_{2\text{-seg}} = -1.36891 \times 10^{-2} \text{ in} = -0.013689'' \text{ (contraction)}$

4-segment ($n = 4$)

$$h = \frac{-108 - 80}{4} = -47$$

i	T_i	$g(T_i)$	weight
0	80	6.4319408×10^{-6}	1
1	33	6.2060313×10^{-6}	4
2	-14	5.9258775×10^{-6}	2
3	-61	5.5914796×10^{-6}	4
4	-108	5.2028374×10^{-6}	1

Weighted sum:

$$6.4319408 + 4(6.2060313) + 2(5.9258775) + 4(5.5914796) + 5.2028374 = 70.6765768 \ (\times 10^{-6})$$

$$I_4 = \frac{-47}{3} \times 70.6765768 \times 10^{-6} = -15.66667 \times 70.6765768 \times 10^{-6} = -1.1072664 \times 10^{-3}$$

$$\Delta D_{4\text{-seg}} = -1.36891 \times 10^{-2} \text{ in} = -0.013689'' \text{ (contraction)}$$

Physical reading. Both estimates say the trunnion contracts by about 0.0137'' in diameter — enough clearance for the shrink fit.

3.2 Question 1(b) [4+3, CO3, PO2]

Problem. Find the relative approximate error between the result of 2-segment and 4-segment Simpson's 1/3 rule. Analyze the reason for getting such low error.

3.2.1 Solution

$$|\epsilon_a| = \left| \frac{\Delta D_4 - \Delta D_2}{\Delta D_4} \right| \times 100 = \left| \frac{-0.0136891340749 - (-0.0136891340749)}{-0.0136891340749} \right| \times 100$$

$$|\epsilon_a| = 0\% \text{ (zero to machine precision)}$$

3.2.2 Why the error is (exactly) zero

The integrand is a quadratic polynomial, and Simpson's 1/3 rule is exact for all polynomials of degree ≤ 3 .

The truncation error of the single-application rule is

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\xi),$$

and for the composite rule

$$E_t = -\frac{(b-a)h^4}{180} f^{(4)}.$$

Here

$$g(T) = -1.2278 \times 10^{-11}T^2 + 6.194 \times 10^{-9}T + 6.015 \times 10^{-6} \implies \frac{d^4g}{dT^4} \equiv 0.$$

Every derivative of g beyond the second vanishes identically, so $E_t = 0$ **regardless of h** . Both the 2-segment and the 4-segment rules therefore return the *exact* value of the integral, and any residual difference between them is pure round-off (below 10^{-16} relative here).

Confirming analytically:

$$\int_{80}^{-108} g dT = \left[-\frac{1.2278 \times 10^{-11}}{3} T^3 + \frac{6.194 \times 10^{-9}}{2} T^2 + 6.015 \times 10^{-6} T \right]_{80}^{-108} = -1.1072664 \times 10^{-3}$$

$$\Delta D_{\text{exact}} = 12.363 \times (-1.1072664 \times 10^{-3}) = -0.0136891 \text{ in } \checkmark$$

exactly matching both numerical results.

The lesson. Refining h buys nothing once the rule's *degree of precision* already exceeds the degree of the integrand. Simpson's 1/3 rule has degree of precision 3; the integrand has degree 2; so both segment counts are already exact and the extra function evaluations of the 4-segment rule are wasted work. Had the integrand contained a T^4 term, the error would have dropped by a factor of $2^4 = 16$ on going from $n = 2$ to $n = 4$.

3.3 Question 2(a) [12+4, CO2, PO1]

Problem. To simplify a model for a diode, it is approximated by a forward bias model consisting of a DC voltage, V_d , and a resistor, R_d . Table 1 contains the voltage vs current data that is collected for a small signal:

V (volts)	I (amps)
0.6	0.01
0.7	0.05
0.8	0.2
0.9	0.7
1.0	2.0
1.1	4.0

Current can be linearly regressed from voltage data as $I = B_0 + B_1 V$, where B_0 and B_1 are the constants of regression. Once B_0 and B_1 are known, the voltage and resistance for the diode can be computed as $V_d = -\frac{B_0}{B_1}$ and $R_d = \frac{1}{B_1}$. Find the voltage and resistance for the diode.

3.3.1 Solution

$n = 6$. Build the summation table:

i	V_i	I_i	$V_i I_i$	V_i^2
1	0.6	0.01	0.006	0.36
2	0.7	0.05	0.035	0.49
3	0.8	0.20	0.160	0.64
4	0.9	0.70	0.630	0.81
5	1.0	2.00	2.000	1.00
6	1.1	4.00	4.400	1.21
Σ	5.1	6.96	7.231	4.51

Slope.

$$B_1 = \frac{n \sum V_i I_i - \sum V_i \sum I_i}{n \sum V_i^2 - (\sum V_i)^2} = \frac{6(7.231) - (5.1)(6.96)}{6(4.51) - (5.1)^2} = \frac{43.386 - 35.496}{27.06 - 26.01} = \frac{7.890}{1.05}$$

$$B_1 = 7.514286 \text{ A/V}$$

Intercept.

$$B_0 = \bar{I} - B_1 \bar{V} = \frac{6.96}{6} - 7.514286 \left(\frac{5.1}{6} \right) = 1.16 - 7.514286(0.85) = 1.16 - 6.387143$$

$$B_0 = -5.227143 \text{ A}$$

Regression line. $I = -5.227143 + 7.514286 V$

Diode parameters.

$$V_d = -\frac{B_0}{B_1} = -\frac{-5.227143}{7.514286} = \frac{5.227143}{7.514286}$$

$$V_d = 0.6956 \text{ V}$$

$$R_d = \frac{1}{B_1} = \frac{1}{7.514286}$$

$$R_d = 0.1331 \text{ } \Omega$$

Interpretation and a caution. $V_d \approx 0.70 \text{ V}$ is the familiar turn-on/knee voltage of a silicon diode, and $R_d \approx 0.13 \text{ } \Omega$ is the small bulk resistance of the forward-biased device — both physically sensible. One should note, however, that the true I - V characteristic is *exponential* ($I = I_s(e^{V/nV_T} - 1)$), so the straight line is a deliberate piecewise-linear simplification valid only over the sampled forward-bias window; extrapolating it below V_d would predict a negative current.

3.4 Question 2(b) [9, CO3, PO2]

Problem. Explain how relative error can provide more useful insight compared to true error when analyzing the results of a numerical method.

3.4.1 Solution

Definitions.

$$E_t = \text{true value} - \text{approximate value} \quad (\text{true error, an } \textit{absolute} \text{ quantity})$$

$$\epsilon_t = \frac{\text{true value} - \text{approximate value}}{\text{true value}} \times 100\% \quad (\text{true relative error})$$

1. True error carries units and is therefore not interpretable alone. $E_t = 1$ cm is a triumph when measuring the span of a bridge (10^4 cm) and a disaster when measuring the diameter of a rivet (1 cm). The *same* number signals opposite verdicts. Relative error normalises by the magnitude of the quantity, so $\epsilon_t = 0.01\%$ and $\epsilon_t = 100\%$ tell the story directly.

2. Relative error is dimensionless, so results become comparable. Errors from methods applied to different quantities — a temperature in $^{\circ}\text{C}$, a contraction in inches, a velocity in m/s — cannot be ranked by E_t ; expressed as percentages they can. This is what makes it possible to say “Simpson’s rule beat the Trapezoidal rule on this problem” at all.

3. It connects directly to significant digits. The number of correct significant digits m satisfies

$$|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$$

so $|\epsilon_a| < 5\% \Rightarrow$ at least 1 correct digit, $|\epsilon_a| < 0.5\% \Rightarrow$ at least 2, and so on. There is no comparable statement for E_t without also knowing the magnitude of the answer.

4. It supports a usable stopping criterion. In practice the true value is unknown — if we knew it we would not be computing. So one uses the **approximate relative error** between successive iterates/refinements,

$$|\epsilon_a| = \left| \frac{\text{present} - \text{previous}}{\text{present}} \right| \times 100\%,$$

and stops when $|\epsilon_a| < \epsilon_s$, the pre-specified tolerance. This is exactly how the number of segments in Q1, or the iterations of Newton–Raphson, are decided in real code. A “stop when $E_t < \text{tol}$ ” rule would require the answer in advance and would need retuning for every new problem scale.

5. It matches how floating-point arithmetic actually behaves. Machine epsilon, round-off and conditioning are all *relative* notions: a double stores about 16 significant digits regardless of exponent. Relative error therefore speaks the same language as the hardware; absolute error does not.

Where true error still matters. When there is an externally imposed absolute tolerance — a manufacturing clearance of $\pm 0.001''$, say — E_t is the relevant measure. And relative error is ill-defined or misleading when the true value is zero or near zero. The practical rule is to quote both.

3.5 Question 3(a) [10+5, CO1, PO2]

Problem. Derive the Trapezoidal rule of Integration from both Calculus and Method of Coefficients. Analyze the geometrical interpretation of these two derivations.

3.5.1 Solution

Derivation 1 — from Calculus (integrate the interpolant) Approximate f on $[a, b]$ by the **first-order Lagrange (straight-line) interpolant** through the endpoints:

$$f(x) \approx P_1(x) = f(a)\frac{x-b}{a-b} + f(b)\frac{x-a}{b-a}$$

Integrate term by term:

$$\begin{aligned} \int_a^b f \, dx &\approx \int_a^b \left[\frac{f(a)}{a-b}(x-b) + \frac{f(b)}{b-a}(x-a) \right] dx \\ &= \frac{f(a)}{a-b} \left[\frac{(x-b)^2}{2} \right]_a^b + \frac{f(b)}{b-a} \left[\frac{(x-a)^2}{2} \right]_a^b \\ &= \frac{f(a)}{a-b} \left[0 - \frac{(a-b)^2}{2} \right] + \frac{f(b)}{b-a} \left[\frac{(b-a)^2}{2} - 0 \right] \\ &= -\frac{f(a)(a-b)}{2} + \frac{f(b)(b-a)}{2} = \frac{(b-a)}{2} [f(a) + f(b)] \end{aligned}$$

$$\boxed{\int_a^b f(x) \, dx \approx (b-a) \frac{f(a) + f(b)}{2}}$$

Derivation 2 — from the Method of Coefficients Postulate a two-point formula with unknown weights:

$$\int_a^b f(x) \, dx \approx c_0 f(a) + c_1 f(b)$$

and force it to be **exact for** $f = 1$ **and** $f = x$ (two conditions, two unknowns).

$$f = 1: \quad \int_a^b dx = b - a \implies c_0 + c_1 = b - a \quad (\text{i})$$

$$f = x: \quad \int_a^b x \, dx = \frac{b^2 - a^2}{2} \implies c_0 a + c_1 b = \frac{b^2 - a^2}{2} = \frac{(b-a)(b+a)}{2} \quad (\text{ii})$$

From (i), $c_0 = (b-a) - c_1$. Substituting into (ii):

$$[(b-a) - c_1]a + c_1 b = \frac{(b-a)(b+a)}{2} \implies a(b-a) + c_1(b-a) = \frac{(b-a)(b+a)}{2}$$

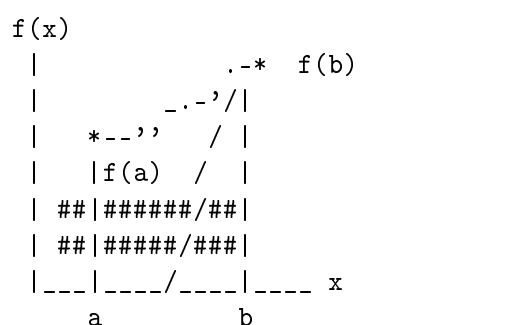
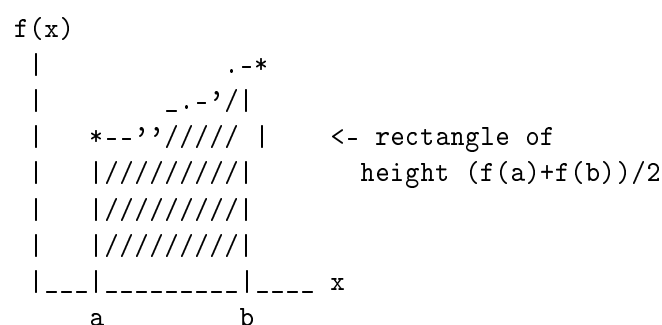
Divide by $(b-a) \neq 0$:

$$a + c_1 = \frac{a+b}{2} \implies c_1 = \frac{b-a}{2}, \quad c_0 = (b-a) - \frac{b-a}{2} = \frac{b-a}{2}$$

$$\boxed{\int_a^b f(x) \, dx \approx \frac{b-a}{2} [f(a) + f(b)]}$$

— identical to Derivation 1.

Geometrical interpretation of the two derivations

$f(x)$ 	$f(x)$ 
CALCULUS view: area under the straight-line interpolant = area of a trapezium	COEFFICIENTS view: area of a rectangle of width (b-a) and height = average of the ordinates

Calculus derivation — “integrate the interpolant”. The picture is a **trapezium** with parallel sides $f(a)$ and $f(b)$ and width $(b - a)$; its area is width $\times$ average of parallel sides, which is literally the formula. The approximation error is the region between the true curve and the chord — positive when f is concave down, negative when concave up, consistent with $E_t = -\frac{(b-a)^3}{12}f''(\xi)$.

Method of coefficients — “match moments”. Here the picture is a **rectangle** of width $(b - a)$ whose height $\frac{f(a)+f(b)}{2}$ is the mean of the two ordinates. The exactness conditions say: *reproduce the zeroth moment (the area of a constant) and the first moment (the area of a ramp) exactly*. Since any straight line is a combination of these two, the formula integrates every linear function exactly — which is a second route to the same conclusion, and a purely algebraic one that never mentions interpolation.

Why they must agree. Both formulae are exact on the 2-dimensional space of linear functions and both use the same two nodes; a linear functional determined by its action on a basis is unique, so the two derivations cannot disagree. The calculus route explains *why the error looks like a second derivative*; the coefficients route generalises immediately to any node set (see Q5(a) and Session A Q6(b)).

3.6 Question 3(b) [10, CO2, PO1]

Problem. Taking up to 5 terms of the Taylor series, find the value of $\cos(\pi)$ given that we know $\cos(\frac{\pi}{2}) = 0$.

3.6.1 Solution

Expand about $x = \frac{\pi}{2}$ with step $h = \pi - \frac{\pi}{2} = \frac{\pi}{2}$:

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \frac{f^{(4)}(x)}{4!}h^4 + \dots$$

Derivatives of $f(x) = \cos x$ evaluated at $x = \pi/2$:

term	derivative	value at $\pi/2$
1	$f(x) = \cos x$	$\cos \frac{\pi}{2} = 0$
2	$f'(x) = -\sin x$	$-\sin \frac{\pi}{2} = -1$
3	$f''(x) = -\cos x$	$-\cos \frac{\pi}{2} = 0$
4	$f'''(x) = \sin x$	$\sin \frac{\pi}{2} = 1$
5	$f^{(4)}(x) = \cos x$	$\cos \frac{\pi}{2} = 0$

Assemble the 5 terms with $h = \pi/2 = 1.5707963$:

$$\begin{aligned} \cos(\pi) &\approx \underbrace{0}_{1\text{st}} + \underbrace{(-1)h}_{2\text{nd}} + \underbrace{\frac{0}{2!}h^2}_{3\text{rd}} + \underbrace{\frac{1}{3!}h^3}_{4\text{th}} + \underbrace{\frac{0}{4!}h^4}_{5\text{th}} \\ &= -1.5707963 + \frac{(1.5707963)^3}{6} = -1.5707963 + \frac{3.8757846}{6} = -1.5707963 + 0.6459641 \end{aligned}$$

$\cos(\pi) \approx -0.9248322$

Error analysis. The true value is $\cos \pi = -1$, so

$$\begin{aligned} E_t &= -1 - (-0.9248322) = -0.0751678 \\ |\epsilon_t| &= \left| \frac{-1 - (-0.9248322)}{-1} \right| \times 100 = \boxed{7.517\%} \end{aligned}$$

Comment. The convergence is slow because $h = \pi/2 \approx 1.571 > 1$: successive terms scale as $h^n/n!$, and with $h > 1$ the factorial has to work hard before it wins. The next nonzero term is $-\frac{h^5}{120} = -0.0797$, which would bring the estimate to -1.0046 — the series is alternating in its nonzero terms and converging, just not quickly. Expanding about a nearer point (or exploiting $\cos \pi = -\cos 0$ with $h = 0$) would be far more efficient.

3.7 Question 4(a) [11, CO2, PO1]

Problem. Thermistors are used to measure the temperature of bodies. These instruments are based on materials' change in resistance with temperature. To measure temperature, manufacturers provide you with a temperature vs resistance calibration curve. If you measure resistance, you can find the temperature. Several observations with a thermistor are shown in Table 2:

R (ohm)	T ($^{\circ}\text{C}$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the temperature for **754.8 ohms** using a **1st order Lagrange polynomial**.

3.7.1 Solution

Select the two nodes that bracket $R = 754.8$ most closely:

$$R_0 = 911.3 \ (T = 30.131), \quad R_1 = 636 \ (T = 40.12)$$

$$T_1(R) = \frac{R - R_1}{R_0 - R_1} T(R_0) + \frac{R - R_0}{R_1 - R_0} T(R_1)$$

Weights at $R = 754.8$:

$$L_0 = \frac{754.8 - 636}{911.3 - 636} = \frac{118.8}{275.3} = 0.4315292$$

$$L_1 = \frac{754.8 - 911.3}{636 - 911.3} = \frac{-156.5}{-275.3} = 0.5684708$$

(Check: $L_0 + L_1 = 1.0000000 \checkmark$)

$$T_1(754.8) = 0.4315292(30.131) + 0.5684708(40.12) = 13.00223 + 22.80722$$

$T_1(754.8) = 35.8095 \text{ } ^\circ\text{C}$

3.8 Question 4(b) [11+3, CO3, PO2]

Problem. Using the previous data, determine the result for the same resistance (754.8 ohms) using a **2nd order Lagrange polynomial** and find the absolute relative approximate error.

3.8.1 Solution

Add the next closest node. Distances from 754.8: $|636 - 754.8| = 118.8$, $|911.3 - 754.8| = 156.5$, $|451.1 - 754.8| = 303.7$, $|1101 - 754.8| = 346.2$. So take

$$R_0 = 911.3 \ (30.131), \quad R_1 = 636 \ (40.12), \quad R_2 = 451.1 \ (50.128)$$

$$T_2(R) = \sum_{i=0}^2 L_i(R) T_i, \quad L_i(R) = \prod_{\substack{j=0 \\ j \neq i}}^2 \frac{R - R_j}{R_i - R_j}$$

Weights at $R = 754.8$:

$$L_0 = \frac{(754.8 - 636)(754.8 - 451.1)}{(911.3 - 636)(911.3 - 451.1)} = \frac{(118.8)(303.7)}{(275.3)(460.2)} = \frac{36079.56}{126692.06} = 0.2847793$$

$$L_1 = \frac{(754.8 - 911.3)(754.8 - 451.1)}{(636 - 911.3)(636 - 451.1)} = \frac{(-156.5)(303.7)}{(-275.3)(184.9)} = \frac{-47529.05}{-50902.97} = 0.9337186$$

$$L_2 = \frac{(754.8 - 911.3)(754.8 - 636)}{(451.1 - 911.3)(451.1 - 636)} = \frac{(-156.5)(118.8)}{(-460.2)(-184.9)} = \frac{-18592.20}{85090.98} = -0.2184979$$

(Check: $0.2847793 + 0.9337186 - 0.2184979 = 1.0000000$ ✓)

$$\begin{aligned} T_2(754.8) &= 0.2847793(30.131) + 0.9337186(40.12) - 0.2184979(50.128) \\ &= 8.58003 + 37.45079 - 10.95221 \end{aligned}$$

$$\boxed{T_2(754.8) = 35.0886 \text{ }^\circ\text{C}}$$

Absolute relative approximate error.

$$|\epsilon_a| = \left| \frac{T_2 - T_1}{T_2} \right| \times 100 = \left| \frac{35.0886 - 35.8095}{35.0886} \right| \times 100 = \frac{0.7209}{35.0886} \times 100$$

$$\boxed{|\epsilon_a| = 2.054\%}$$

Significant digits. $|\epsilon_a| = 2.054\% < 5\% = 0.5 \times 10^{2-1}$, so **at least 1 significant digit** is correct in T_2 . (It falls short of 0.5%, so 2 digits are not yet guaranteed.)

Comment. The drop from 35.81 to 35.09 on adding one node reflects the genuine curvature of the thermistor's T vs R characteristic — it is logarithmic-like, not linear — so the quadratic is the better estimate and a cubic through all four points would refine it further.

3.9 Question 5(a) [3+10, CO2, PO1]

Problem. What is the minimum number of points needed to evaluate the integral for Simpson's 3/8 rule? Derive the Multiple Segment Simpson's 3/8 rule.

3.9.1 Solution

Minimum number of points Simpson's 3/8 rule fits a **cubic** $P_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ through the data. A cubic has **four** coefficients, so four independent conditions — and hence **four points** — are required to determine it uniquely.

$$\boxed{\text{Minimum} = 4 \text{ points } (= 3 \text{ segments}), \quad h = \frac{b-a}{3}}$$

Equivalently: the segment count n must be a **multiple of 3**, and the smallest such positive multiple is $n = 3$, which uses $n + 1 = 4$ points.

Derivation of the Multiple Segment (composite) Simpson's 3/8 rule Step 0 — the single application. For four equispaced points spanning an interval of width $3h$:

$$\int_{x_0}^{x_3} f(x) dx \approx \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

Step 1 — partition. Divide $[a, b]$ into n equal segments with n a multiple of 3:

$$h = \frac{b-a}{n}, \quad x_i = a + ih \quad (i = 0, 1, \dots, n)$$

and split the integral into $n/3$ consecutive blocks of three segments each:

$$\int_a^b f dx = \int_{x_0}^{x_3} f dx + \int_{x_3}^{x_6} f dx + \dots + \int_{x_{n-3}}^{x_n} f dx$$

Step 2 — apply the single rule to each block.

$$\begin{aligned} \int_a^b f dx &\approx \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] + \frac{3h}{8} [f(x_3) + 3f(x_4) + 3f(x_5) + f(x_6)] + \dots \\ &\dots + \frac{3h}{8} [f(x_{n-3}) + 3f(x_{n-2}) + 3f(x_{n-1}) + f(x_n)] \end{aligned}$$

Step 3 — collect coefficients. The end nodes x_0, x_n appear once (weight 1). Nodes whose index is *not* a multiple of 3 appear once with weight 3. Interior nodes whose index *is* a multiple of 3 are shared between two adjacent blocks and so pick up weight $1 + 1 = 2$.

$$\int_a^b f(x) dx \approx \frac{3h}{8} \left[f(x_0) + 3 \sum_{\substack{i=1 \\ i \bmod 3 \neq 0}}^{n-1} f(x_i) + 2 \sum_{\substack{i=3 \\ i \bmod 3 = 0}}^{n-3} f(x_i) + f(x_n) \right]$$

with weight pattern $1, 3, 3, 2, 3, 3, 2, \dots, 2, 3, 3, 1$.

Step 4 — error. Each block contributes $-\frac{(3h)^5}{6480} f^{(4)}(\xi_j) = -\frac{3h^5}{80} f^{(4)}(\xi_j)$. Summing over $n/3$ blocks and using the mean-value form,

$$E_t = -\frac{(b-a)h^4}{80} \overline{f^{(4)}} = O(h^4)$$

so halving h reduces the error by a factor of 16.

(This is the same derivation as Session A Q4(b), which asked only for the formula.)

3.10 Question 5(b) [3×4, CO2, PO1]

Problem. Suppose you are given the following data:

x	0	0.5	1.0	1.5	2.0
$F(x)$	0	0.19	0.26	0.29	0.31

Find the Forward Difference, Backward Difference and Central Difference approximations of $F'(1)$ using $h = 0.5$.

3.10.1 Solution

Take $x_i = 1$, so $x_i - h = 0.5$, $x_i + h = 1.5$, with

$$F(0.5) = 0.19, \quad F(1.0) = 0.26, \quad F(1.5) = 0.29.$$

Forward Difference.

$$F'(1) \approx \frac{F(1+h) - F(1)}{h} = \frac{F(1.5) - F(1.0)}{0.5} = \frac{0.29 - 0.26}{0.5} = \frac{0.03}{0.5}$$

$$\boxed{F'(1)|_{\text{FD}} = 0.06}$$

Backward Difference.

$$F'(1) \approx \frac{F(1) - F(1-h)}{h} = \frac{F(1.0) - F(0.5)}{0.5} = \frac{0.26 - 0.19}{0.5} = \frac{0.07}{0.5}$$

$$\boxed{F'(1)|_{\text{BD}} = 0.14}$$

Central Difference.

$$F'(1) \approx \frac{F(1+h) - F(1-h)}{2h} = \frac{F(1.5) - F(0.5)}{2(0.5)} = \frac{0.29 - 0.19}{1.0} = \frac{0.10}{1.0}$$

$$\boxed{F'(1)|_{\text{CD}} = 0.10}$$

Observation. $\text{CD} = \frac{\text{FD} + \text{BD}}{2} = \frac{0.06 + 0.14}{2} = 0.10$ exactly — a direct illustration of the cancellation argument in Session A Q1(a): the forward and backward errors are $\mp \frac{h}{2} F''(\xi)$, equal in magnitude and opposite in sign, so their average kills the $O(h)$ term.

The wide spread between 0.06 and 0.14 also warns that F'' is large here (the data are strongly curved, flattening out towards $x = 2$); a central-difference estimate of $F''(1)$ gives $\frac{0.29 - 2(0.26) + 0.19}{0.5^2} = \frac{-0.04}{0.25} = -0.16$, and indeed $\frac{h}{2}|F''| = \frac{0.5}{2}(0.16) = 0.04$, exactly the half-spread observed.

3.11 Question 6 — preamble

Suppose you are given the following matrix A :

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

3.12 Question 6(a)(i) [4, CO1, PO2]

Problem. Analyze the properties of the matrix A to determine whether it can have complex eigenvalues.

3.12.1 Solution

Observe that A is real and symmetric: $a_{ij} = a_{ji}$ for all i, j ($a_{12} = a_{21} = 1$, $a_{13} = a_{31} = 2$, $a_{23} = a_{32} = 3$), and every entry is real. Hence $A^T = A$.

Theorem (spectral theorem for real symmetric matrices). *Every eigenvalue of a real symmetric matrix is real.*

Proof. Let $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, allowing $\lambda, \mathbf{v}$ to be complex. Pre-multiply by the conjugate transpose $\bar{\mathbf{v}}^T$:

$$\bar{\mathbf{v}}^T A\mathbf{v} = \lambda \bar{\mathbf{v}}^T \mathbf{v}.$$

Now take the conjugate transpose of the left side. Since A is real and symmetric, $\bar{A}^T = A$, so

$$\overline{(\bar{\mathbf{v}}^T A\mathbf{v})} = \mathbf{v}^T \bar{A} \bar{\mathbf{v}} = (\bar{\mathbf{v}}^T A\mathbf{v})^T \Big|_{\text{scalar}} = \bar{\mathbf{v}}^T A\mathbf{v},$$

i.e. the scalar $\bar{\mathbf{v}}^T A\mathbf{v}$ equals its own conjugate and is therefore **real**. Also $\bar{\mathbf{v}}^T \mathbf{v} = \sum |v_i|^2 > 0$ is real and positive. Hence

$$\lambda = \frac{\bar{\mathbf{v}}^T A\mathbf{v}}{\bar{\mathbf{v}}^T \mathbf{v}} = \frac{\text{real}}{\text{real} > 0} \in \mathbb{R}. \quad \blacksquare$$

A is real symmetric $\implies$ it **cannot** have complex eigenvalues.

Two further consequences worth stating: the eigenvectors belonging to distinct eigenvalues are mutually **orthogonal**, and A is **orthogonally diagonalisable**, $A = Q\Lambda Q^T$. Part (a)(ii) confirms all of this explicitly.

3.13 Question 6(a)(ii) [12, CO2, PO1]

Problem. Find the eigenvalues and eigenvectors of the matrix A .

3.13.1 Solution

Characteristic polynomial. For a 3×3 matrix,

$$\det(A - \lambda I) = -\left[\lambda^3 - c_1\lambda^2 + c_2\lambda - c_3\right]$$

where $c_1 = \text{tr } A$, $c_2 = \text{sum of the principal } 2 \times 2 \text{ minors}$, $c_3 = \det A$.

$$c_1 = 3 + 2 + 1 = 6$$

$$c_2 = \det \begin{bmatrix} 2 & 3 \\ 3 & 1 \end{bmatrix} + \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} + \det \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = (2 - 9) + (3 - 4) + (6 - 1) = -7 - 1 + 5 = -3$$

$$c_3 = \det A = 3(2 \cdot 1 - 3 \cdot 3) - 1(1 \cdot 1 - 3 \cdot 2) + 2(1 \cdot 3 - 2 \cdot 2) = 3(-7) - 1(-5) + 2(-1) = -21 + 5 - 2 = -18$$

$$\implies \lambda^3 - 6\lambda^2 - 3\lambda + 18 = 0$$

Factorise. Trying $\lambda = 6$: $216 - 216 - 18 + 18 = 0$ ✓, so $(\lambda - 6)$ is a factor. Dividing,

$$(\lambda - 6)(\lambda^2 - 3) = 0$$

$$\boxed{\lambda_1 = 6, \quad \lambda_2 = \sqrt{3} = 1.73205, \quad \lambda_3 = -\sqrt{3} = -1.73205}$$

All three are real, confirming part (i). (Check: $\lambda_1\lambda_2\lambda_3 = 6(3)(-1) = -18 = \det A$ ✓, and $\sum \lambda_i = 6 = \text{tr } A$ ✓.)

Eigenvector for $\lambda_1 = 6$

$$(A - 6I) = \begin{bmatrix} -3 & 1 & 2 \\ 1 & -4 & 3 \\ 2 & 3 & -5 \end{bmatrix}$$

Row 2: $x - 4y + 3z = 0 \Rightarrow x = 4y - 3z$. Substituting in Row 1:

$$-3(4y - 3z) + y + 2z = -12y + 9z + y + 2z = -11y + 11z = 0 \Rightarrow y = z$$

Then $x = 4z - 3z = z$. Taking $z = 1$:

$$\boxed{\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}$$

(Check: row sums of A are $3 + 1 + 2 = 6$, $1 + 2 + 3 = 6$, $2 + 3 + 1 = 6$ — A has constant row sum 6, so $\mathbf{1}$ is an eigenvector with eigenvalue 6 by inspection.)

Eigenvectors for $\lambda = \pm\sqrt{3}$ For a general λ , an eigenvector is the cross product of any two rows of $(A - \lambda I)$:

$$\mathbf{r}_1 = (3 - \lambda, 1, 2), \quad \mathbf{r}_2 = (1, 2 - \lambda, 3)$$

$$\mathbf{v} = \mathbf{r}_1 \times \mathbf{r}_2 = (3 - 2(2 - \lambda), 2 - 3(3 - \lambda), (3 - \lambda)(2 - \lambda) - 1) = (2\lambda - 1, 3\lambda - 7, \lambda^2 - 5\lambda + 5)$$

Using $\lambda^2 = 3$, the third component simplifies to $8 - 5\lambda$:

$$\mathbf{v}(\lambda) = (2\lambda - 1, 3\lambda - 7, 8 - 5\lambda)$$

For $\lambda_2 = \sqrt{3}$:

$$\boxed{\mathbf{v}_2 = \begin{bmatrix} 2\sqrt{3} - 1 \\ 3\sqrt{3} - 7 \\ 8 - 5\sqrt{3} \end{bmatrix} = \begin{bmatrix} 2.46410 \\ -1.80385 \\ -0.66025 \end{bmatrix}}$$

For $\lambda_3 = -\sqrt{3}$:

$$\boxed{\mathbf{v}_3 = \begin{bmatrix} -2\sqrt{3} - 1 \\ -3\sqrt{3} - 7 \\ 8 + 5\sqrt{3} \end{bmatrix} = \begin{bmatrix} -4.46410 \\ -12.19615 \\ 16.66025 \end{bmatrix}}$$

Verification (for λ_2): $A\mathbf{v}_2$ gives $3(2.46410) + 1(-1.80385) + 2(-0.66025) = 7.39230 - 1.80385 - 1.32050 = 4.26795$, and $\sqrt{3}(2.46410) = 4.26795$ ✓

Orthogonality check (predicted by part (i)): $\mathbf{v}_1 \cdot \mathbf{v}_2 = 2.46410 - 1.80385 - 0.66025 = 0.0000$ ✓

3.14 Question 6(b) [9, CO2, PO1]

Problem. Find the LU Decomposition of the matrix $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$.

3.14.1 Solution

Doolittle's method (Gaussian elimination, recording the multipliers in L).

Step 1 — eliminate column 1. Pivot $u_{11} = 3$.

$$m_{21} = \frac{1}{3}: \quad R_2 \leftarrow R_2 - \frac{1}{3}R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 3 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 2 - \frac{1}{3} & 3 - \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 0 & \frac{5}{3} & \frac{7}{3} \end{bmatrix}$$

$$m_{31} = \frac{2}{3}: \quad R_3 \leftarrow R_3 - \frac{2}{3}R_1$$

$$\begin{bmatrix} 2 & 3 & 1 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 3 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 3 - \frac{2}{3} & 1 - \frac{4}{3} \end{bmatrix} = \begin{bmatrix} 0 & \frac{7}{3} & -\frac{1}{3} \end{bmatrix}$$

$$A \longrightarrow \begin{bmatrix} 3 & 1 & 2 \\ 0 & \frac{5}{3} & \frac{7}{3} \\ 0 & \frac{7}{3} & -\frac{1}{3} \end{bmatrix}$$

Step 2 — eliminate column 2. Pivot $\frac{5}{3}$.

$$m_{32} = \frac{7/3}{5/3} = \frac{7}{5} = 1.4: \quad R_3 \leftarrow R_3 - \frac{7}{5}R_2$$

$$-\frac{1}{3} - \frac{7}{5} \left(\frac{7}{3} \right) = -\frac{1}{3} - \frac{49}{15} = -\frac{5}{15} - \frac{49}{15} = -\frac{54}{15} = -\frac{18}{5}$$

Result.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{2}{3} & \frac{7}{5} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & \frac{5}{3} & \frac{7}{3} \\ 0 & 0 & -\frac{18}{5} \end{bmatrix}$$

In decimals:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0.33333 & 1 & 0 \\ 0.66667 & 1.40000 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 1.66667 & 2.33333 \\ 0 & 0 & -3.60000 \end{bmatrix}$$

Verification. Row 3 of LU :

$$(LU)_{31} = \frac{2}{3}(3) = 2 \checkmark$$

$$(LU)_{32} = \frac{2}{3}(1) + \frac{7}{5}\left(\frac{5}{3}\right) = \frac{2}{3} + \frac{7}{3} = 3 \checkmark$$

$$(LU)_{33} = \frac{2}{3}(2) + \frac{7}{5}\left(\frac{7}{3}\right) + \left(-\frac{18}{5}\right) = \frac{4}{3} + \frac{49}{15} - \frac{54}{15} = \frac{20}{15} + \frac{49}{15} - \frac{54}{15} = \frac{15}{15} = 1 \checkmark$$

Cross-check with Q6(a)(ii).

$$\det A = \prod u_{ii} = 3 \cdot \frac{5}{3} \cdot \left(-\frac{18}{5}\right) = -18$$

which agrees exactly with $c_3 = \det A = -18$ found from the characteristic polynomial, and with $\lambda_1 \lambda_2 \lambda_3 = 6 \cdot \sqrt{3} \cdot (-\sqrt{3}) = -18$ ✓

4 Session C — Math 4641, Summer Semester 2021–2022 (15 May 2023)

B.Sc. Engg. CSE, 6th Semester — Duration 3 hours — Full marks 150

4.1 Question 1(a) — preamble

Suppose you are given the data shown in Table 1:

t	$f(t)$
0	0
0.5	0.19
1.0	0.26
1.5	0.29
2.0	0.31

4.2 Question 1(a)(i) [7, CO1, PO1]

Problem. Using Taylor series, derive the Central Difference Approximation formula for $f''(x_i)$.

4.2.1 Solution

Expand f about x_i in both directions:

$$f(x_i + h) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f'''(x_i)}{3!}h^3 + \frac{f^{(4)}(x_i)}{4!}h^4 + \dots \quad (1)$$

$$f(x_i - h) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 - \frac{f'''(x_i)}{3!}h^3 + \frac{f^{(4)}(x_i)}{4!}h^4 - \dots \quad (2)$$

Add (1) and (2). All **odd**-order terms cancel:

$$\begin{aligned} f(x_i + h) + f(x_i - h) &= 2f(x_i) + f''(x_i)h^2 + \frac{2f^{(4)}(x_i)}{4!}h^4 + \dots \\ &= 2f(x_i) + f''(x_i)h^2 + \frac{f^{(4)}(x_i)}{12}h^4 + \dots \end{aligned}$$

Solve for $f''(x_i)$:

$$f''(x_i) = \frac{f(x_i + h) - 2f(x_i) + f(x_i - h)}{h^2} - \frac{f^{(4)}(x_i)}{12}h^2 - \dots$$

Truncating and applying the mean-value form of the remainder:

$$\boxed{f''(x_i) \approx \frac{f(x_i + h) - 2f(x_i) + f(x_i - h)}{h^2}, \quad E_t = -\frac{f^{(4)}(\xi)}{12}h^2 = O(h^2)}$$

Note that the *cubic* term cancelled along with all other odd terms, which is why a three-point formula achieves second-order — not first-order — accuracy for the second derivative.

4.3 Question 1(a)(ii) [2×3, CO2, PO1]

Problem. Find $f'(1.5)$ and $f''(1.5)$.

■ *Related to Session B Q5(b), which uses the same table but evaluates the three difference formulae at $x = 1$.*

4.3.1 Solution

The point $t = 1.5$ is interior, with $h = 0.5$, so central differences are available in both cases:

$$f(1.0) = 0.26, \quad f(1.5) = 0.29, \quad f(2.0) = 0.31$$

First derivative (central difference).

$$f'(1.5) \approx \frac{f(2.0) - f(1.0)}{2h} = \frac{0.31 - 0.26}{2(0.5)} = \frac{0.05}{1.0}$$

$$\boxed{f'(1.5) \approx 0.05}$$

Second derivative (central difference, from part (i)).

$$f''(1.5) \approx \frac{f(2.0) - 2f(1.5) + f(1.0)}{h^2} = \frac{0.31 - 2(0.29) + 0.26}{(0.5)^2} = \frac{0.31 - 0.58 + 0.26}{0.25} = \frac{-0.01}{0.25}$$

$$\boxed{f''(1.5) \approx -0.04}$$

Interpretation. $f' > 0$ and $f'' < 0$: the data are increasing but concave down, i.e. the curve is flattening out towards a plateau — consistent with the visible trend 0, 0.19, 0.26, 0.29, 0.31, where the successive increments 0.19, 0.07, 0.03, 0.02 are shrinking.

4.4 Question 1(b) [7+5, CO3, PO1]

Problem. Prove whether Central Difference Approximation produces less error than Forward Difference or Backward Difference Approximation. Explain why Forward Difference and Backward Difference formulae may still be needed even though Central Difference formula produces less error than both.

4.4.1 Solution

Part 1 — proof that the Central Difference has smaller error From the Taylor expansions (1) and (2) above:

Forward. Rearranging (1):

$$\underbrace{\frac{f(x_i + h) - f(x_i)}{h}}_{\text{FD estimate}} = f'(x_i) + \frac{f''(x_i)}{2}h + \frac{f'''(x_i)}{6}h^2 + \dots$$

$$\implies E_{\text{FD}} = f'(x_i) - \text{FD} = -\frac{f''(\xi_1)}{2}h = O(h)$$

Backward. Rearranging (2):

$$\underbrace{\frac{f(x_i) - f(x_i - h)}{h}}_{\text{BD estimate}} = f'(x_i) - \frac{f''(x_i)}{2}h + \frac{f'''(x_i)}{6}h^2 - \dots$$

$$\implies E_{\text{BD}} = +\frac{f''(\xi_2)}{2}h = O(h)$$

Central. Subtracting (2) from (1), the h^2 terms **cancel identically**:

$$f(x_i + h) - f(x_i - h) = 2f'(x_i)h + \frac{2f'''(x_i)}{6}h^3 + \dots$$

$$\underbrace{\frac{f(x_i + h) - f(x_i - h)}{2h}}_{\text{CD estimate}} = f'(x_i) + \frac{f'''(x_i)}{6}h^2 + \dots$$

$$\implies E_{\text{CD}} = -\frac{f'''(\xi_3)}{6}h^2 = O(h^2)$$

Conclusion. For h small enough (and $f'' \neq 0$),

$$\left| \frac{E_{\text{CD}}}{E_{\text{FD}}} \right| = \left| \frac{h^2 f'''/6}{h f''/2} \right| = \frac{h}{3} \left| \frac{f'''}{f''} \right| \xrightarrow{h \rightarrow 0} 0$$

so $|E_{\text{CD}}| \ll |E_{\text{FD}}|, |E_{\text{BD}}|$. ■

The structural reason — note that E_{FD} and E_{BD} are $\mp \frac{h}{2}f''$: *equal magnitude, opposite sign*. And indeed

$$\frac{\text{FD} + \text{BD}}{2} = \frac{1}{2} \left[\frac{f(x_i + h) - f(x_i)}{h} + \frac{f(x_i) - f(x_i - h)}{h} \right] = \frac{f(x_i + h) - f(x_i - h)}{2h} = \text{CD}$$

The central difference **is** the average of the other two, and averaging cancels their leading errors exactly. (Session B Q5(b) shows this numerically: $\frac{0.06+0.14}{2} = 0.10$.)

Practical consequence. Halving h reduces the FD/BD error by 2 but the CD error by 4.

Part 2 — why FD and BD are still needed

1. **At the boundaries of a data set, the central stencil does not exist.** The central formula at x_i requires *both* $x_i - h$ and $x_i + h$. At the first data point there is no $x_i - h$, and at the last there is no $x_i + h$. In Table 1 above, $f'(0)$ can only be estimated forward and $f'(2)$ only backward. Any scheme that must produce derivatives at every node — gradient computation over a grid, boundary conditions of a PDE, edge detection over an image — needs one-sided formulae at the edges.
2. **Marching (initial-value) problems are one-sided by nature.** Euler, Runge–Kutta and all explicit ODE integrators advance from known past values to an unknown future one. Using y_{i+1} to compute the slope that *produces* y_{i+1} would make the scheme implicit, requiring a nonlinear solve at every step. The forward difference is what makes explicit time-stepping possible.
3. **Data may exist on one side only.** Physical measurements often start at $t = 0$ (a switch-on, an impact, an injection) with nothing before it; or the function may be undefined past a boundary (a wall, a phase change, the edge of a domain).
4. **Discontinuities and one-sided derivatives.** If f has a kink or a jump at x_i — the thermocline of Session A Q6(a), a shock in a flow, a material interface — straddling it with a central stencil averages across the discontinuity and produces a meaningless number. The correct object is the one-sided derivative, computed with FD from the right or BD from the left.
5. **Upwinding for stability.** In convection-dominated problems, the *centred* scheme is notoriously unstable/oscillatory, and one deliberately uses the one-sided difference taken from the upwind direction, accepting $O(h)$ accuracy in exchange for a stable solution.

Summary. Central difference is the accuracy champion in the interior; forward and backward differences are what make the method *applicable* at the edges, in time-marching, across discontinuities, and where stability demands it.

4.5 Question 2(a)(i) [5, CO1, PO1]

Problem. Rewrite the following third-order differential equation as a set of simultaneous first-order differential equations:

$$\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = e^{-t}, \quad y(0) = 1, \quad \frac{dy}{dx}(0) = 2, \quad \frac{d^2y}{dx^2}(0) = 2$$

4.5.1 Solution

Step 1 — introduce state variables, one for each derivative below the highest:

$$z_1 = y, \quad z_2 = \frac{dy}{dx}, \quad z_3 = \frac{d^2y}{dx^2}$$

Step 2 — write the trivial chain relations:

$$\frac{dz_1}{dx} = \frac{dy}{dx} = z_2, \quad \frac{dz_2}{dx} = \frac{d^2y}{dx^2} = z_3$$

Step 3 — isolate the highest derivative from the original ODE:

$$\frac{d^3y}{dx^3} = e^{-t} - 2\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - y \implies \frac{dz_3}{dx} = e^{-t} - 2z_3 - 2z_2 - z_1$$

The equivalent first-order system (taking $t \equiv x$ as the single independent variable, as the initial conditions imply):

$\frac{dz_1}{dx} = z_2,$	$z_1(0) = 1$
$\frac{dz_2}{dx} = z_3,$	$z_2(0) = 2$
$\frac{dz_3}{dx} = e^{-x} - 2z_3 - 2z_2 - z_1,$	$z_3(0) = 2$

or in vector form $\frac{d\mathbf{z}}{dx} = \mathbf{F}(x, \mathbf{z})$ with $\mathbf{z}(0) = (1, 2, 2)^T$ — exactly the form every standard IVP solver (Euler, Heun, RK4) accepts.

4.6 Question 2(a)(ii) [14, CO2, PO1]

Problem. For the differential equation in Question 2.a)i, find $y(0.75)$ using **Euler's method**. Consider step size $h = 0.25$.

4.6.1 Solution

Euler's method applied componentwise:

$$\mathbf{z}_{i+1} = \mathbf{z}_i + \mathbf{F}(x_i, \mathbf{z}_i) h$$

i.e.

$$z_{1,i+1} = z_{1,i} + z_{2,i}h, \quad z_{2,i+1} = z_{2,i} + z_{3,i}h, \quad z_{3,i+1} = z_{3,i} + (e^{-x_i} - 2z_{3,i} - 2z_{2,i} - z_{1,i})h$$

with $h = 0.25$ and $\mathbf{z}_0 = (1, 2, 2)$ at $x_0 = 0$. Three steps reach $x = 0.75$.

Step 1: $x_0 = 0 \rightarrow x_1 = 0.25$

$$\begin{aligned} z'_1 &= z_2 = 2, & z'_2 &= z_3 = 2 \\ z'_3 &= e^0 - 2(2) - 2(2) - 1 = 1 - 4 - 4 - 1 = -8 \end{aligned}$$

$$z_1(0.25) = 1 + 2(0.25) = 1.5$$

$$z_2(0.25) = 2 + 2(0.25) = 2.5$$

$$z_3(0.25) = 2 + (-8)(0.25) = 2 - 2 = 0$$

Step 2: $x_1 = 0.25 \rightarrow x_2 = 0.5$

$$z'_1 = z_2 = 2.5, \quad z'_2 = z_3 = 0$$

$$z'_3 = e^{-0.25} - 2(0) - 2(2.5) - 1.5 = 0.7788008 - 0 - 5 - 1.5 = -5.7211992$$

$$z_1(0.5) = 1.5 + 2.5(0.25) = 2.125$$

$$z_2(0.5) = 2.5 + 0(0.25) = 2.5$$

$$z_3(0.5) = 0 + (-5.7211992)(0.25) = -1.4302998$$

Step 3: $x_2 = 0.5 \rightarrow x_3 = 0.75$

$$z'_1 = z_2 = 2.5, \quad z'_2 = z_3 = -1.4302998$$

$$z'_3 = e^{-0.5} - 2(-1.4302998) - 2(2.5) - 2.125 = 0.6065307 + 2.8605996 - 5 - 2.125 = -3.6578697$$

$$z_1(0.75) = 2.125 + 2.5(0.25) = 2.75$$

$$z_2(0.75) = 2.5 + (-1.4302998)(0.25) = 2.1424251$$

$$z_3(0.75) = -1.4302998 + (-3.6578697)(0.25) = -2.3447672$$

Summary table

i	x_i	$z_1 = y$	$z_2 = y'$	$z_3 = y''$	z'_1	z'_2	z'_3
0	0.00	1.000000	2.000000	2.000000	2.000000	2.000000	-8.000000
1	0.25	1.500000	2.500000	0.000000	2.500000	0.000000	-5.721199
2	0.50	2.125000	2.500000	-1.430300	2.500000	-1.430300	-3.657870
3	0.75	2.750000	2.142425	-2.344767	—	—	—

$y(0.75) \approx 2.75$

(As by-products, $y'(0.75) \approx 2.1424$ and $y''(0.75) \approx -2.3448$.)

Accuracy note. Euler is $O(h)$ and $h = 0.25$ is coarse; the z_3 column swings from +2 to -2.34 in three steps, which signals that the local truncation error is not small. Halving h would roughly halve the error; RK4 (Q6(a)) would be far better for the same number of f -evaluations.

4.7 Question 2(b) [2×3, CO1, PO1]

Problem. Show how the variables are transformed when linearization is applied on Exponential function, Power function and Growth function.

4.7.1 [!] EXACT DUPLICATE — SKIPPED

This question is an exact duplicate of **Session A — Question 3(b)** (“Describe how linearization transform is applied on the following nonlinear models: i. Exponential model, ii. Power model, iii. Growth model”, 15 April 2022). The full solution — including the transformation tables for $y = ae^{bx}$, $y = ax^b$ and $y = \frac{ax}{b+x}$, and the recovery of a and b — is given there and is not repeated.

4.8 Question 3(a) [7, CO3, PO1]

Problem. Explain how LU Decomposition is more computationally efficient for solving linear systems compared to Inverse matrix multiplication or Gaussian Elimination.

4.8.1 Solution

Let A be $n \times n$ and suppose m different right-hand sides $\mathbf{b}_1, \dots, \mathbf{b}_m$ must be solved against the same A .

Against Gaussian Elimination Gaussian elimination fuses the factorisation with the right-hand side: the forward elimination sweep, costing $\approx \frac{n^3}{3}$ flops, operates on the augmented matrix $[A | \mathbf{b}]$. If $\mathbf{b}$ changes, **the entire $O(n^3)$ sweep must be redone** even though A is untouched — all the factorisation work is discarded.

LU separates the two concerns:

$$A = LU \quad (\text{once, } \frac{n^3}{3} \text{ flops, depends only on } A)$$

$$\text{then for each } \mathbf{b}: \quad L\mathbf{z} = \mathbf{b} \text{ (forward, } \frac{n^2}{2}) \rightarrow U\mathbf{x} = \mathbf{z} \text{ (back, } \frac{n^2}{2})$$

$$\text{Total:} \quad \text{GE} = m \cdot \frac{n^3}{3} \quad \text{vs} \quad \text{LU} = \frac{n^3}{3} + m n^2$$

For $n = 100$, $m = 50$: $\text{GE} \approx 1.67 \times 10^7$ flops, $\text{LU} \approx 8.3 \times 10^5$ — a **20× saving**, growing linearly with m .

Against explicit inverse multiplication Computing A^{-1} and then forming $\mathbf{x} = A^{-1}\mathbf{b}$ is worse on every axis:

- **Cost.** Forming A^{-1} requires solving $A\mathbf{x}_j = \mathbf{e}_j$ for $j = 1..n$, i.e. one LU factorisation **plus n pairs of substitutions**: $\frac{n^3}{3} + n \cdot n^2 = \frac{4n^3}{3}$ flops — about **four times** an LU factorisation. The matrix–vector product then costs a further n^2 per right-hand side, the same as the two substitutions, so nothing is gained back.
- **Numerical stability.** Explicit inversion is less accurate. The computed A^{-1} carries round-off from n separate solves, and multiplying by it amplifies the error. Backward-error analysis shows LU with partial pivoting to be backward stable, while $\hat{A}^{-1}\mathbf{b}$ is not.
- **Storage and sparsity.** A^{-1} is essentially always **dense**, even when A is sparse or banded. For a banded A , L and U inherit the band structure and cost $O(n)$ storage, whereas A^{-1} needs $O(n^2)$ — fatal at large n .

Free by-products of LU

$$\det A = \prod_{i=1}^n u_{ii} \quad (\text{no extra work})$$

and A^{-1} , when genuinely required, is available column by column at $O(n^2)$ each.

LU: factor once $O(n^3/3)$, reuse forever at $O(n^2)$ per RHS.

Question 3(b) below is a direct application: two missiles, **one** shared coefficient matrix.

4.9 Question 3(b) [18, CO2, PO1]

Problem. Suppose you are given the time vs velocity data for two separate missiles in Table 2:

t	V_1	V_2
0	0	0
10	227.04	152.35
15	362.78	316.62
20	517.35	502.71
22.5	602.97	620.25
30	901.67	978.87

Using **Direct method of Interpolation with a third order polynomial**, find the velocity of both missiles at $t = 21$ seconds. For solving the simultaneous equations, you are **not allowed to use inverse matrix multiplication or Gaussian Elimination more than once**.

4.9.1 Solution

Choosing the nodes A third-order polynomial $v(t) = a_0 + a_1t + a_2t^2 + a_3t^3$ needs **four** points. Choose the four that bracket $t = 21$ most tightly:

$$t_0 = 15, \quad t_1 = 20, \quad t_2 = 22.5, \quad t_3 = 30$$

The key observation (satisfying the constraint) Both missiles are sampled at the **same times**, so the coefficient matrix is **identical** for both systems:

$$\underbrace{\begin{bmatrix} 1 & 15 & 225 & 3375 \\ 1 & 20 & 400 & 8000 \\ 1 & 22.5 & 506.25 & 11390.625 \\ 1 & 30 & 900 & 27000 \end{bmatrix}}_{M \text{ (same for both)}} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 362.78 \\ 517.35 \\ 602.97 \\ 901.67 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 316.62 \\ 502.71 \\ 620.25 \\ 978.87 \end{bmatrix}$$

So we perform **one** elimination — an LU decomposition of M — and then run **two cheap forward/back substitutions**, never touching elimination again. This is exactly the scenario of Q3(a).

Step 1 — LU decomposition of M (done once) Doolittle elimination on M :

Column 1 ($m_{21} = m_{31} = m_{41} = 1$): subtract R_1 from R_2, R_3, R_4 :

$$\begin{bmatrix} 1 & 15 & 225 & 3375 \\ 0 & 5 & 175 & 4625 \\ 0 & 7.5 & 281.25 & 8015.625 \\ 0 & 15 & 675 & 23625 \end{bmatrix}$$

Column 2 ($m_{32} = \frac{7.5}{5} = 1.5$, $m_{42} = \frac{15}{5} = 3$):

$$\begin{bmatrix} 1 & 15 & 225 & 3375 \\ 0 & 5 & 175 & 4625 \\ 0 & 0 & 18.75 & 1078.125 \\ 0 & 0 & 150 & 9750 \end{bmatrix}$$

Column 3 ($m_{43} = \frac{150}{18.75} = 8$):

$$9750 - 8(1078.125) = 9750 - 8625 = 1125$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1.5 & 1 & 0 \\ 1 & 3 & 8 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 15 & 225 & 3375 \\ 0 & 5 & 175 & 4625 \\ 0 & 0 & 18.75 & 1078.125 \\ 0 & 0 & 0 & 1125 \end{bmatrix}$$

Step 2 — Missile 1: forward then back substitution Forward, $Lz = \mathbf{V}_1$:

$$z_1 = 362.78$$

$$z_2 = 517.35 - 1(362.78) = 154.57$$

$$z_3 = 602.97 - 1(362.78) - 1.5(154.57) = 602.97 - 362.78 - 231.855 = 8.335$$

$$z_4 = 901.67 - 1(362.78) - 3(154.57) - 8(8.335) = 901.67 - 362.78 - 463.71 - 66.68 = 8.50$$

Back, $U\mathbf{a} = \mathbf{z}$:

$$a_3 = \frac{8.50}{1125} = 7.555556 \times 10^{-3}$$

$$a_2 = \frac{8.335 - 1078.125(7.555556 \times 10^{-3})}{18.75} = \frac{8.335 - 8.145833}{18.75} = 1.008889 \times 10^{-2}$$

$$a_1 = \frac{154.57 - 175(1.008889 \times 10^{-2}) - 4625(7.555556 \times 10^{-3})}{5} = \frac{154.57 - 1.765556 - 34.944444}{5} = 23.572$$

$$\begin{aligned} a_0 &= 362.78 - 15(23.572) - 225(1.008889 \times 10^{-2}) - 3375(7.555556 \times 10^{-3}) \\ &= 362.78 - 353.58 - 2.27 - 25.50 = -18.57 \end{aligned}$$

$$v_1(t) = -18.57 + 23.572t + 1.008889 \times 10^{-2}t^2 + 7.555556 \times 10^{-3}t^3$$

Evaluate at $t = 21$:

$$\begin{aligned} v_1(21) &= -18.57 + 23.572(21) + 1.008889 \times 10^{-2}(441) + 7.555556 \times 10^{-3}(9261) \\ &= -18.57 + 495.012 + 4.44920 + 69.97200 \end{aligned}$$

$$\boxed{v_1(21) = 550.8632 \text{ m/s}}$$

Step 3 — Missile 2: reuse the same L, U Forward, $Lz = V_2$:

$$z_1 = 316.62$$

$$z_2 = 502.71 - 316.62 = 186.09$$

$$z_3 = 620.25 - 316.62 - 1.5(186.09) = 620.25 - 316.62 - 279.135 = 24.495$$

$$z_4 = 978.87 - 316.62 - 3(186.09) - 8(24.495) = 978.87 - 316.62 - 558.27 - 195.96 = -91.98$$

Back, $Ua = z$:

$$a_3 = \frac{-91.98}{1125} = -8.176 \times 10^{-2}$$

$$a_2 = \frac{24.495 - 1078.125(-8.176 \times 10^{-2})}{18.75} = \frac{24.495 + 88.14750}{18.75} = 6.0076$$

$$a_1 = \frac{186.09 - 175(6.0076) - 4625(-8.176 \times 10^{-2})}{5} = \frac{186.09 - 1051.33 + 378.14}{5} = -97.42$$

$$\begin{aligned} a_0 &= 316.62 - 15(-97.42) - 225(6.0076) - 3375(-8.176 \times 10^{-2}) \\ &= 316.62 + 1461.30 - 1351.71 + 275.94 = 702.15 \end{aligned}$$

$$v_2(t) = 702.15 - 97.42t + 6.0076t^2 - 8.176 \times 10^{-2}t^3$$

Evaluate at $t = 21$:

$$\begin{aligned} v_2(21) &= 702.15 - 97.42(21) + 6.0076(441) - 8.176 \times 10^{-2}(9261) \\ &= 702.15 - 2045.82 + 2649.35 - 757.18 \end{aligned}$$

$$\boxed{v_2(21) = 548.5022 \text{ m/s}}$$

Answer summary

Missile	Cubic coefficients (a_0, a_1, a_2, a_3)	$v(21)$
1	(-18.57, 23.572, 0.01008889, 0.0553632)	550.8632 m/s
2	(702.15, -97.42, 6.0076, -0.08176)	548.5022 m/s

Missile 1 is marginally ahead at $t = 21$ s, by about 2.36 m/s — consistent with the raw table, where V_2 overtakes V_1 somewhere between $t = 20$ and $t = 22.5$. **Elimination was performed exactly once**, as required.

4.10 Question 4(a)(i) [6, CO1, PO1]

Problem. What is the geometrical interpretation of Eigenvalues and Eigenvectors for a given matrix? Explain with an example.

4.10.1 Solution

A square matrix A acting on a vector $\mathbf{x}$ generally does two things: **rotates** it and **rescales** it. The eigen-equation

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}$$

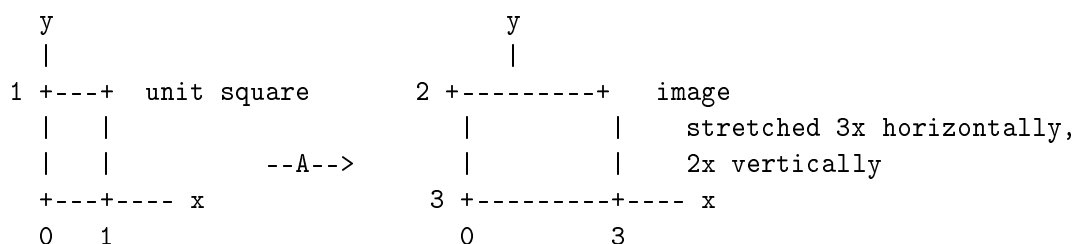
picks out the exceptional directions where the rotation part disappears.

- **Eigenvector** $\mathbf{v}$ — a direction that the transformation leaves **invariant**. $A\mathbf{v}$ lies along the *same line* through the origin as $\mathbf{v}$; the vector is not turned off its own axis.
- **Eigenvalue** λ — the **stretch factor** along that invariant direction. $|\lambda| > 1$ stretches, $|\lambda| < 1$ compresses, $\lambda < 0$ additionally flips the vector to point the opposite way, and $\lambda = 0$ collapses that direction to the origin (so A is singular).
- $\det A = \prod \lambda_i$ is the factor by which the transformation scales **area (2D) or volume (3D)**.

Example. Take

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Then $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, so $\lambda_1 = 3$ along the x -axis and $\lambda_2 = 2$ along the y -axis.



A general vector such as $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ maps to $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ — **its direction changes** (from 45° to 33.7°), so it is *not* an eigenvector. Only the two axis directions survive untilted. The area of the unit square goes from 1 to $6 = \det A = 3 \times 2$.

Non-diagonal example. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ the eigen-directions are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ($\lambda = 3$, stretched) and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ($\lambda = 1$, unchanged): the map stretches the plane along the 45° diagonal and leaves the anti-diagonal alone.

Where this matters numerically. The eigenvalues of an iteration matrix govern convergence (spectral radius $\rho < 1 \Rightarrow$ convergence); the ratio $\lambda_{\max}/\lambda_{\min}$ is the condition number that predicts how much round-off will be amplified; and the largest-magnitude eigenvalue is what the power method converges to.

4.11 Question 4(a)(ii) [12, CO2, PO1]

Problem. Provide mathematical reasoning as to whether $\lambda = 3$ is one of the four eigenvalues of the following matrix A :

$$A = \begin{bmatrix} 4 & 3 & 6 & 3 \\ 3 & 4 & 6 & 2 \\ 4 & 2 & 3 & 2 \\ 2 & 3 & 6 & 5 \end{bmatrix}$$

4.11.1 Solution

The criterion. λ is an eigenvalue of A if and only if $(A - \lambda I)$ is singular, i.e.

$$\det(A - \lambda I) = 0.$$

(Reason: $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$ means $(A - \lambda I)\mathbf{v} = \mathbf{0}$ has a nontrivial solution, which requires the matrix to have a nonzero null space, i.e. zero determinant.)

So we test whether $\det(A - 3I) = 0$.

$$A - 3I = \begin{bmatrix} 1 & 3 & 6 & 3 \\ 3 & 1 & 6 & 2 \\ 4 & 2 & 0 & 2 \\ 2 & 3 & 6 & 2 \end{bmatrix}$$

Evaluate the determinant by forward elimination (a determinant is unchanged by adding a multiple of one row to another).

Column 1, pivot = 1: $R_2 \leftarrow R_2 - 3R_1$, $R_3 \leftarrow R_3 - 4R_1$, $R_4 \leftarrow R_4 - 2R_1$:

$$\begin{bmatrix} 1 & 3 & 6 & 3 \\ 0 & -8 & -12 & -7 \\ 0 & -10 & -24 & -10 \\ 0 & -3 & -6 & -4 \end{bmatrix}$$

Column 2, pivot = -8 : $R_3 \leftarrow R_3 - \frac{-10}{-8}R_2 = R_3 - \frac{5}{4}R_2$, $R_4 \leftarrow R_4 - \frac{-3}{-8}R_2 = R_4 - \frac{3}{8}R_2$:

$$R_3: (-24 - \frac{5}{4}(-12), -10 - \frac{5}{4}(-7)) = (-24 + 15, -10 + 8.75) = (-9, -1.25)$$

$$R_4: (-6 - \frac{3}{8}(-12), -4 - \frac{3}{8}(-7)) = (-6 + 4.5, -4 + 2.625) = (-1.5, -1.375)$$

$$\begin{bmatrix} 1 & 3 & 6 & 3 \\ 0 & -8 & -12 & -7 \\ 0 & 0 & -9 & -\frac{5}{4} \\ 0 & 0 & -\frac{3}{2} & -\frac{11}{8} \end{bmatrix}$$

Column 3, pivot = -9 : $R_4 \leftarrow R_4 - \frac{-3/2}{-9}R_3 = R_4 - \frac{1}{6}R_3$:

$$-\frac{11}{8} - \frac{1}{6}(-\frac{5}{4}) = -\frac{11}{8} + \frac{5}{24} = \frac{-33 + 5}{24} = -\frac{28}{24} = -\frac{7}{6}$$

$$\begin{bmatrix} 1 & 3 & 6 & 3 \\ 0 & -8 & -12 & -7 \\ 0 & 0 & -9 & -\frac{5}{4} \\ 0 & 0 & 0 & -\frac{7}{6} \end{bmatrix}$$

Determinant = product of the pivots:

$$\det(A - 3I) = (1)(-8)(-9)\left(-\frac{7}{6}\right) = 72 \times \left(-\frac{7}{6}\right)$$

$$\boxed{\det(A - 3I) = -84 \neq 0}$$

4.11.2 Conclusion

$$\boxed{\lambda = 3 \text{ is NOT an eigenvalue of } A.}$$

Since $\det(A - 3I) = -84 \neq 0$, the matrix $(A - 3I)$ is nonsingular, so $(A - 3I)\mathbf{v} = \mathbf{0}$ admits **only** the trivial solution $\mathbf{v} = \mathbf{0}$ — and the zero vector is excluded by definition. No eigenvector can correspond to $\lambda = 3$.

Cross-check. The characteristic polynomial factors as

$$\det(A - \lambda I) = (\lambda - 2)(\lambda^3 - 14\lambda^2 - 2\lambda + 21)$$

whose roots are

$$\lambda \approx 14.0359, \quad 2, \quad 1.2054, \quad -1.2413$$

— and 3 is not among them. Substituting $\lambda = 3$ into the polynomial gives $(3 - 2)(27 - 126 - 6 + 21) = 1 \times (-84) = -84$ ✓, matching the determinant computed above.

4.12 Question 4(b) [7, CO2, PO1]

Problem. Find the eigenvalues and eigenvectors of the following matrix B :

$$B = \begin{bmatrix} 6 & -3 \\ 1 & 4 \end{bmatrix}$$

4.12.1 Solution

Characteristic equation.

$$\det(B - \lambda I) = \det \begin{bmatrix} 6 - \lambda & -3 \\ 1 & 4 - \lambda \end{bmatrix} = (6 - \lambda)(4 - \lambda) - (-3)(1) = \lambda^2 - 10\lambda + 24 + 3$$

$$\lambda^2 - 10\lambda + 27 = 0$$

(Consistent with $\text{tr } B = 10$ and $\det B = 24 + 3 = 27$.)

Roots.

$$\lambda = \frac{10 \pm \sqrt{100 - 4(27)}}{2} = \frac{10 \pm \sqrt{-8}}{2} = \frac{10 \pm 2i\sqrt{2}}{2}$$

$$\lambda_{1,2} = 5 \pm i\sqrt{2} = 5 \pm 1.41421i$$

The discriminant is negative, so the eigenvalues form a **complex-conjugate pair** — possible here because B is **not symmetric** ($b_{12} = -3 \neq 1 = b_{21}$). Compare Session B Q6(a)(i), where symmetry forced real eigenvalues.

Eigenvectors. Solve $(B - \lambda I)\mathbf{v} = \mathbf{0}$. Using the first row,

$$(6 - \lambda)x - 3y = 0 \implies \frac{x}{y} = \frac{3}{6 - \lambda}$$

For $\lambda_1 = 5 + i\sqrt{2}$: $6 - \lambda_1 = 1 - i\sqrt{2}$, so

$$\frac{x}{y} = \frac{3}{1 - i\sqrt{2}} = \frac{3(1 + i\sqrt{2})}{(1)^2 + (\sqrt{2})^2} = \frac{3(1 + i\sqrt{2})}{3} = 1 + i\sqrt{2}$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 + i\sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + 1.41421i \\ 1 \end{bmatrix}$$

For $\lambda_2 = 5 - i\sqrt{2}$ (conjugate):

$$\mathbf{v}_2 = \begin{bmatrix} 1 - i\sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 - 1.41421i \\ 1 \end{bmatrix}$$

Verification for λ_1 .

$$B\mathbf{v}_1 = \begin{bmatrix} 6(1 + i\sqrt{2}) - 3(1) \\ 1(1 + i\sqrt{2}) + 4(1) \end{bmatrix} = \begin{bmatrix} 3 + 6i\sqrt{2} \\ 5 + i\sqrt{2} \end{bmatrix}$$

$$\lambda_1 \mathbf{v}_1 = (5 + i\sqrt{2}) \begin{bmatrix} 1 + i\sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 5 + 5i\sqrt{2} + i\sqrt{2} + i^2 \cdot 2 \\ 5 + i\sqrt{2} \end{bmatrix} = \begin{bmatrix} 3 + 6i\sqrt{2} \\ 5 + i\sqrt{2} \end{bmatrix} \quad \checkmark$$

Geometric reading. Complex eigenvalues $\lambda = re^{\pm i\theta}$ mean the map has **no real invariant direction** — it rotates the plane by $\theta = \arctan(\sqrt{2}/5) = 15.79^\circ$ while scaling by $r = |\lambda| = \sqrt{25 + 2} = \sqrt{27} = 5.196$. Every real vector is turned, which is why no real eigenvector exists.

4.13 Question 5(a) [13, CO2, PO1]

Problem. Find the following integral using **Simpson's 3/8 Rule**:

$$I = \int_0^{100} f(x) dx$$

where

$$f(x) = \begin{cases} 0, & 0 < x < 30 \\ -9.1688 \times 10^{-6}x^3 + 2.796 \times 10^{-3}x^2 - 0.28487x + 9.6778, & 30 < x < 172 \end{cases}$$

4.13.1 Solution

Step 0 — handle the discontinuity first f is piecewise defined with a jump at $x = 30$: it is identically zero to the left and takes the value

$$f(30^+) = -9.1688 \times 10^{-6}(27000) + 2.796 \times 10^{-3}(900) - 0.28487(30) + 9.6778 = 3.40054$$

to the right. A quadrature rule assumes a *smooth* integrand over each application; straddling $x = 30$ with a single stencil would let the cubic fit smear the jump across the interval and corrupt the result. So split at the breakpoint — this is the essential step:

$$I = \underbrace{\int_0^{30} 0 \, dx}_{=0} + \int_{30}^{100} f_c(x) \, dx, \quad f_c(x) = -9.1688 \times 10^{-6}x^3 + 2.796 \times 10^{-3}x^2 - 0.28487x + 9.6778$$

Step 1 — Simpson's 3/8 on $[30, 100]$

$$h = \frac{b-a}{3} = \frac{100-30}{3} = \frac{70}{3} = 23.33333$$

$$x_0 = 30, \quad x_1 = 53.33333, \quad x_2 = 76.66667, \quad x_3 = 100$$

Function values.

$$f(x_0) = f(30):$$

$$\begin{aligned} & -9.1688 \times 10^{-6}(27,000) + 2.796 \times 10^{-3}(900) - 0.28487(30) + 9.6778 \\ & = -0.247558 + 2.516400 - 8.546100 + 9.677800 = 3.400542 \end{aligned}$$

$$f(x_1) = f(53.33333):$$

$$\begin{aligned} & -9.1688 \times 10^{-6}(151,703.7) + 2.796 \times 10^{-3}(2844.444) - 0.28487(53.33333) + 9.6778 \\ & = -1.390940 + 7.953067 - 15.192533 + 9.677800 = 1.046859 \end{aligned}$$

$$f(x_2) = f(76.66667):$$

$$\begin{aligned} & -9.1688 \times 10^{-6}(450,629.6) + 2.796 \times 10^{-3}(5877.778) - 0.28487(76.66667) + 9.6778 \\ & = -4.131734 + 16.434267 - 21.840033 + 9.677800 = 0.140300 \end{aligned}$$

$$f(x_3) = f(100):$$

$$\begin{aligned} & -9.1688 \times 10^{-6}(1,000,000) + 2.796 \times 10^{-3}(10,000) - 0.28487(100) + 9.6778 \\ & = -9.168800 + 27.960000 - 28.487000 + 9.677800 = -0.018000 \end{aligned}$$

i	x_i	$f(x_i)$	weight
0	30.00000	3.400542	1
1	53.33333	1.046859	3

i	x_i	$f(x_i)$	weight
2	76.66667	0.140300	3
3	100.00000	-0.018000	1

Apply the rule.

$$I \approx \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] = \frac{3(23.33333)}{8} [3.400542 + 3(1.046859) + 3(0.140300) - 0.018000]$$

$$= 8.75 [3.400542 + 3.140577 + 0.420900 - 0.018000] = 8.75 \times 6.944019$$

$$I = \int_0^{100} f(x) dx \approx 60.7602$$

Accuracy Because f_c is a **cubic** and Simpson's $3/8$ rule has degree of precision 3, this result is **exact** (up to round-off). Analytically:

$$\int_{30}^{100} f_c dx = \left[-\frac{9.1688 \times 10^{-6}}{4} x^4 + \frac{2.796 \times 10^{-3}}{3} x^3 - \frac{0.28487}{2} x^2 + 9.6778x \right]_{30}^{100} = 60.76018$$

✓ — the same phenomenon as Session B Q1(b).

Warning about the naive route. Applying the rule blindly on $[0, 100]$ with $h = 100/3$ puts nodes at $x = 0, 33.33, 66.67, 100$ and gives $I \approx 125.24$ — more than **twice** the true value, because the cubic fitted through those four ordinates never “sees” the jump and wrongly extends positive area back over $[0, 30]$ where $f \equiv 0$. **Always split a quadrature at a discontinuity.**

4.14 Question 5(b) [12, CO2, PO1]

Problem. Solve the integral given in Question 5.a) using **Euler's method** considering step size $h = 25$.

4.14.1 Solution

Recasting the integral as an initial-value problem

$$I = \int_0^{100} f(x) dx \iff \frac{dI}{dx} = f(x), \quad I(0) = 0,$$

and $I(100)$ is the answer. Euler's method:

$$I_{i+1} = I_i + f(x_i)h$$

Note the right-hand side depends on x only, so Euler here is exactly the **left-endpoint rectangle (Riemann) rule**.

With $h = 25$ and $x_i = 0, 25, 50, 75$ (four steps reach $x = 100$):

Function values needed

$$f(0) = 0 \quad (\text{since } 0 < 0 < 30 \text{ region: } f \equiv 0)$$

$$f(25) = 0 \quad (25 < 30)$$

$$\begin{aligned} f(50) &= -9.1688 \times 10^{-6}(125,000) + 2.796 \times 10^{-3}(2500) - 0.28487(50) + 9.6778 \\ &= -1.146100 + 6.990000 - 14.243500 + 9.677800 = 1.278200 \end{aligned}$$

$$\begin{aligned} f(75) &= -9.1688 \times 10^{-6}(421,875) + 2.796 \times 10^{-3}(5625) - 0.28487(75) + 9.6778 \\ &= -3.868065 + 15.727500 - 21.365250 + 9.677800 = 0.171985 \end{aligned}$$

Marching

step	x_i	$f(x_i)$	I_i	$I_{i+1} = I_i + f(x_i)h$
1	0	0	0	$0 + 25(0) = 0$
2	25	0	0	$0 + 25(0) = 0$
3	50	1.278200	0	$0 + 25(1.278200) = 31.95500$
4	75	0.171985	31.95500	$31.95500 + 25(0.171985) = 36.25462$

$I(100) \approx 36.2546$

Comparison with Q5(a)

$$E_t = 60.7602 - 36.2546 = 24.5056, \quad |\epsilon_t| = \left| \frac{60.7602 - 36.2546}{60.7602} \right| \times 100 = \boxed{40.33\%}$$

Why Euler does so badly here. Three reasons compound:

1. **It is only $O(h)$ accurate**, versus $O(h^4)$ for Simpson's 3/8.
2. **$h = 25$ is enormous** relative to the scale over which f varies — f falls from 3.40 at $x = 30$ to -0.018 at $x = 100$, and the whole rise-and-fall is sampled at just two nonzero points.
3. **The grid misses the discontinuity entirely.** The node at $x = 25$ reports $f = 0$, so Euler credits *zero* area to the whole interval $[25, 50]$ even though f is positive and largest over $[30, 50]$. Being a left-endpoint rule applied to a decreasing function, it also systematically under-counts thereafter.

Placing a node exactly at $x = 30$ (e.g. by integrating $[30, 100]$ with $h = 17.5$) would remove the worst of the error.

4.15 Question 6(a) [12, CO2, PO1]

Problem. When a voltage input of 20V is provided, the speed of a motor is obtained by the following differential equation:

$$20 = 0.02 \frac{dv}{dt} + 0.06v$$

If the initial speed of the motor was 0, then find the speed at $t = 0.8$ using **Runge–Kutta 4th order method** taking step size $h = 0.4$.

4.15.1 Solution

Put the ODE in standard form.

$$0.02 \frac{dv}{dt} = 20 - 0.06v \implies \frac{dv}{dt} = \frac{20 - 0.06v}{0.02} = 1000 - 3v$$

$$\boxed{f(t, v) = 1000 - 3v, \quad v(0) = 0, \quad h = 0.4}$$

RK4 formulae.

$$v_{i+1} = v_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$$

$$k_1 = f(t_i, v_i), \quad k_2 = f\left(t_i + \frac{h}{2}, v_i + \frac{h}{2}k_1\right), \quad k_3 = f\left(t_i + \frac{h}{2}, v_i + \frac{h}{2}k_2\right), \quad k_4 = f(t_i + h, v_i + hk_3)$$

Here f does not depend on t , so only the v -argument matters. With $\frac{h}{2} = 0.2$:

Step 1: $t_0 = 0 \rightarrow t_1 = 0.4, \quad v_0 = 0$

$$k_1 = 1000 - 3(0) = 1000$$

$$k_2 = 1000 - 3(0 + 0.2(1000)) = 1000 - 3(200) = 1000 - 600 = 400$$

$$k_3 = 1000 - 3(0 + 0.2(400)) = 1000 - 3(80) = 1000 - 240 = 760$$

$$k_4 = 1000 - 3(0 + 0.4(760)) = 1000 - 3(304) = 1000 - 912 = 88$$

$$\frac{k_1 + 2k_2 + 2k_3 + k_4}{6} = \frac{1000 + 2(400) + 2(760) + 88}{6} = \frac{1000 + 800 + 1520 + 88}{6} = \frac{3408}{6} = 568$$

$$v_1 = 0 + 568(0.4) = 227.2$$

Step 2: $t_1 = 0.4 \rightarrow t_2 = 0.8, \quad v_1 = 227.2$

$$k_1 = 1000 - 3(227.2) = 1000 - 681.6 = 318.4$$

$$k_2 = 1000 - 3(227.2 + 0.2(318.4)) = 1000 - 3(227.2 + 63.68) = 1000 - 3(290.88) = 1000 - 872.64 = 127.36$$

$$k_3 = 1000 - 3(227.2 + 0.2(127.36)) = 1000 - 3(227.2 + 25.472) = 1000 - 3(252.672) = 1000 - 758.016 = 241.984$$

$$k_4 = 1000 - 3(227.2 + 0.4(241.984)) = 1000 - 3(227.2 + 96.7936) = 1000 - 3(323.9936) = 1000 - 971.9808 = 28.0192$$

$$\frac{k_1 + 2k_2 + 2k_3 + k_4}{6} = \frac{318.4 + 2(127.36) + 2(241.984) + 28.0192}{6} = \frac{1085.1072}{6} = 180.8512$$

$$v_2 = 227.2 + 180.8512(0.4) = 227.2 + 72.34048$$

$$v(0.8) \approx 299.5405 \text{ (speed units, e.g. rad/s)}$$

Summary and accuracy check

i	t_i	v_i	k_1	k_2	k_3	k_4	slope
0	0.0	0.0000	1000.000	400.000	760.000	88.019	568.0000
1	0.4	227.2000	318.400	127.360	241.984	28.0192	180.8512
2	0.8	299.5405					

Exact solution. The ODE is linear:

$$v(t) = \frac{1000}{3} (1 - e^{-3t}) \implies v(0.8) = 333.3333 (1 - e^{-2.4}) = 303.0940$$

$$|\epsilon_t| = \left| \frac{303.0940 - 299.5405}{303.0940} \right| \times 100 = 1.172\%$$

Only 1.17% error from two steps with a very coarse $h = 0.4$ (note $3h = 1.2$, so the step is comparable to the system time constant $\tau = 1/3$). Halving h would cut the error by $2^4 = 16$. The steady-state speed is $v_\infty = 1000/3 = 333.33$, and the numerical solution is correctly converging towards it.

4.16 Question 6(b) [13, CO2, PO1]

Problem. For drawing road networks from aerial images, light intensities are measured at different pixel locations. The following intensities are given as a function of pixel location in Table 3:

Pixel (k)	Intensity (y)
-3	119
-2	165
-1	231
0	243
1	244
2	214
3	136

Suppose we want to perform polynomial regression on the given data using a 2nd order polynomial:

$$y = a_0 + a_1 k + a_2 k^2$$

Find the corresponding polynomial model parameters.

4.16.1 Solution

Normal equations Minimising $S_r = \sum_{i=1}^n (y_i - a_0 - a_1 k_i - a_2 k_i^2)^2$ and setting $\partial S_r / \partial a_j = 0$ gives the linear system

$$\begin{bmatrix} n & \sum k_i & \sum k_i^2 \\ \sum k_i & \sum k_i^2 & \sum k_i^3 \\ \sum k_i^2 & \sum k_i^3 & \sum k_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum k_i y_i \\ \sum k_i^2 y_i \end{bmatrix}$$

Summation table

i	k_i	y_i	k_i^2	k_i^3	k_i^4	$k_i y_i$	$k_i^2 y_i$
1	-3	119	9	-27	81	-357	1071
2	-2	165	4	-8	16	-330	660
3	-1	231	1	-1	1	-231	231
4	0	243	0	0	0	0	0
5	1	244	1	1	1	244	244
6	2	214	4	8	16	428	856
7	3	136	9	27	81	408	1224
$\sum$	0	1352	28	0	196	162	4286

with $n = 7$.

The odd-power sums vanish ($\sum k_i = 0$, $\sum k_i^3 = 0$) because the pixel locations are symmetric about $k = 0$. This decouples the system beautifully.

The system

$$\begin{bmatrix} 7 & 0 & 28 \\ 0 & 28 & 0 \\ 28 & 0 & 196 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1352 \\ 162 \\ 4286 \end{bmatrix}$$

Row 2 is decoupled:

$$28a_1 = 162 \implies a_1 = \frac{162}{28} = \frac{81}{14}$$

$$\boxed{a_1 = 5.785714}$$

Rows 1 and 3 form a 2×2 system in a_0, a_2 :

$$7a_0 + 28a_2 = 1352 \quad (\text{i})$$

$$28a_0 + 196a_2 = 4286 \quad (\text{ii})$$

$$(\text{ii}) - 4 \times (\text{i}): \quad 196a_2 - 112a_2 = 4286 - 5408$$

$$84a_2 = -1122 \implies a_2 = -\frac{1122}{84} = -\frac{187}{14}$$

$$\boxed{a_2 = -13.357143}$$

From (i): $7a_0 = 1352 - 28\left(-\frac{187}{14}\right) = 1352 + 374 = 1726$

$$a_0 = \frac{1726}{7}$$

$$a_0 = 246.571429$$

The fitted model

$$y = 246.5714 + 5.7857k - 13.3571k^2$$

Verification and interpretation

k	y (measured)	$\hat{y}$ (model)	residual
-3	119	109.0	+10.0
-2	165	181.6	-16.6
-1	231	227.4	+3.6
0	243	246.6	-3.6
1	244	239.0	+5.0
2	214	204.7	+9.3
3	136	143.7	-7.7

The residuals sum to ≈ 0 ✓ (a property guaranteed by the first normal equation).

Since $a_2 < 0$ the parabola opens downward, as an intensity profile across a bright road against a dark background should. The **peak** of the fitted profile is at

$$k^* = -\frac{a_1}{2a_2} = -\frac{5.785714}{2(-13.357143)} = 0.2166 \text{ pixels}$$

with peak intensity $\hat{y}(k^*) = 247.2$ — i.e. the sub-pixel centre of the road is estimated to lie about 0.22 pixel to the right of pixel 0. This sub-pixel localisation is precisely what the quadratic fit is used for in road-extraction algorithms; simply picking the brightest pixel would give the integer answer $k = 1$, a 0.78-pixel bias.

5 Session D — Math 4543, Winter Semester 2024–2025 (08 April 2026, Morning)

B.Sc. Engg. SWE, 5th Semester — Duration 3 hours — Full marks 150

5.1 Question 1(a) [3+2, CO2, PO1]

Problem. What are the steps of problem-solving in engineering? With a suitable example, briefly explain how numerical methods are used to solve an engineering problem.

5.1.1 Solution

The steps

1. **Describe the physical problem.** State what is happening, what is known, what is required, and what accuracy is acceptable.
2. **Formulate the mathematical model.** Apply the governing physical laws (conservation of mass/momentum/energy, Newton's laws, Ohm's law, Fourier's law) to obtain equations — algebraic, transcendental, differential or integral — together with boundary and/or initial conditions. Simplifying assumptions are declared here.
3. **Choose a solution method.** Attempt an exact/analytical solution first. If none exists in closed form, or the geometry/data make it impractical, select an appropriate **numerical method**.
4. **Implement and compute.** Discretise, code the algorithm, choose the step size / tolerance / number of segments, and run it.
5. **Verify and validate.** Check convergence (does refining h change the answer?), estimate $|\epsilon_a|$, check limiting/steady-state behaviour, compare with experiment or with an analytically solvable special case.
6. **Interpret and act.** Translate the number back into an engineering decision, and iterate on the model if the result is unsatisfactory.

Worked example — the shrink-fit trunnion (Session B Q1)

Step	Content
1. Physical problem	A trunnion of diameter 12.363" must be cooled in dry ice (-108°F) from 80°F so it contracts enough to slide into its hub. How much does the diameter shrink?
2. Mathematical model	The coefficient of thermal expansion α varies with temperature, so $\Delta D = D \int_{80}^{-108} \alpha(T) dT$ with $\alpha(T) = -1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6}$.

Step	Content
3. Choose a method	Here the integral does happen to be elementary — but in practice $\alpha(T)$ is supplied as a table of measurements , with no formula to integrate. So a numerical quadrature (Simpson's 1/3 rule) is used.
4. Compute	2-segment and 4-segment Simpson's 1/3: $\Delta D = -0.013689''$ in both cases.
5. Verify	$ \epsilon_a = 0\%$ between the two segment counts $\Rightarrow$ converged. Cross-checked against the analytical value: identical.
6. Interpret	A contraction of $0.0137''$; compare with the required interference fit. If insufficient, the engineer returns to step 1 — try liquid nitrogen, or heat the hub instead.

The general point. Numerical methods enter at step 3, whenever the model produced at step 2 has no tractable closed-form solution — integrals of tabulated data, nonlinear root-finding, large linear systems, ODEs with nonlinear right-hand sides. They convert a mathematical statement that cannot be solved symbolically into a finite sequence of arithmetic operations a computer can execute, at the price of a **truncation error** that the engineer must then bound and control (step 5).

5.2 Question 1(b) [6, CO4, PO1]

Problem. What are the drawbacks of the Bisection method? Explain with proper examples and illustrations.

5.2.1 Solution

Bisection is unconditionally convergent given a valid bracket — that is its virtue. Its drawbacks are these:

1. Slow (linear) convergence The error is exactly halved each iteration:

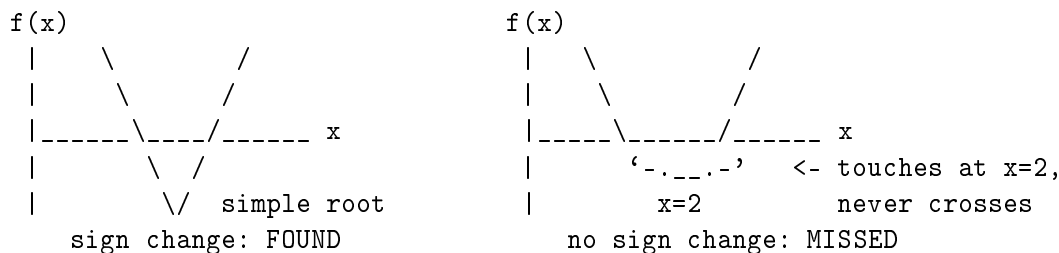
$$|\epsilon_n| = \frac{b-a}{2^{n+1}}$$

so it gains only about **0.3 decimal digits per iteration** and has order of convergence 1. Newton–Raphson, by contrast, is quadratic (doubling correct digits per step) and the Secant method is superlinear (≈ 1.618).

Example. On $[1, 2]$, reaching 10^{-6} accuracy needs $n \approx \log_2(10^6) \approx 20$ iterations; Newton typically needs 4–5.

2. It requires a sign-change bracket, so it misses roots of even multiplicity The method needs $f(x_\ell)f(x_u) < 0$. A root where the curve **touches** the axis without crossing is invisible to it.

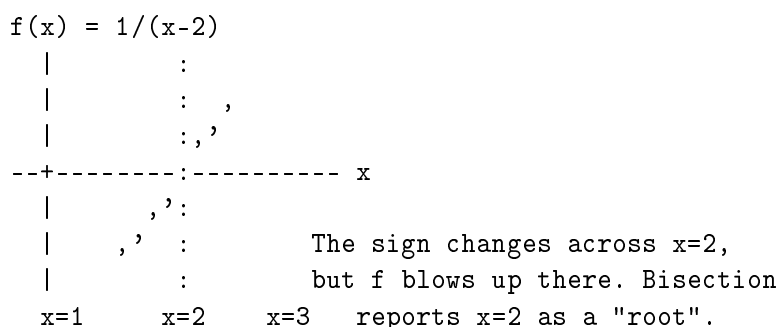
Example. $f(x) = (x - 2)^2$ has the double root $x = 2$, but $f(x) \geq 0$ everywhere, so **no** bracket with a sign change exists.



The same failure occurs for any root of even multiplicity, e.g. $(x - 1)^4$.

3. A singularity inside the bracket is mistaken for a root Bisection only inspects the *sign* of f , never its magnitude. A pole also flips the sign.

Example. $f(x) = \frac{1}{x-2}$ on $[1, 3]$: $f(1) = -1$, $f(3) = +1$, so $f(x_\ell)f(x_u) < 0$ and the method happily “converges” to $x = 2$ — which is a **vertical asymptote where $|f| \rightarrow \infty$** , not a root.



Remedy: always check that $|f(x_r)|$ is small at the returned point, not just that the interval has shrunk.

4. Only one root is found, even when several lie in the bracket If the interval contains an odd number of roots the method finds *one* of them (unpredictably which); if it contains an **even** number, the endpoints have the same sign and the method refuses to start at all.

Example. $f(x) = (x - 1)(x - 2)(x - 3)$ on $[0, 4]$: $f(0) = -6$, $f(4) = 6$ — a valid bracket — but the three roots 1, 2, 3 cannot all be recovered from it.

5. It cannot find complex roots The method is built on real sign changes, so complex roots (e.g. of $x^2 + 1 = 0$) are entirely out of reach — unlike Newton–Raphson with a complex initial guess or Müller’s method.

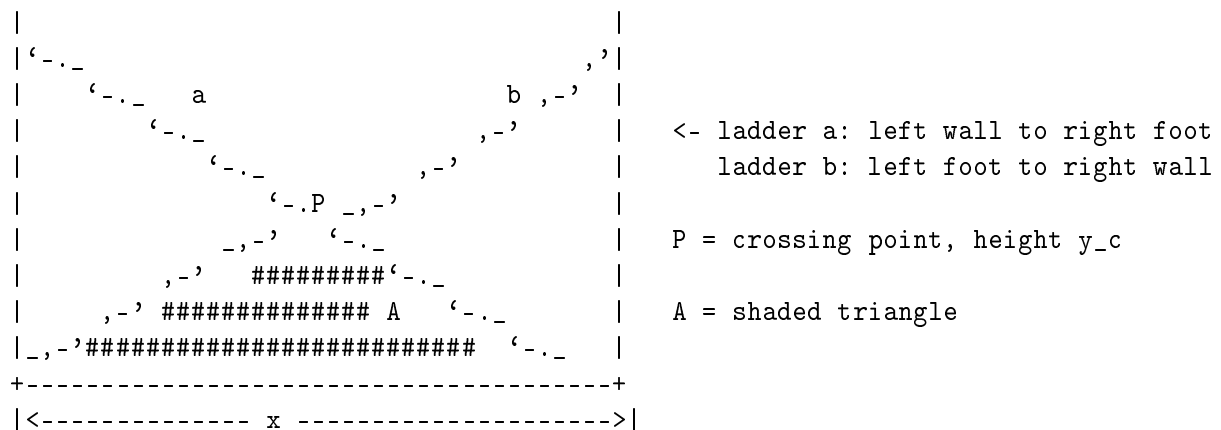
6. It ignores all information beyond the sign f is evaluated at every iteration, yet only one bit of the result (its sign) is used. The magnitude of f — which False Position and the Secant method exploit — is thrown away. A good initial guess is also wasted: if the root happens to lie at the very first midpoint, bisection cannot know and grinds on regardless.

7. An initial bracket must be supplied For a function whose roots are not roughly known in advance, finding x_ℓ, x_u with $f(x_\ell)f(x_u) < 0$ is itself a search problem that the method does not solve.

In balance. These drawbacks are why bisection is usually used as a **safeguard** — to obtain a small reliable bracket — with a fast open method (Newton, secant) taking over inside it. Hybrid schemes such as Brent’s method do exactly this.

5.3 Question 1(c) — preamble

Consider two ladders of known lengths $a > 0$ and $b > 0$ leaning across a straight alley of unknown width $x > 0$, touching opposite vertical walls, as portrayed in Figure 1. The triangle bounded by the ground and the lower sections of the two ladders has a known area $A > 0$. Assume the configuration is physically realizable and that $0 < x < \min(a, b)$.



5.4 Question 1(c)(i) [12, CO1, PO1]

Problem. From the given scenario, formulate a function $f(x)$ (in terms of a , b , and A) whose positive root yields the width x of the alley.

5.4.1 Solution

Step 1 — heights reached on the walls. Each ladder, its wall, and the ground form a right triangle with hypotenuse the ladder length and base the alley width x . By Pythagoras:

$$p \equiv \text{height reached by ladder } a = \sqrt{a^2 - x^2}$$

$$q \equiv \text{height reached by ladder } b = \sqrt{b^2 - x^2}$$

(Both are real because $0 < x < \min(a, b)$.)

Step 2 — height of the crossing point. Put the origin at the foot of the left wall, x -axis along the ground. Ladder a runs from $(0, p)$ down to $(x, 0)$; ladder b runs from $(0, 0)$ up to (x, q) . Their lines are

$$y_a(t) = p \left(1 - \frac{t}{x}\right), \quad y_b(t) = q \frac{t}{x}$$

Setting $y_a = y_b$:

$$p - \frac{pt}{x} = \frac{qt}{x} \implies px = t(p+q) \implies t^* = \frac{px}{p+q}$$

and the crossing height is

$$y_c = q \frac{t^*}{x} = \frac{pq}{p+q} \iff \frac{1}{y_c} = \frac{1}{p} + \frac{1}{q}$$

(the classical “crossed-ladders” harmonic relation — note it is independent of where along the ground the crossing occurs).

Step 3 — area of the shaded triangle. The triangle bounded by the ground and the *lower* sections of the two ladders has base x (the full alley width, from wall to wall) and apex at the crossing point, so its height is y_c :

$$A = \frac{1}{2} x y_c = \frac{1}{2} \cdot \frac{x pq}{p+q}$$

Step 4 — clear denominators to obtain a root-finding form. Multiply through by $2(p+q)$:

$$x pq = 2A (p+q)$$

$$f(x) = x\sqrt{a^2 - x^2}\sqrt{b^2 - x^2} - 2A \left(\sqrt{a^2 - x^2} + \sqrt{b^2 - x^2} \right) = 0$$

The positive root of f on $(0, \min(a, b))$ is the alley width x .

Sanity check of the formulation.

- **Limits.** As $x \rightarrow 0^+$: $f \rightarrow 0 - 2A(a+b) < 0$. As $x \rightarrow \min(a, b)^-$: one of p, q vanishes, the first term $\rightarrow 0$ and $f \rightarrow -2A\sqrt{|a^2 - b^2|} < 0 \dots$ so f must be *positive* somewhere in between for a root to exist — which is exactly the condition that A is not too large. Physically: for given a, b there is a maximum achievable triangle area, and A must not exceed it. (Generically two positive roots exist; the physically intended one is selected by the initial guess.)
- **Numerical check.** Take $a = 10, b = 12, x = 6$: $p = 8, q = \sqrt{108} = 10.3923$, $y_c = \frac{8(10.3923)}{18.3923} = 4.5203$, $A = \frac{1}{2}(6)(4.5203) = 13.5609$. Substituting these back, $f(6) = 6(8)(10.3923) - 2(13.5609)(8 + 10.3923) = 498.83 - 498.83 = 0 \checkmark$

Non-dimensional form (useful in practice, avoids catastrophic cancellation): divide by pq to get the equivalent, better-conditioned equation

$$g(x) = x - 2A \left(\frac{1}{p} + \frac{1}{q} \right) = x - 2A \left(\frac{1}{\sqrt{a^2 - x^2}} + \frac{1}{\sqrt{b^2 - x^2}} \right) = 0.$$

5.5 Question 1(c)(ii) [6, CO2, PO1]

Problem. Derive the Newton–Raphson formula for iteratively solving $f(x) = 0$, using the function f formulated in the preceding Question 1.c)i.

5.5.1 Solution

General Newton–Raphson Expand f about the current iterate x_i and truncate after the linear term:

$$f(x_{i+1}) \approx f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

Demanding $f(x_{i+1}) = 0$ gives

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

(Geometrically: follow the tangent at $(x_i, f(x_i))$ down to the x -axis.)

Differentiating our f Write $p = \sqrt{a^2 - x^2}$, $q = \sqrt{b^2 - x^2}$, so that

$$f(x) = x p q - 2A(p + q)$$

$$\frac{dp}{dx} = \frac{-x}{\sqrt{a^2 - x^2}} = -\frac{x}{p}, \quad \frac{dq}{dx} = -\frac{x}{q}$$

Apply the product rule to xpq :

$$\frac{d}{dx}(xpq) = pq + x(p'q + pq') = pq + x\left(-\frac{x}{p}q - p\frac{x}{q}\right) = pq - x^2\left(\frac{q}{p} + \frac{p}{q}\right)$$

And

$$\frac{d}{dx}\left[-2A(p + q)\right] = -2A\left(-\frac{x}{p} - \frac{x}{q}\right) = 2Ax\left(\frac{1}{p} + \frac{1}{q}\right)$$

$$f'(x) = pq - x^2\left(\frac{q}{p} + \frac{p}{q}\right) + 2Ax\left(\frac{1}{p} + \frac{1}{q}\right)$$

The iteration formula

$$x_{i+1} = x_i - \frac{x_i p_i q_i - 2A(p_i + q_i)}{p_i q_i - x_i^2\left(\frac{q_i}{p_i} + \frac{p_i}{q_i}\right) + 2Ax_i\left(\frac{1}{p_i} + \frac{1}{q_i}\right)}$$

where

$$p_i = \sqrt{a^2 - x_i^2}, \quad q_i = \sqrt{b^2 - x_i^2}.$$

Written out fully in terms of a, b, A :

$$x_{i+1} = x_i - \frac{x_i \sqrt{a^2 - x_i^2} \sqrt{b^2 - x_i^2} - 2A(\sqrt{a^2 - x_i^2} + \sqrt{b^2 - x_i^2})}{\sqrt{a^2 - x_i^2} \sqrt{b^2 - x_i^2} - x_i^2 \left(\sqrt{\frac{b^2 - x_i^2}{a^2 - x_i^2}} + \sqrt{\frac{a^2 - x_i^2}{b^2 - x_i^2}} \right) + 2Ax_i \left(\frac{1}{\sqrt{a^2 - x_i^2}} + \frac{1}{\sqrt{b^2 - x_i^2}} \right)}$$

Practical notes.

- **Initial guess.** Take x_0 somewhere in $(0, \min(a, b))$; a reasonable choice is $x_0 = \frac{1}{2} \min(a, b)$.

- **Guard the domain.** If an iterate leaves $(0, \min(a, b))$, p or q becomes imaginary. Clamp the step (or damp it, $x_{i+1} = x_i - \omega f/f'$ with $\omega < 1$) to keep the iterate inside.
- **Stopping criterion.** $|\epsilon_a| = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100\% < \epsilon_s$.
- **Convergence.** Quadratic near a simple root; the method may stall near $\min(a, b)$ where $f' \rightarrow \pm\infty$, another reason to prefer the better-conditioned $g(x)$ of part (i) in code.

5.6 Question 2(a) [5, CO4, PO1]

Problem. Define Round-off Error. What are the different ways through which Round-off Error can be introduced? Explain with proper examples.

5.6.1 Solution

Definition. *Round-off error* is the error introduced because a computer represents and manipulates numbers with a **finite number of digits**, so the stored value differs from the exact mathematical value:

$$E_{\text{round-off}} = \text{exact value} - \text{stored/computed value}$$

It is distinct from **truncation error**, which arises from replacing an exact mathematical procedure (a limit, a series, an integral) by a finite approximation.

Ways in which round-off error is introduced 1. **Finite word-length representation of real numbers.** A `double` holds about 16 significant decimal digits (machine epsilon $\epsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$). Irrationals and non-terminating decimals must be truncated. *Example.* $\pi = 3.14159265358979\dots$ stored as 3.141593 (7 digits) carries $E \approx -3.5 \times 10^{-7}$.

2. **Base-2 representation of exact decimals.** Numbers that terminate in base 10 may not terminate in base 2. *Example.* $0.1_{10} = 0.0001100110011\overline{0011}_2$ — a repeating binary fraction. Hence in floating point $0.1 + 0.2 = 0.30000000000000004 \neq 0.3$, and $0.1+0.2 == 0.3$ evaluates to **false**.

3. **Chopping versus rounding.** *Chopping* discards trailing digits; *rounding* adjusts the last retained digit. *Example.* Storing $\frac{2}{3} = 0.6666\dots$ to 4 digits: chopping $\rightarrow 0.6666$ ($E = 6.67 \times 10^{-5}$), rounding $\rightarrow 0.6667$ ($E = -3.33 \times 10^{-5}$). Chopping is biased (always toward zero) and accumulates faster; rounding is unbiased.

4. **Subtractive cancellation** — the most dangerous source. Subtracting two nearly equal numbers annihilates the leading digits and promotes the round-off noise into significance. *Example.* With 5-digit arithmetic, $\sqrt{9.01} - 3 = 3.0017 - 3 = 0.0017$ — only 2 significant digits survive out of 5. The relative error jumps from $\sim 10^{-5}$ to $\sim 10^{-2}$. *Classic case:* the quadratic formula for $x^2 + 10^6x + 1 = 0$; the root $\frac{-b+\sqrt{b^2-4ac}}{2a}$ cancels catastrophically, and one must use the algebraically equivalent $\frac{2c}{-b-\sqrt{b^2-4ac}}$ instead.

5. **Accumulation over many operations.** Each of N operations contributes $O(\epsilon_{\text{mach}})$; errors accumulate as $O(N\epsilon)$ in the worst case, $O(\sqrt{N}\epsilon)$ statistically. *Example.* Adding 0.1 to itself 10^6 times in single precision gives ≈ 100958.34 instead of 100000 — an error of nearly 1%.

6. **Swamping** — adding a small number to a large one. *Example.* In 5-digit arithmetic, $10000 + 0.0001 = 10000$ exactly: the small addend is entirely lost. Summing a long series

from largest to smallest term loses the tail; summing smallest-first (or using Kahan summation) recovers it.

7. Overflow and underflow. Exceeding the exponent range yields $\pm\infty$ or flushes to zero. *Example.* e^{1000} overflows a `double` ($\approx 1.8 \times 10^{308}$ maximum).

8. Non-associativity of floating-point arithmetic. $(a+b)+c \neq a+(b+c)$ in general, so the *order* of operations changes the answer — which is why parallel reductions can give run-to-run variation.

Practical implication. Round-off error *grows* as h decreases (more operations, more cancellation), while truncation error *shrinks*. Their sum has a minimum at some optimal h — pushing h below it makes results **worse**, a central lesson of numerical differentiation.

5.7 Question 2(b) — preamble

The Maclaurin series for $\sin(x)$ is given as follows:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

5.8 Question 2(b)(i) [2, CO4, PO1]

Problem. Why is the sinusoid function $\sin(x)$ categorized as a transcendental function?

5.8.1 Solution

A function is **algebraic** if it satisfies a polynomial equation

$$P_n(x)y^n + P_{n-1}(x)y^{n-1} + \dots + P_0(x) = 0$$

whose coefficients $P_k(x)$ are themselves polynomials in x — equivalently, if it can be built from x by a **finite** number of additions, subtractions, multiplications, divisions and root extractions. A function that is **not** algebraic is called **transcendental** — it “transcends” algebra.

$\sin(x)$ satisfies **no such polynomial relation**, for the following reason: an algebraic function has only finitely many zeros on any bounded interval (a nonzero polynomial $P(x, y)$ can vanish on at most finitely many isolated points for fixed behaviour), whereas

$$\sin(x) = 0 \quad \text{at} \quad x = 0, \pm\pi, \pm2\pi, \pm3\pi, \dots$$

i.e. at **infinitely many** points, with no accumulation. No non-trivial polynomial equation can be satisfied by a function with infinitely many isolated zeros of this kind. Hence $\sin$ is transcendental.

The numerical consequence — and the reason the question is asked in this course — is that $\sin(x)$ **cannot be evaluated by a finite sequence of arithmetic operations**. It must be represented by an *infinite* process, such as the Maclaurin series above. Any computer evaluation therefore truncates that series and necessarily incurs **truncation error**, which part (ii) quantifies. The same is true of e^x , $\ln x$, $\cos x$, $\tan^{-1} x$ — all transcendental, all approximated by series or rational (Padé/CORDIC) schemes in practice.

5.9 Question 2(b)(ii) [5, CO2, PO1]

Problem. Use the Remainder Theorem to find bounds on the truncation error in the representation of $\sin(2)$ if only the first 3 terms of the Maclaurin series are used.

5.9.1 Solution

The approximation. The first 3 terms of the series are

$$S_3(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

At $x = 2$:

$$S_3(2) = 2 - \frac{8}{6} + \frac{32}{120} = 2 - 1.333333 + 0.266667$$

$$\boxed{\sin(2) \approx 0.933333}$$

Taylor's Remainder Theorem. For a Taylor polynomial P_n of degree n about 0,

$$R_n(x) = f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1}, \quad c \in (0, x)$$

Identifying n . The polynomial $S_3(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ has degree 5, but the x^6 coefficient of $\sin$ is **zero**, so S_3 is simultaneously the Taylor polynomial of degree **6**. Taking $n = 6$ gives the sharper bound.

Bounding the derivative. Every derivative of $\sin$ is $\pm \sin$ or $\pm \cos$, so

$$\left| f^{(7)}(c) \right| = |-\cos c| \leq 1 \quad \text{for all } c$$

The bound.

$$|R_6(2)| = \left| \frac{f^{(7)}(c)}{7!} 2^7 \right| \leq \frac{1 \cdot 2^7}{7!} = \frac{128}{5040}$$

$$\boxed{|R_6(2)| \leq 0.025397}$$

i.e. the truncation error lies in

$$-0.025397 \leq R_6(2) \leq 0.025397$$

Verification against the actual error.

$$\sin(2) = 0.909297, \quad E_t = 0.909297 - 0.933333 = -0.024036$$

$$|E_t| = 0.024036 \leq 0.025397 \checkmark$$

The bound is satisfied and is remarkably tight here (within 5.4% of the actual error), because the next term of the alternating series, $-\frac{2^7}{7!} = -0.025397$, dominates the tail.

Guaranteed accuracy. Since $|R_6| \leq 0.0254$ and $\sin(2) \approx 0.909$,

$$|\epsilon_t| \leq \frac{0.025397}{0.909297 - 0.025397} \times 100 \approx 2.87\%$$

so **at least one significant digit** of $S_3(2) = 0.9333$ is guaranteed correct.

Cruder alternative. Using $n = 5$ instead gives $|R_5(2)| \leq \frac{2^6}{6!} = \frac{64}{720} = 0.08889$ — valid, but 3.5 times looser.

5.10 Question 2(c) [15, CO2, PO1]

Problem. The Taylor series for a function $f(x)$ is

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \cdots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of $f(x) = \operatorname{cosec}(x)$.

5.10.1 Solution

First, an essential observation Setting $x = 0$ in the Taylor series turns it into the Maclaurin series

$$f(h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + \cdots$$

which requires f and all its derivatives to **exist and be finite at 0**.

But

$$\operatorname{cosec}(x) = \frac{1}{\sin x}, \quad \lim_{x \rightarrow 0} \frac{1}{\sin x} = \pm\infty$$

so cosec has a **simple pole at** $x = 0$: it is not defined there, let alone differentiable.

A Maclaurin series for $\operatorname{cosec}(x)$ does not exist.

What *does* exist — and what is unmistakably intended — is the **Laurent series about** $x = 0$: the same kind of power-series expansion, extended to allow a finite number of negative powers. Since the pole is simple, exactly one negative power (x^{-1}) appears, and $x \operatorname{cosec}(x)$ **does** have a genuine Maclaurin series. We derive it below.

Step 1 — Maclaurin series of the denominator, from the Taylor series Apply the given series to $f = \sin$ at $x = 0$ (so h becomes the variable), using $\sin(0) = 0$, $\cos(0) = 1$:

term	derivative	value at 0	contribution
1	$\sin$	0	0
2	$\cos$	1	h
3	$-\sin$	0	0
4	$-\cos$	-1	$-h^3/3!$
5	$\sin$	0	0
6	$\cos$	1	$h^5/5!$
7	$-\sin$	0	0
8	$-\cos$	-1	$-h^7/7!$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots = x \underbrace{\left(1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040} + \cdots\right)}_{\equiv u(x)}$$

Step 2 — postulate the reciprocal series Because the pole is simple and cosec is an **odd** function, write

$$\operatorname{cosec}(x) = \frac{1}{x u(x)} = \frac{1}{x} (1 + c_2 x^2 + c_4 x^4 + c_6 x^6 + \dots)$$

(only even powers inside the bracket, since $\frac{1}{x} \times \text{even}$ is odd).

Step 3 — determine c_2, c_4, c_6 by matching coefficients The defining condition is $u(x) \cdot (1 + c_2 x^2 + c_4 x^4 + c_6 x^6) = 1$:

$$\left(1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040}\right) (1 + c_2 x^2 + c_4 x^4 + c_6 x^6) = 1 + 0 \cdot x^2 + 0 \cdot x^4 + 0 \cdot x^6 + \dots$$

Coefficient of x^0 : $1 = 1 \checkmark$

Coefficient of x^2 :

$$c_2 - \frac{1}{6} = 0 \implies \boxed{c_2 = \frac{1}{6}}$$

Coefficient of x^4 :

$$\begin{aligned} c_4 - \frac{c_2}{6} + \frac{1}{120} &= 0 \implies c_4 = \frac{1}{6} \cdot \frac{1}{6} - \frac{1}{120} = \frac{1}{36} - \frac{1}{120} \\ &= \frac{10}{360} - \frac{3}{360} \implies \boxed{c_4 = \frac{7}{360}} \end{aligned}$$

Coefficient of x^6 :

$$\begin{aligned} c_6 - \frac{c_4}{6} + \frac{c_2}{120} - \frac{1}{5040} &= 0 \\ c_6 &= \frac{7}{360 \cdot 6} - \frac{1}{6 \cdot 120} + \frac{1}{5040} = \frac{7}{2160} - \frac{1}{720} + \frac{1}{5040} \end{aligned}$$

Common denominator 15120:

$$= \frac{49}{15120} - \frac{21}{15120} + \frac{3}{15120} = \frac{31}{15120} \implies \boxed{c_6 = \frac{31}{15120}}$$

Result

$$\operatorname{cosec}(x) = \frac{1}{x} + \frac{x}{6} + \frac{7x^3}{360} + \frac{31x^5}{15120} + \frac{127x^7}{604800} + \dots$$

valid for $0 < |x| < \pi$ (the radius is set by the nearest other pole, at $x = \pm\pi$).

Equivalently, the honest **Maclaurin series** of the regularised function is

$$x \operatorname{cosec}(x) = 1 + \frac{x^2}{6} + \frac{7x^4}{360} + \frac{31x^6}{15120} + \dots$$

Numerical check at $x = 0.5$

$$\frac{1}{0.5} + \frac{0.5}{6} + \frac{7(0.125)}{360} + \frac{31(0.03125)}{15120} = 2 + 0.0833333 + 0.0024306 + 0.0000641 = 2.0858280$$

True value: $\frac{1}{\sin 0.5} = \frac{1}{0.4794255} = 2.0858296$. $|\epsilon_t| = 7.7 \times 10^{-5}\% \checkmark$

5.11 Question 3(a) [6, CO4, PO1]

Problem. Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique.

5.11.1 Solution

Statement. Let $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ be $n + 1$ data points with **distinct** abscissae $x_i \neq x_j$ for $i \neq j$. Then there is **at most one** polynomial of degree $\leq n$ satisfying $P(x_i) = y_i$ for all i .

Proof (by contradiction).

Suppose two such polynomials exist:

$$P(x) \text{ and } Q(x), \quad \deg P \leq n, \quad \deg Q \leq n$$

$$P(x_i) = y_i \quad \text{and} \quad Q(x_i) = y_i \quad (i = 0, 1, \dots, n)$$

Define their difference

$$R(x) = P(x) - Q(x).$$

Step 1 — the degree of R . The difference of two polynomials of degree at most n is a polynomial of degree at most n :

$$\deg R \leq n.$$

Step 2 — the roots of R . At each data point,

$$R(x_i) = P(x_i) - Q(x_i) = y_i - y_i = 0, \quad i = 0, 1, \dots, n$$

so R has $n + 1$ **distinct roots** $x_0, x_1, \dots, x_n$ (distinct because the abscissae were assumed distinct).

Step 3 — apply the Fundamental Theorem of Algebra. A nonzero polynomial of degree m has **at most m roots**. If R were not identically zero, it would have degree $m \leq n$ and hence at most n roots — contradicting Step 2, which exhibits $n + 1$ of them.

$$\implies R(x) \equiv 0 \implies P(x) = Q(x) \text{ for all } x.$$

The two polynomials are identical, so the interpolating polynomial is **unique**. ■

Alternative proof (linear algebra). Writing $P(x) = a_0 + a_1x + \dots + a_nx^n$, the interpolation conditions form the linear system $V\mathbf{a} = \mathbf{y}$ with the **Vandermonde matrix**

$$V = \begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & & & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix}, \quad \det V = \prod_{0 \leq i < j \leq n} (x_j - x_i)$$

Since all x_i are distinct, every factor is nonzero, so $\det V \neq 0$, V is invertible, and $\mathbf{a} = V^{-1}\mathbf{y}$ is the **unique** coefficient vector. ■

Existence is supplied constructively by the Lagrange form $P(x) = \sum_{i=0}^n y_i L_i(x)$, so together the interpolating polynomial exists and is unique.

Practical significance. This is why Lagrange interpolation, Newton's divided differences and the direct (Vandermonde) method **always give the same polynomial** for the same data — they are three constructions of one object, differing only in cost and conditioning. It is also why the direct method of Session C Q3(b) has a well-posed solution.

5.12 Question 3(b)(i) [2, CO3, PO1]

Problem. What is the Runge phenomenon?

5.12.1 Solution

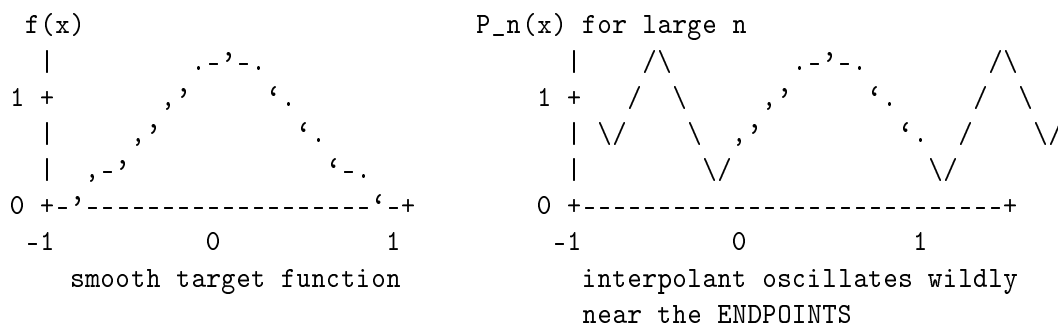
The **Runge phenomenon** is the failure of high-degree polynomial interpolation on **equally spaced nodes**: as the degree n increases, the interpolant develops violent oscillations near the **ends** of the interval, and the maximum error **diverges** rather than converging to zero.

Runge's classic example.

$$f(x) = \frac{1}{1 + 25x^2} \quad \text{on } [-1, 1]$$

with $n + 1$ equispaced nodes. Then

$$\lim_{n \rightarrow \infty} \max_{-1 \leq x \leq 1} |f(x) - P_n(x)| = \infty$$



Why it happens. The interpolation error is

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

For equispaced nodes the node polynomial $\prod (x - x_i)$ is far larger near the ends of the interval than in the middle, and for the Runge function the growth of $f^{(n+1)}$ outruns $(n+1)!$. The two effects combine into divergence at the edges. (The deeper reason is that f has complex poles at $x = \pm i/5$, too close to the interval for the equispaced Lebesgue constant — which grows like $2^n/(n \ln n)$ — to be tamed.)

Remedies.

1. **Piecewise low-order interpolation / splines** — keep the degree fixed at 1, 2 or 3 and add more pieces. Cubic splines converge for any smooth f . (*This is what part (ii) below is about.*)

2. **Chebyshev nodes** — cluster points near the ends, $x_i = \cos\left(\frac{i\pi}{n}\right)$, which flattens the node polynomial and restores convergence.
3. **Least-squares regression** with a low-degree polynomial instead of exact interpolation.

Moral. More data does **not** automatically mean a better interpolant. Raising the degree of a single global polynomial is the wrong way to use extra data.

5.13 Question 3(b)(ii) [5, CO2, PO1]

Problem. A natural cubic spline $S_3(x)$ on $[0, 2]$ is defined by

$$S_3(x) = \begin{cases} 1 + 2x - x^3; & 0 \leq x \leq 1 \\ 2 + a(x-1) + b(x-1)^2 + c(x-1)^3; & 1 \leq x \leq 2 \end{cases}$$

Given that $S_3(x)$ is a natural cubic spline, it must satisfy the following conditions:

- $S_3(x)$ is continuous at the interior point $x = 1$,
- The second derivative $S_3''(x)$ vanishes at the exterior points $x = 0$ and $x = 2$, as it exhibits no curvature at those points.

Find the values of the constants a , b , and c .

5.13.1 Solution

Write the two pieces as

$$S_1(x) = 1 + 2x - x^3, \quad S_2(x) = 2 + a(x-1) + b(x-1)^2 + c(x-1)^3$$

Derivatives.

$$\begin{aligned} S_1'(x) &= 2 - 3x^2, & S_1''(x) &= -6x \\ S_2'(x) &= a + 2b(x-1) + 3c(x-1)^2, & S_2''(x) &= 2b + 6c(x-1) \end{aligned}$$

Condition 1 — continuity of S_3 at $x = 1$

$$S_1(1) = 1 + 2(1) - 1^3 = 2, \quad S_2(1) = 2 + 0 + 0 + 0 = 2$$

$$S_1(1) = S_2(1) = 2 \quad \checkmark$$

Automatically satisfied for any a, b, c — the form of S_2 was written with this built in. It gives no equation, so the remaining spline smoothness conditions must supply them.

Condition 2 — continuity of the first derivative at $x = 1$ (C^1)

$$S_1'(1) = 2 - 3(1)^2 = -1, \quad S_2'(1) = a$$

$$\implies \boxed{a = -1}$$

Condition 3 — continuity of the second derivative at $x = 1$ (C^2)

$$S_1''(1) = -6(1) = -6, \quad S_2''(1) = 2b$$

$$2b = -6 \quad \implies \quad \boxed{b = -3}$$

Condition 4 — natural (free) end condition at $x = 2$

$$S_2''(2) = 2b + 6c(2 - 1) = 2b + 6c = 0$$

$$2(-3) + 6c = 0 \implies 6c = 6 \implies \boxed{c = 1}$$

Check the other natural end condition at $x = 0$

$$S_1''(0) = -6(0) = 0 \quad \checkmark$$

Consistent — the left piece was already constructed to be natural at $x = 0$, confirming the data are self-consistent.

5.13.2 Answer

$$\boxed{a = -1, \quad b = -3, \quad c = 1}$$

so the spline is

$$S_3(x) = \begin{cases} 1 + 2x - x^3, & 0 \leq x \leq 1 \\ 2 - (x - 1) - 3(x - 1)^2 + (x - 1)^3, & 1 \leq x \leq 2 \end{cases}$$

Verification of all four spline conditions.

condition	left value	right value	✓
$S(1)$	2	2	✓
$S'(1)$	-1	-1	✓
$S''(1)$	-6	-6	✓
$S''(0) = 0$	0	—	✓
$S''(2) = 0$	—	$2(-3) + 6(1) = 0$	✓

At the right end, $S_3(2) = 2 - 1 - 3 + 1 = -1$.

5.14 Question 3(c) [8, CO2, PO1]

Problem. A robot arm with a rapid laser scanner is performing a quick quality check on holes drilled in a $15'' \times 10''$ rectangular plate. The centers of the holes describe the path the arm needs to follow. The hole centers are located on a Cartesian coordinate system (with the origin at the bottom left corner of the plate) as given in Table 1.

Abscissa, x (in)	2	4.25	5.25	7.81	9.2	10.6
Ordinate, y (in)	7.2	7.1	6	5	3.5	5

If the laser is traversing from $x = 2$ to $x = 4.25$ to $x = 5.25$ along a quadratic path, determine the value of y at $x = 4$ using the **Newton's Divided Difference method** of interpolation with a **second order polynomial**.

5.14.1 Solution

The problem statement prescribes the three nodes:

$$x_0 = 2 \ (y_0 = 7.2), \quad x_1 = 4.25 \ (y_1 = 7.1), \quad x_2 = 5.25 \ (y_2 = 6.0)$$

Newton's divided-difference form (2nd order):

$$y(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

where

$$b_0 = f(x_0), \quad b_1 = f[x_1, x_0], \quad b_2 = f[x_2, x_1, x_0]$$

Step 1 — zeroth divided difference

$$b_0 = f(x_0) = 7.2$$

Step 2 — first divided differences

$$f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{7.1 - 7.2}{4.25 - 2} = \frac{-0.1}{2.25} = -0.0444444$$

$$f[x_2, x_1] = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{6.0 - 7.1}{5.25 - 4.25} = \frac{-1.1}{1.0} = -1.1$$

$$\implies b_1 = -0.0444444$$

Step 3 — second divided difference

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{-1.1 - (-0.0444444)}{5.25 - 2} = \frac{-1.0555556}{3.25} = -0.3247863$$

Divided-difference table

x_i	$f(x_i)$	1st D.D.	2nd D.D.
2.00	7.2		
		-0.0444444	
4.25	7.1		-0.3247863
		-1.1000000	
5.25	6.0		

Step 4 — the interpolating polynomial

$$y(x) = 7.2 - 0.04444444(x - 2) - 0.3247863(x - 2)(x - 4.25)$$

Step 5 — evaluate at $x = 4$

$$\begin{aligned} y(4) &= 7.2 - 0.04444444(4 - 2) - 0.3247863(4 - 2)(4 - 4.25) \\ &= 7.2 - 0.04444444(2) - 0.3247863(2)(-0.25) \\ &= 7.2 - 0.0888889 + 0.1623932 \end{aligned}$$

$y(4) = 7.2735$ in

Discussion **First-order estimate for comparison.** Truncating after b_1 :

$$y_1(4) = 7.2 - 0.04444444(2) = 7.1111$$

$$|\epsilon_a| = \left| \frac{7.2735 - 7.1111}{7.2735} \right| \times 100 = 2.233\%$$

so at least **one significant digit** is correct, and the quadratic term contributes +0.162 in — not negligible for a laser-positioning tolerance.

Efficiency note. The great advantage of Newton's form over Lagrange here is that the first-order estimate is **contained inside** the second-order one: adding the node $x_2 = 5.25$ only required computing b_2 and appending one term. Nothing already computed was discarded. With the Lagrange form, every weight L_i must be recomputed from scratch when a node is added (compare Session B Q4(a) vs Q4(b)).

Sanity check. $y(4) = 7.2735$ lies between $y(2) = 7.2$ and $y(4.25) = 7.1$? It does **not** — it sits slightly above both. That is legitimate: the fitted parabola opens downward ($b_2 < 0$) with a local maximum in the interval, so a modest overshoot above the bracketing ordinates is expected. Whether the physical hole pattern really behaves that way is exactly the kind of question the quality check is meant to answer.

5.15 Question 4(a) — preamble

The general linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is

$$y = a_0 + a_1x$$

where a_0 and a_1 are the two parameters of the regression model.

5.16 Question 4(a)(i) [7, CO4, PO1]

Problem. Derive the formulae for finding the optimal values of a_0 and a_1 .

5.16.1 Solution

Step 1 — define the residual and the optimization criterion.

For each data point, the residual is the vertical distance between the measured value and the model:

$$E_i = y_i - (a_0 + a_1 x_i)$$

The **least-squares criterion** minimises the sum of the squares of the residuals:

$$S_r(a_0, a_1) = \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

(Why squares, and not $\sum E_i$ or $\sum |E_i|$? Because $\sum E_i$ can be zero for infinitely many bad lines — positive and negative residuals cancel — and $\sum |E_i|$ is not differentiable at $E_i = 0$, so no closed-form solution exists. Squaring penalises large residuals, is always positive, and is smooth.)

Step 2 — set the partial derivatives to zero.

At the minimum, S_r is stationary with respect to both parameters:

$$\frac{\partial S_r}{\partial a_0} = \sum_{i=1}^n 2(y_i - a_0 - a_1 x_i)(-1) = 0 \implies \sum y_i - \sum a_0 - a_1 \sum x_i = 0$$

$$\frac{\partial S_r}{\partial a_1} = \sum_{i=1}^n 2(y_i - a_0 - a_1 x_i)(-x_i) = 0 \implies \sum x_i y_i - a_0 \sum x_i - a_1 \sum x_i^2 = 0$$

Step 3 — the normal equations. Noting $\sum a_0 = n a_0$:

$$\boxed{n a_0 + a_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i} \quad (\text{N1})$$

$$\boxed{a_0 \sum_{i=1}^n x_i + a_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i} \quad (\text{N2})$$

Two linear equations in the two unknowns a_0, a_1 .

Step 4 — solve by elimination (Cramer's rule).

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Determinant: $D = n \sum x_i^2 - (\sum x_i)^2$

$$a_1 = \frac{\begin{vmatrix} n & \sum y_i \\ \sum x_i & \sum x_i y_i \end{vmatrix}}{D} \implies \boxed{a_1 = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}}$$

Substituting into (N1) and dividing by n :

$$a_0 = \bar{y} - a_1 \bar{x} = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n}$$

Equivalent (numerically preferable) forms:

$$a_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

which avoids the subtractive cancellation in $n \sum x_i^2 - (\sum x_i)^2$.

Geometric reading of (N1). Dividing by n gives $\bar{y} = a_0 + a_1 \bar{x}$: the least-squares line always passes through the centroid $(\bar{x}, \bar{y})$ of the data.

(Worked numerically in Session B Q2(a) for the diode data.)

5.17 Question 4(a)(ii) [5, CO4, PO1]

Problem. Prove that the values of a_0 and a_1 obtained using the formulae derived in Question 4.a)i correspond to the absolute minimum of the optimization criterion used.

■ Session A Q5(b) asks essentially the same question; the proof is given in full in both places for self-containedness.

5.17.1 Solution

Setting the first derivatives to zero locates a **stationary point** — which could in principle be a minimum, a maximum, or a saddle. We must show it is a minimum, and moreover the **global** one.

Step 1 — second-derivative (Hessian) test

$$\frac{\partial^2 S_r}{\partial a_0^2} = \frac{\partial}{\partial a_0} \left[-2 \sum (y_i - a_0 - a_1 x_i) \right] = 2 \sum_{i=1}^n 1 = 2n$$

$$\frac{\partial^2 S_r}{\partial a_1^2} = \frac{\partial}{\partial a_1} \left[-2 \sum (y_i - a_0 - a_1 x_i) x_i \right] = 2 \sum_{i=1}^n x_i^2$$

$$\frac{\partial^2 S_r}{\partial a_0 \partial a_1} = 2 \sum_{i=1}^n x_i$$

$$H = \begin{bmatrix} \frac{\partial^2 S_r}{\partial a_0^2} & \frac{\partial^2 S_r}{\partial a_0 \partial a_1} \\ \frac{\partial^2 S_r}{\partial a_1 \partial a_0} & \frac{\partial^2 S_r}{\partial a_1^2} \end{bmatrix} = 2 \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

Leading principal minor:

$$H_{11} = 2n > 0 \quad (\text{since } n \geq 1)$$

Determinant:

$$\det H = 4 \left[n \sum x_i^2 - \left(\sum x_i \right)^2 \right]$$

Apply the algebraic identity

$$n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 = n \sum_{i=1}^n (x_i - \bar{x})^2$$

(Proof: $n \sum (x_i - \bar{x})^2 = n \left[\sum x_i^2 - 2\bar{x} \sum x_i + n\bar{x}^2 \right] = n \sum x_i^2 - 2n\bar{x} \sum x_i + n^2 \bar{x}^2$, and with $\bar{x} = \frac{\sum x_i}{n}$ the last two terms are $-2(\sum x_i)^2 + (\sum x_i)^2 = -(\sum x_i)^2$.)

Hence

$$\det H = 4n \sum_{i=1}^n (x_i - \bar{x})^2 > 0$$

strictly, **provided the x_i are not all equal** — which is precisely the condition under which a line is determinable at all.

Since $H_{11} > 0$ and $\det H > 0$, the Hessian is **positive definite**, so the stationary point is a **strict local minimum**. (It cannot be a maximum: $H_{11} > 0$; nor a saddle: $\det H > 0$.)

Step 2 — from local minimum to absolute minimum

$$S_r(a_0, a_1) = \sum (y_i - a_0 - a_1 x_i)^2$$

is a **quadratic function** of (a_0, a_1) — expanding it gives

$$S_r = \underbrace{\binom{n}{2} a_0^2}_{\text{quadratic}} + \underbrace{\left(\sum x_i^2 \right) a_1^2}_{\text{quadratic}} + 2 \left(\sum x_i \right) a_0 a_1 - 2 \left(\sum y_i \right) a_0 - 2 \left(\sum x_i y_i \right) a_1 + \sum y_i^2$$

whose quadratic part has the matrix $\frac{1}{2}H$, positive definite by Step 1.

A quadratic form with positive-definite Hessian is **strictly convex** on $\mathbb{R}^2$: for any (a_0, a_1) and the stationary point (a_0^*, a_1^*) , Taylor's theorem is *exact* after the second-order term, so

$$S_r(a_0, a_1) = S_r(a_0^*, a_1^*) + \frac{1}{2} \delta^T H \delta, \quad \delta = \begin{bmatrix} a_0 - a_0^* \\ a_1 - a_1^* \end{bmatrix}$$

(the linear term vanishes because the gradient is zero there). Since H is positive definite,

$$\delta^T H \delta > 0 \quad \text{for every } \delta \neq \mathbf{0}$$

$$\implies S_r(a_0, a_1) > S_r(a_0^*, a_1^*) \quad \text{for all } (a_0, a_1) \neq (a_0^*, a_1^*)$$

The point is the strict **absolute (global)** minimum of S_r , and it is unique.

Uniqueness, independently. The normal equations are linear with determinant $D = n \sum (x_i - \bar{x})^2 \neq 0$, so they have exactly one solution — there is no other stationary point to compete.

Degenerate case. If $x_1 = x_2 = \dots = x_n$ then $\det H = 0$, S_r becomes a degenerate parabolic trough, and infinitely many (a_0, a_1) achieve the same minimum. Physically the data are vertical and the model $y = a_0 + a_1 x$ is inappropriate.

5.18 Question 4(b) [5, CO3, PO1]

Problem. In the case of the Logistic Growth model, determine the values of y that correspond to $x = 0$ and $x \rightarrow \infty$.

5.18.1 Solution

The model. The logistic growth model is

$$y = \frac{a}{1 + b e^{-cx}}, \quad a > 0, b > 0, c > 0$$

where a is the saturation level (carrying capacity), b sets the position of the curve along the x -axis, and c the growth rate.

At $x = 0$

$$e^{-c(0)} = e^0 = 1$$

$$y(0) = \frac{a}{1 + b(1)}$$

$$\boxed{y(0) = \frac{a}{1 + b}}$$

This is the **initial population** / **starting value**. Note $0 < y(0) < a$ for $b > 0$, so the curve always starts below the ceiling. (Conversely, given a measured $y(0) = y_0$, the parameter b is recovered as $b = \frac{a}{y_0} - 1$.)

As $x \rightarrow \infty$ Since $c > 0$,

$$\lim_{x \rightarrow \infty} e^{-cx} = 0 \implies \lim_{x \rightarrow \infty} (1 + b e^{-cx}) = 1$$

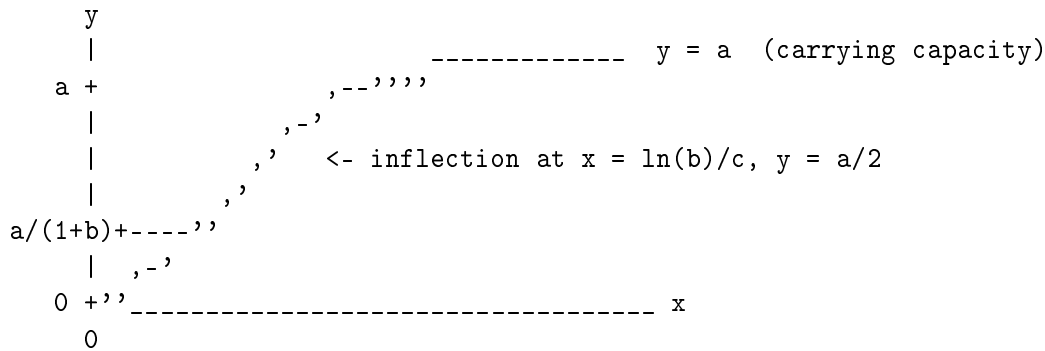
$$\lim_{x \rightarrow \infty} y = \frac{a}{1}$$

$$\boxed{\lim_{x \rightarrow \infty} y = a}$$

This is the **carrying capacity** / **saturation level** — the horizontal asymptote the curve approaches from below but never reaches.

Summary and interpretation

x	y	meaning
0	$\frac{a}{1 + b}$	initial value
$\rightarrow \infty$	a	saturation level (upper asymptote)
$\rightarrow -\infty$	0	lower asymptote (for completeness)



The characteristic S-shape arises because growth is nearly exponential while $y \ll a$ but is throttled as $y \rightarrow a$. The **inflection point** (fastest growth) occurs where $be^{-cx} = 1$, i.e.

$$x_{\text{inf}} = \frac{\ln b}{c}, \quad y_{\text{inf}} = \frac{a}{2}$$

— exactly half the carrying capacity.

Relevance to regression. The two limits give excellent *initial estimates* for a nonlinear fit: read a off the plateau of the data, then b from the first data point via $b = a/y_0 - 1$.

5.19 Question 4(c) [5, CO3, PO2]

Problem. Suppose you are given n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ and you want to regress them to the Exponential model $y = e^{bx}$.

- Without performing any data transformation, you obtain the regression model $y = e^{k_1 x}$.
- With data transformation via linearization, you obtain the regression model $y = e^{k_2 x}$.

With proper reasoning, explain whether k_1 and k_2 are equal or not.

5.19.1 Solution

In general, $k_1 \neq k_2$.

The reason: the two procedures minimise different objective functions.

The direct (nonlinear) fit minimises error in y

$$S_r^{(1)}(k) = \sum_{i=1}^n (y_i - e^{kx_i})^2$$

$$\frac{dS_r^{(1)}}{dk} = -2 \sum_{i=1}^n (y_i - e^{kx_i}) x_i e^{kx_i} = 0 \implies \sum_{i=1}^n x_i y_i e^{k_1 x_i} = \sum_{i=1}^n x_i e^{2k_1 x_i} \quad (\alpha)$$

a **transcendental** equation in k_1 , requiring Newton–Raphson or bisection.

The linearized fit minimises error in $\ln y$ Taking logarithms, $\ln y = bx$, so with $Y = \ln y$ the criterion becomes

$$S_r^{(2)}(k) = \sum_{i=1}^n (\ln y_i - kx_i)^2$$

$$\frac{dS_r^{(2)}}{dk} = -2 \sum_{i=1}^n (\ln y_i - kx_i) x_i = 0 \implies k_2 = \frac{\sum_{i=1}^n x_i \ln y_i}{\sum_{i=1}^n x_i^2} \quad (\beta)$$

a **closed-form, linear** result.

Why they differ 1. **Different residuals are being squared.** (α) squares $y_i - \hat{y}_i$; (β) squares $\ln y_i - \ln \hat{y}_i$. These are different distances in different spaces, and there is no reason for their minimisers to coincide.

2. **The logarithm re-weights the data.** For small δ ,

$$\ln y_i - \ln \hat{y}_i = \ln \left(\frac{y_i}{\hat{y}_i} \right) \approx \frac{y_i - \hat{y}_i}{\hat{y}_i}$$

so the linearized fit is effectively minimising **relative** (percentage) errors,

$$S_r^{(2)} \approx \sum \left(\frac{y_i - \hat{y}_i}{\hat{y}_i} \right)^2,$$

whereas the direct fit minimises **absolute** errors. Consequently the linearized fit gives **much more weight to small y values** and comparatively little to large ones — the opposite emphasis of the direct fit. With exponential data spanning several decades, this systematically pulls k_2 away from k_1 .

3. **Minimisation and nonlinear transformation do not commute.** For a nonlinear monotone map T , in general

$$\arg \min_k \sum (y_i - f_k(x_i))^2 \neq \arg \min_k \sum (T(y_i) - T(f_k(x_i)))^2$$

Only an *affine* T would preserve the minimiser, and $\ln$ is not affine.

The exception If the data lie **exactly** on some exponential curve, $y_i = e^{kx_i}$ for all i , then both criteria attain their absolute minimum value 0 at the same k , so

$$k_1 = k_2 = k \quad (\text{exact-fit case only}).$$

For real, noisy data this never happens, and $k_1 \neq k_2$.

Practical guidance

	Direct (nonlinear)	Linearized
Minimises	absolute error in y	relative error in y
Cost	iterative root-finding, needs initial guess	closed form, instant

	Direct (nonlinear)	Linearized
Statistically optimal when	noise is additive with constant variance	noise is multiplicative (constant % error)
Requires	nothing special	$y_i > 0$ for all i

Standard practice: use the linearized fit to obtain a cheap, robust starting value, then refine it with the direct nonlinear fit if additive noise is the right model. Also note that linearization is impossible if any $y_i \leq 0$, whereas the direct fit is unaffected.

(This is the quantitative counterpart of the caveat noted in Session A Q3(b).)

5.20 Question 5(a)(i) [7, CO2, PO1]

Problem. Using the method of coefficients, derive the formula for the Single-segment Trapezoidal rule.

Session B Q3(a) asks for the same derivation plus the calculus route and the geometric interpretation; the algebra is repeated here for self-containedness.

5.20.1 Solution

Step 1 — postulate the form. Assume the integral can be written as a weighted sum of the function values at the two endpoints, with weights to be determined:

$$\int_a^b f(x) dx \approx c_0 f(a) + c_1 f(b) \quad (*)$$

Step 2 — impose exactness conditions. Two unknowns require two conditions. Demand that (*) be **exact** for the two lowest monomials, $f(x) = 1$ and $f(x) = x$ — equivalently, that the rule integrate every straight line exactly.

Condition A: $f(x) = 1$.

$$\begin{aligned} \text{LHS} &= \int_a^b 1 dx = b - a, & \text{RHS} &= c_0(1) + c_1(1) = c_0 + c_1 \\ \implies & & c_0 + c_1 &= b - a \end{aligned} \quad (\text{A})$$

Condition B: $f(x) = x$.

$$\begin{aligned} \text{LHS} &= \int_a^b x dx = \frac{b^2 - a^2}{2}, & \text{RHS} &= c_0 a + c_1 b \\ \implies & & c_0 a + c_1 b &= \frac{b^2 - a^2}{2} = \frac{(b - a)(b + a)}{2} \end{aligned} \quad (\text{B})$$

Step 3 — solve the 2×2 system.

From (A): $c_0 = (b - a) - c_1$. Substitute into (B):

$$[(b - a) - c_1]a + c_1 b = \frac{(b - a)(b + a)}{2}$$

$$a(b-a) - c_1a + c_1b = \frac{(b-a)(b+a)}{2}$$

$$a(b-a) + c_1(b-a) = \frac{(b-a)(b+a)}{2}$$

Divide through by $(b-a) \neq 0$:

$$a + c_1 = \frac{a+b}{2} \implies c_1 = \frac{a+b}{2} - a = \frac{b-a}{2}$$

and from (A):

$$c_0 = (b-a) - \frac{b-a}{2} = \frac{b-a}{2}$$

$$c_0 = c_1 = \frac{b-a}{2}$$

Step 4 — the rule. Substituting back into (*):

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

— the single-segment (single-application) Trapezoidal rule.

Degree of precision. The rule was constructed to be exact for degree ≤ 1 . Testing $f = x^2$:

$$\int_a^b x^2 dx = \frac{b^3 - a^3}{3} \quad \text{vs} \quad \frac{b-a}{2}(a^2 + b^2)$$

which differ (e.g. on $[0, 1]$: $\frac{1}{3}$ vs $\frac{1}{2}$). So the **degree of precision is exactly 1**, and the error term involves the second derivative:

$$E_t = -\frac{(b-a)^3}{12} f''(\xi), \quad \xi \in (a, b)$$

5.21 Question 5(a)(ii) [6, CO2, PO1]

Problem. Applying the formula from Question 5.a)i, derive the formula for the Multi-segment Trapezoidal rule.

5.21.1 Solution

Step 1 — partition the interval. Divide $[a, b]$ into n equal segments:

$$h = \frac{b-a}{n}, \quad x_i = a + ih \quad (i = 0, 1, \dots, n), \quad x_0 = a, \quad x_n = b$$

Step 2 — split the integral (additivity).

$$\int_a^b f(x) dx = \int_{x_0}^{x_1} f dx + \int_{x_1}^{x_2} f dx + \dots + \int_{x_{n-1}}^{x_n} f dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x) dx$$

Step 3 — apply the single-segment rule of part (i) to each sub-interval. Each has width $x_{i+1} - x_i = h$, so

$$\int_{x_i}^{x_{i+1}} f dx \approx \frac{h}{2} [f(x_i) + f(x_{i+1})]$$

$$\int_a^b f dx \approx \frac{h}{2} [f(x_0) + f(x_1)] + \frac{h}{2} [f(x_1) + f(x_2)] + \cdots + \frac{h}{2} [f(x_{n-1}) + f(x_n)]$$

Step 4 — collect like terms. Write out the sum:

$$= \frac{h}{2} \left[f(x_0) + \underbrace{f(x_1) + f(x_1)}_{2f(x_1)} + \underbrace{f(x_2) + f(x_2)}_{2f(x_2)} + \cdots + \underbrace{f(x_{n-1}) + f(x_{n-1})}_{2f(x_{n-1})} + f(x_n) \right]$$

Each **interior** node $x_1, \dots, x_{n-1}$ is the right endpoint of one segment *and* the left endpoint of the next, so it appears **twice**; the two **end** nodes x_0 and x_n appear only **once**.

$$\boxed{\int_a^b f(x) dx \approx \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right], \quad h = \frac{b-a}{n}}$$

Equivalently, in “average height $\times$ width” form:

$$\int_a^b f(x) dx \approx (b-a) \frac{f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}$$

Weight pattern: 1, 2, 2, $\dots$, 2, 1.

Step 5 — error. Each segment contributes $-\frac{h^3}{12} f''(\xi_i)$; summing over n segments and applying the mean-value theorem for the average of f'' :

$$E_t = -\frac{h^3}{12} \sum_{i=0}^{n-1} f''(\xi_i) = -\frac{(b-a)^3}{12n^2} \overline{f''} = O(h^2)$$

So doubling the number of segments cuts the error by a factor of **4** — the composite rule is $O(h^2)$ even though the single application is $O(h^3)$, because the number of segments grows as h shrinks.

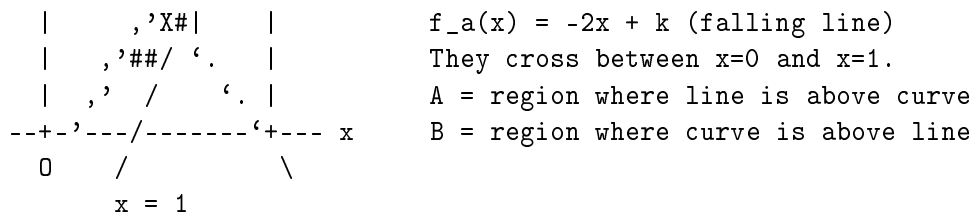
Sanity check ($n = 1$). The sum $\sum_{i=1}^0$ is empty, so the formula collapses to $\frac{h}{2} [f(x_0) + f(x_1)] = \frac{b-a}{2} [f(a) + f(b)] \checkmark$ — part (i).

5.22 Question 5(b) — preamble

In Figure 2, the area of the shaded regions A and B , portrayed beneath the two functions $f_a(x)$ and $f_b(x)$ respectively, are equal. The aforementioned functions are

$$f_a(x) = -2x + k, \quad f_b(x) = e^{2x}$$

y		/		
		/		
		A // B		$f_b(x) = e^{(2x)}$ (rising curve)



5.23 Question 5(b)(i) [8, CO2, PO1]

Problem. Using the Composite Simpson's 3/8 Rule, with $n = 9$ segments, approximate the value of k .

5.23.1 Solution

Setting up the equation for k Let $x^* \in (0, 1)$ be the crossing point where $f_a = f_b$. From the figure, over $[0, 1]$:

- On $[0, x^*]$ the **line is above the curve**, and region A has area $\int_0^{x^*} (f_a - f_b) dx$.
- On $[x^*, 1]$ the **curve is above the line**, and region B has area $\int_{x^*}^1 (f_b - f_a) dx$.

Setting $\text{area}(A) = \text{area}(B)$:

$$\int_0^{x^*} (f_a - f_b) dx = \int_{x^*}^1 (f_b - f_a) dx = - \int_{x^*}^1 (f_a - f_b) dx$$

Bringing everything to one side, the two contributions **cancel exactly**:

$$\boxed{\int_0^1 (f_a(x) - f_b(x)) dx = 0}$$

This is the key simplification — the crossing point x^* never has to be found.

Substituting the functions:

$$\int_0^1 (-2x + k - e^{2x}) dx = 0 \implies k \int_0^1 dx = \int_0^1 (2x + e^{2x}) dx$$

$$\boxed{k = \int_0^1 g(x) dx, \quad g(x) = 2x + e^{2x}}$$

Composite Simpson's 3/8 rule with $n = 9$ $n = 9$ is a multiple of 3 ✓. Step size:

$$h = \frac{b - a}{n} = \frac{1 - 0}{9} = 0.1111111$$

Weight pattern 1, 3, 3, 2, 3, 3, 2, 3, 3, 1:

i	x_i	$g(x_i) = 2x_i + e^{2x_i}$	weight w_i	$w_i g(x_i)$
0	0.0000000	1.0000000	1	1.0000000
1	0.1111111	1.4710711	3	4.4132133
2	0.2222222	2.0040679	3	6.0122038
3	0.3333333	2.6144007	2	5.2288014
4	0.4444444	3.3213143	3	9.9639430
5	0.5555556	4.1488429	3	12.4465287
6	0.6666667	5.1270012	2	10.2540025
7	0.7777778	6.2932734	3	18.8798203
8	0.8888889	7.6944714	3	23.0834141
9	1.0000000	9.3890561	1	9.3890561
			Σ	100.6709832

(sample: $g(0.1111111) = 2(0.1111111) + e^{0.2222222} = 0.2222222 + 1.2488489 = 1.4710711$)

$$k \approx \frac{3h}{8} \sum_i w_i g(x_i) = \frac{3(0.1111111)}{8} \times 100.6709832 = 0.0416667 \times 100.6709832$$

$k \approx 4.194624$

Cross-check. With this k , the crossing point is $-2x + 4.194624 = e^{2x} \Rightarrow x^* = 0.5612$, comfortably inside $(0, 1)$ ✓, consistent with Figure 2.

5.24 Question 5(b)(ii) [6, CO2, PO1]

Problem. Analytically determine the exact value of k , denoted by k_{exact} , and use it to calculate the true error (E_t) and the absolute relative true error ($|\epsilon_t|\%$).

5.24.1 Solution

Exact value

$$\begin{aligned} k_{\text{exact}} &= \int_0^1 (2x + e^{2x}) dx = \left[x^2 + \frac{e^{2x}}{2} \right]_0^1 \\ &= \left(1^2 + \frac{e^2}{2} \right) - \left(0 + \frac{e^0}{2} \right) = 1 + \frac{e^2}{2} - \frac{1}{2} = 1 + \frac{e^2 - 1}{2} = \frac{e^2 + 1}{2} \end{aligned}$$

With $e^2 = 7.3890561$:

$$k_{\text{exact}} = \frac{7.3890561 + 1}{2} = \frac{8.3890561}{2}$$

$k_{\text{exact}} = \frac{e^2 + 1}{2} = 4.1945280$

(Equivalent verification via the original equation: $\int_0^1 (-2x + k - e^{2x}) dx = -1 + k - \frac{e^2 - 1}{2} = 0 \Rightarrow k = 1 + \frac{e^2 - 1}{2}$ ✓)

True error

$$E_t = k_{\text{exact}} - k_{\text{approx}} = 4.1945280 - 4.1946243$$

$$E_t = -9.6247 \times 10^{-5}$$

The negative sign means the composite Simpson's 3/8 rule **over-estimates** k here — expected, since $g'''' > 0$ for $g = 2x + e^{2x}$ and the error term is $E_t = -\frac{(b-a)h^4}{80}g^{(4)} < 0$.

Absolute relative true error

$$|\epsilon_t| = \left| \frac{k_{\text{exact}} - k_{\text{approx}}}{k_{\text{exact}}} \right| \times 100 = \left| \frac{-9.6247 \times 10^{-5}}{4.1945280} \right| \times 100 = 2.2946 \times 10^{-5} \times 100$$

$$|\epsilon_t| = 0.0022946\% \approx 2.29 \times 10^{-3}\%$$

Significant digits Since

$$|\epsilon_t| = 0.0022946\% < 0.5 \times 10^{-5}\% = 0.005\%$$

at least 5 significant digits of the approximation are correct: $k_{\text{approx}} = 4.1946$ vs $k_{\text{exact}} = 4.1945$ — agreement to 4.1945 ✓

Comment on the accuracy The error is tiny because (a) Simpson's 3/8 is $O(h^4)$, (b) $h = 1/9$ is small so $h^4 = 1.52 \times 10^{-4}$, and (c) $g(x) = 2x + e^{2x}$ is smooth with a modest fourth derivative, $\max_{[0,1]} g^{(4)} = 16e^2 = 118.2$. The predicted error bound

$$|E_t| \leq \frac{(b-a)h^4}{80} \max |g^{(4)}| = \frac{1 \times 1.5242 \times 10^{-4}}{80} \times 118.2 = 2.25 \times 10^{-4}$$

comfortably contains the observed 9.62×10^{-5} ✓

5.25 Question 6(a) [8, CO2, PO1]

Problem. A ball at 1500 K is allowed to cool down in air at an ambient temperature of 500 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -1.2 \times 10^{-12} (\theta^4 - 625 \times 10^8)$$

where θ is in K and t in seconds. Find the temperature at $t = 200$ seconds using **Runge–Kutta 2nd order method (Heun's version)**. Assume a step size of $h = 100$ seconds.

5.25.1 Solution

The ODE.

$$f(t, \theta) = -1.2 \times 10^{-12} (\theta^4 - 6.25 \times 10^{10}), \quad \theta(0) = 1500, \quad h = 100$$

(Note $625 \times 10^8 = 6.25 \times 10^{10} = (500)^4$ — the ambient temperature to the fourth power, as Stefan–Boltzmann radiation requires. So $\frac{d\theta}{dt} = 0$ when $\theta = 500$: the ball cools asymptotically to ambient.)

Heun's method (RK2 with $a_1 = a_2 = \frac{1}{2}$):

$$\begin{aligned}\theta_{i+1} &= \theta_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h \\ k_1 &= f(t_i, \theta_i), \quad k_2 = f(t_i + h, \theta_i + k_1h)\end{aligned}$$

Two steps of $h = 100$ reach $t = 200$.

Step 1: $t_0 = 0 \rightarrow t_1 = 100$, $\theta_0 = 1500$ **Slope k_1 :**

$$\theta_0^4 = 1500^4 = 5.0625 \times 10^{12}$$

$$k_1 = -1.2 \times 10^{-12} (5.0625 \times 10^{12} - 6.25 \times 10^{10}) = -1.2 \times 10^{-12} (5.0000 \times 10^{12}) = -6.0000$$

Predictor:

$$\theta^{\text{pred}} = \theta_0 + k_1h = 1500 + (-6.0000)(100) = 900.0000$$

Slope k_2 :

$$(900)^4 = 6.561 \times 10^{11}$$

$$k_2 = -1.2 \times 10^{-12} (6.561 \times 10^{11} - 6.25 \times 10^{10}) = -1.2 \times 10^{-12} (5.936 \times 10^{11}) = -0.712320$$

Corrector:

$$\theta_1 = 1500 + \frac{-6.0000 + (-0.712320)}{2}(100) = 1500 + (-3.356160)(100) = 1500 - 335.616$$

$$\theta(100) = 1164.384 \text{ K}$$

Step 2: $t_1 = 100 \rightarrow t_2 = 200$, $\theta_1 = 1164.384$ **Slope k_1 :**

$$(1164.384)^4 = 1.8381668 \times 10^{12}$$

$$k_1 = -1.2 \times 10^{-12} (1.8381668 \times 10^{12} - 6.25 \times 10^{10}) = -1.2 \times 10^{-12} (1.7756668 \times 10^{12}) = -2.130800$$

Predictor:

$$\theta^{\text{pred}} = 1164.384 + (-2.130800)(100) = 1164.384 - 213.080 = 951.304$$

Slope k_2 :

$$(951.304)^4 = 8.1898748 \times 10^{11}$$

$$k_2 = -1.2 \times 10^{-12} (8.1898748 \times 10^{11} - 6.25 \times 10^{10}) = -1.2 \times 10^{-12} (7.5648748 \times 10^{11}) = -0.907785$$

Corrector:

$$\begin{aligned}\theta_2 &= 1164.384 + \frac{-2.130800 + (-0.907785)}{2}(100) = 1164.384 + (-1.519293)(100) \\ &= 1164.384 - 151.929\end{aligned}$$

$\theta(200) \approx 1012.45 \text{ K}$

Summary table

i	t_i	θ_i	k_1	θ^{pred}	k_2	$\frac{k_1+k_2}{2}$
0	0	1500.000	-6.000000	900.000	-0.712320	-3.356160
1	100	1164.384	-2.130800	951.304	-0.907785	-1.519293
2	200	1012.455				

Accuracy check Integrating the ODE numerically to high precision gives $\theta(200) = 1003.15$ K, so

$$E_t = 1003.15 - 1012.45 = -9.30, \quad |\epsilon_t| = \left| \frac{-9.30}{1003.15} \right| \times 100 = 0.927\%$$

Under 1% from only two steps — respectable for RK2 on a strongly nonlinear (θ^4) problem. The error is positive-biased (the estimate cools too slowly) because the true cooling curve is convex and Heun's averaged slope under-predicts the steep initial descent. Halving h would reduce the error by about $2^2 = 4$.

Physical sanity. θ falls from 1500 K towards the ambient 500 K, rapidly at first (radiation $\propto \theta^4$) and then ever more slowly — exactly what the two steps show: -335.6 K in the first 100 s, only -151.9 K in the second.

5.26 Question 6(b)(i) [6, CO2, PO1]

Problem. What is the LU decomposition of A , where

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

5.26.1 [!] EXACT DUPLICATE — SKIPPED

This question is an exact duplicate of **Session A — Question 4(c)** (same matrix, same task; 15 April 2022). The full Doolittle elimination, the resulting factors

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -\frac{1}{3} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & \frac{13}{3} \end{bmatrix}$$

and the verification $LU = A$ are given there and are not repeated.

5.27 Question 6(b)(ii) [0.5×4, CO4, PO1]

Problem. Determine the order and degree of each of these higher-order differential equations:

$$(a) \quad x^2 \frac{dy}{dx} - \left(\frac{d^3y}{dx^3} \right)^3 + 69 = 0 \qquad (b) \quad \left(\frac{dy}{dx} \right)^2 - 420xy = 3$$

$$(c) \frac{d^3y}{dx^3} + x^4 \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^4 = \sin x \qquad (d) \frac{dy}{dx} + xy = x^2$$

5.27.1 Solution

Definitions.

- **Order** = the order of the **highest** derivative appearing in the equation.
- **Degree** = the **power** to which that highest-order derivative is raised, *after* the equation has been made polynomial (free of radicals and fractions) in all its derivatives.

All four equations are already polynomial in the derivatives, so no rationalisation is needed.

$$(a) \ x^2 \frac{dy}{dx} - \left(\frac{d^3y}{dx^3} \right)^3 + 69 = 0$$

Highest derivative: $\frac{d^3y}{dx^3} \Rightarrow$ **order 3**. Its power: $\left(\frac{d^3y}{dx^3} \right)^3 \Rightarrow$ **degree 3**.

Order = 3, Degree = 3

(The x^2 multiplying dy/dx is irrelevant — only derivatives count, not coefficients in x ; and the lower-order dy/dx appears to power 1, which does not affect the degree.)

$$(b) \ \left(\frac{dy}{dx} \right)^2 - 420xy = 3$$

Highest derivative: $\frac{dy}{dx} \Rightarrow$ **order 1**. Its power: $\left(\frac{dy}{dx} \right)^2 \Rightarrow$ **degree 2**.

Order = 1, Degree = 2

$$(c) \ \frac{d^3y}{dx^3} + x^4 \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^4 = \sin x$$

Highest derivative: $\frac{d^3y}{dx^3} \Rightarrow$ **order 3**. It appears to the first power only $\Rightarrow$ **degree 1**.

Order = 3, Degree = 1

Careful: the $\left(\frac{dy}{dx} \right)^4$ factor is tempting but irrelevant — degree is read off the **highest-order** derivative alone. Note that the equation is nevertheless **nonlinear** (products of derivatives), which is a separate classification from degree. The $\sin x$ on the right involves only the independent variable, so it does not affect either count.

(d) $\frac{dy}{dx} + xy = x^2$

Highest derivative: $\frac{dy}{dx} \Rightarrow$ **order 1**. First power $\Rightarrow$ **degree 1**.

Order = 1,	Degree = 1
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(This one is additionally **linear** in y and its derivative — the standard first-order linear ODE, solvable by an integrating factor.)

5.27.2 Summary

Equation	Highest derivative	Order	Degree	Linear?
(a) $x^2y' - (y''')^3 + 69 = 0$	y'''	3	3	No
(b) $(y')^2 - 420xy = 3$	y'	1	2	No
(c) $y''' + x^4y''(y')^4 = \sin x$	y'''	3	1	No
(d) $y' + xy = x^2$	y'	1	1	Yes

Why this matters numerically. The *order* determines how many state variables are needed when converting to a first-order system (see Session C Q2(a)(i)) and hence how many initial conditions must be supplied. The *degree* signals whether the highest derivative can be isolated explicitly as $y^{(n)} = F(x, y, \dots, y^{(n-1)})$ — immediate for degree 1, but requiring the solution of a polynomial equation (with a branch choice) for degree > 1 , as in (a) and (b).

5.28 Question 6(c) [6+2, CO2, PO1]

Problem. Derive the orders of error (in Big O notation) for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for determining the best one of the three aforementioned methods.

5.28.1 [!] EXACT DUPLICATE — SKIPPED

This question is an exact duplicate of **Session A — Question 1(a)** (“Use Taylor’s polynomial to derive the orders of error for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for choosing one of these three.”; 15 April 2022). The full Taylor-series derivations giving

$$E_{\text{FD}} = -\frac{h}{2}f''(\xi) = O(h), \quad E_{\text{BD}} = +\frac{h}{2}f''(\xi) = O(h), \quad E_{\text{CD}} = -\frac{h^2}{6}f'''(\xi) = O(h^2)$$

together with the reasoning that identifies the Central Difference rule as the best of the three, are given there and are not repeated. (See also **Session C Q1(b)**, where the same comparison is made and the continuing necessity of the one-sided formulae is discussed.)